Information Value Is Not Policy Value: When LLM Warm-Starts Help Linear Contextual Bandits

Anonymous Author(s)
Affiliation
Address
email

Abstract

A recent line of work reports "reward multiplication": routing the early rounds of a linear contextual bandit to a frozen LLM policy and then handing control to LinUCB can beat both the LLM and cold-started LinUCB, sometimes by a large factor. We give an exact accounting of when this can happen and of how large it can be. The regret difference between cold LinUCB and the handoff hybrid splits into the expert's direct advantage during the routed rounds plus the value of the data those rounds leave in LinUCB's design matrix; the difference between the expert and the hybrid is the adaptive learner's advantage on the tail. Data has value only through the confidence widths it lowers, widths fall only through coverage, and coverage has a price: the least regret any available action pays per unit of coverage of a direction. Our main theorem bounds the online multiplier over cold LinUCB, on any instance with a finite arm set, by the ratio of the price cold exploration pays for the directions the warm start certifies to the price the expert pays for the same coverage, plus cold LinUCB's regret on the arms the warm start does not certify. A diverging online multiplier therefore requires a diverging price ratio: an action of vanishing gap that covers a direction cold exploration can buy only dearly. The bound is attained, through either channel, on instances where such probes exist; and on the canonical separating instance of the literature, which multiplies offline by a factor of the horizon, no online handoff realizes anything, because every useful observation there costs a constant amount of regret. A catalog of eleven instances, each removing one ingredient, shows that recurrence, cheapness, correct targeting, and the expert's suboptimality on the tail are each necessary. We then show that this price accounting is *complete* under the model it assumes: when rewards are linear, the response sum of a warm start is determined by its Gram matrix, so two experts whose routed rounds cover the same directions leave warm starts with the same law. Any difference between equally covering experts is therefore a departure from the linear model, and we exhibit the departure in closed form: a *transfer* term, the ridge projection of the model's residuals onto the actions the expert chose, which is the only expert-dependent part of the estimate at handoff, and which we bound. A warm start therefore acts through two channels, coverage and transfer, and on real data the transfer is the larger. On AmazonCat-13K three LLM experts of equal accuracy raise the router's accuracy over cold LinUCB by factors of 1.71, 2.72 and 3.29, which is 6.5%, 16% and 21% less regret. Coverage alone saves about 5%, the same for all three experts, as the theory requires; the rest, and every significant difference between the experts, comes from the transfer, and removing it by any of three independent mechanisms collapses those differences. Which expert makes the better warm start is decided by the residuals its choices carry, not by its accuracy.

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1 Introduction

Sequential decision making studies how an agent adapts to an unknown environment through repeated interaction [61, 63], with applications in recommendation [35, 3], healthcare [53, 52], and dialogue [34]. Large pretrained language models supply prior knowledge that can guide early decisions [12, 38, 2, 8, 57].

A striking observation is that combining a frozen LLM with an adaptive linear bandit can strictly dominate either component, sometimes by orders of magnitude [13, 26]. Yet bandit warm-starting [10, 58] and confidence-gated model selection [9, 6, 39] are older objects, and that literature does not report comparable multipliers. The tension dissolves once one separates two things that are easy to conflate: the *policy value* of an expert (how much reward its actions earn) and the *information value* of an expert (how much its actions teach an adaptive learner that keeps their covariance). The two can be wildly different. A separating instance we construct to model the phenomenon documented by [13], stated as Proposition 2, is the cleanest illustration: a constant policy that is wrong on every round, and whose regret is therefore linear, produces a design matrix under which LinUCB's regret collapses from linear to constant. The expert there has no policy value at all; its data has enormous information value.

This paper makes that separation exact and uses it to say when a handoff from a frozen expert to LinUCB can multiply reward. Writing C for cold LinUCB, E for the frozen expert, and H_τ for the hybrid that follows E for τ rounds while feeding every observation to LinUCB and then hands over, we have the identities

$$R_T(\mathsf{C}) - R_T(\mathsf{H}_\tau) = A_\tau + V_\tau, \qquad R_T(\mathsf{E}) - R_T(\mathsf{H}_\tau) = B_\tau,$$

where A_τ is the expert's reward advantage over cold LinUCB on the routed rounds, V_τ is the regret LinUCB saves after the handoff because of the data the expert left behind, and B_τ is the adaptive learner's advantage over the frozen expert on the tail (Proposition 1). Reward multiplication over both base policies means $A_\tau + V_\tau$ and B_τ are simultaneously large. The first is where the trade-off lives: the expert's actions must be cheap (small regret while routed) and informative (large V_τ), and on many instances those two requirements pull in opposite directions.

What governs V_τ is a round-level quantity that is standard in the analysis of optimistic algorithms but rarely isolated. On the confidence event, the instantaneous regret of LinUCB at round t with design matrix V is at most

$$\Phi_t(V) \;:=\; \max_{a \in \mathcal{A}_t} \min\bigl\{\Delta_{t,a},\; 2\beta_t \|x_{t,a}\|_{V^{-1}}\bigr\},$$

the largest gap among competitors whose confidence width still exceeds their gap (Lemma 2). Everything that the prior literature attributes to "dimension," "coherence," or "gap structure" enters through Φ_t: the gap through the first argument of the minimum; the dimension through how many distinct directions can keep a width large; coherence through how much one observation shrinks the width of another direction, which is an exact Sherman–Morrison identity (Lemma 4); and a warm start through the Loewner ordering $(\lambda I + M)^{-1} \preceq (\lambda I)^{-1}$ (Lemma 3). We therefore do not posit separate structural axes. We derive them.

Two consequences follow. First, the determinant-ratio (elliptical potential) bound that underlies every $\widetilde{O}(\sqrt{rT})$-type guarantee cannot by itself certify a multiplicative gain: with $\tau = o(T)$ the ratio of the warm and cold determinant bounds tends to one, so any order-of-magnitude effect must come from competitors whose widths the warm start pushes *below their gaps*, exactly as in Proposition 2 (Proposition 14 and the discussion after it). Second, the offline separation of Proposition 2 has no online counterpart on the same instance: for *every* expert policy and every τ, the handoff hybrid has $V_\tau = 0$, because each round introduces a fresh direction that no earlier observation can cover (Proposition 3). Third, a warm start (M, c) enters LinUCB through two objects, the design matrix and the response sum, and under linear rewards the second is a deterministic function of the first plus noise whose law depends only on the first (Proposition 5). Coverage is therefore the whole of information value in the model our theory assumes, and any difference between equally covering experts is a misspecification effect with an explicit form (Section 3.4). On real data this second channel is not a correction: it carries most of the gain.

Widths fall only through coverage, and coverage has a price: for a direction u, the least regret any action in the current set pays per unit of $\langle x, u\rangle^2$ (Equation (1)). No policy, expert or learner, obtains coverage for less (Lemma 1), and cold LinUCB pays for each direction at the price of the arm

optimism forces it to pull, stopping once that arm's width falls below its gap. Our main theorem (Theorem 1) puts the two facts together: on any instance with a finite arm set, the multiplier of a handoff over cold LinUCB is at most the ratio of cold LinUCB's price to the expert's price on the directions the warm start certifies, plus cold LinUCB's regret on the arms it does not certify, divided by the acquisition regret. The confidence radius cancels. The theorem does not care whether the saving is collected during the routed rounds (A_τ) or afterwards (V_τ), nor whether the routing is fixed or adaptive, since it holds for any comparison policy at any cutoff, and it makes the shape of the answer precise: an online handoff multiplies by a *constant* that is a ratio of prices, and the constant diverges only if the expert has access to actions of vanishing gap that cover directions cold exploration can buy only dearly (Corollary 2). Offline warm starts, whose acquisition is not charged, are the only route to a multiplier of order the horizon.

We then show that the bound is attained. On an instance in which the directions that trap cold LinUCB recur and a cheap "probe" covering them is offered early, a frozen expert that plays the probes and hands off has regret smaller than cold LinUCB's by a factor of order $c^2/(\varepsilon L^2)$, the ratio of the bad arm's price $1/L^2$ to the probe's price ε/c^2, and smaller than the expert's own by a factor that grows with the horizon (Theorem 2); the same ratio is realized on the tail when the routed window keeps cold LinUCB busy elsewhere (Proposition 9). Making the probe dear collapses the multiplier to at most two. A catalog of eleven instances (Section 3.1, Table 1), all built from the canonical one by changing a single ingredient, isolates what each clause of this mechanism contributes, and each experimental configuration of Section 4 has a row in it.

Contributions.

- **A price bound for the online multiplier** (Theorem 1): on any finite-arm instance the advantage of a handoff over cold LinUCB is at most a ratio of coverage prices on the certified directions plus the uncertified part of cold LinUCB's regret, so diverging online multipliers require vanishing prices (Corollary 2).
- **An exact decomposition of the hybrid's advantage** (Proposition 1) into the expert's direct advantage A_τ, the value of the transferred data V_τ, and the adaptive tail advantage B_τ, with the round-level quantity that governs V_τ (Lemma 2) and exact identities for how warm starts and ordinary exploration reduce widths (Lemmas 3 and 4).
- **A separation of information value from policy value.** Proposition 2 (offline transfer by a factor of the horizon) is paired with Proposition 3 (no online transfer on the same instance, for every expert, randomized or adaptive).
- **Instances attaining the price bound** through each channel (Theorem 2, Proposition 9), and a catalog of eleven instances with complete proofs (Section 3.1, Appendix C) that shows which ingredient each negative case is missing and places each experimental configuration.
- **Two channels, and the transfer in closed form** (Section 3.4): under linear rewards the response sum of a warm start is determined by its Gram matrix, so the price ratio is a complete accounting of information value; under misspecification the estimate acquires a *transfer* term (Proposition 6), the ridge projection of the model's residuals onto the expert's chosen actions, which is the only expert-dependent part of the warm start, together with a regret bound that holds against any reference parameter (Proposition 7).
- **Experiments showing that the transfer is the larger channel in practice** (Section 4), on two datasets and twelve dataset/model configurations (ten evaluation seeds per row, with every confidence parameter chosen on a disjoint block of ten tuning seeds). We measure the decomposition directly and report both the accuracy multiplier that the literature calls reward multiplication and the regret ratio that the theorems bound. Coverage delivers a small gain that is the same for every expert; the transfer carries most of the gain and every significant difference between experts of equal accuracy.

A detailed related-work discussion is in Appendix A.

2 Preliminaries

We work in a standard linear contextual bandit [1, 15, 31]. At each round $t = 1, \dots, T$, the learner observes a finite action set $\mathcal{A}_t$ where each $a \in \mathcal{A}_t$ has a feature vector $x_{t,a} \in \mathbb{R}^d$ with $\|x_{t,a}\|_2 \le L$,

selects $A_t \in \mathcal{A}_t$, and observes $r_t = \langle \theta^\star, x_{t,A_t} \rangle + \eta_t$, where $\theta^\star$ is unknown with $\|\theta^\star\|_2 \le S$ and η_t is conditionally mean-zero and σ-sub-Gaussian. Write $\mu_t(a) = \langle \theta^\star, x_{t,a} \rangle$, $a_t^\star \in \arg\max_a \mu_t(a)$, and $\Delta_{t,a} = \mu_t(a_t^\star) - \mu_t(a)$ for the suboptimality gap of a at round t. For a policy π and an interval $I \subseteq [T]$ of rounds, $R_I(\pi) = \sum_{t \in I} \Delta_{t,\pi_t}$ is its pseudo-regret on I, and $R_T(\pi) = R_{[T]}(\pi)$. The language aspect of the applications we target enters only through the feature map: context and action embeddings from pretrained models are combined into $x_{t,a}$, and the frozen expert may be an LLM. Nothing in the theory uses language structure.

LinUCB and its warm start. LinUCB maintains $V_t = V_1 + \sum_{s<t} x_{s,A_s} x_{s,A_s}^\top$, $b_t = b_1 + \sum_{s<t} r_s x_{s,A_s}$, $\widehat{\theta}_t = V_t^{-1} b_t$, and plays $A_t \in \arg\max_a \langle \widehat{\theta}_t, x_{t,a} \rangle + \beta_t \|x_{t,a}\|_{V_t^{-1}}$, where $\|x\|_{V^{-1}} = \sqrt{x^\top V^{-1} x}$ is the confidence width of x and β_t is the confidence radius. *Cold-started* LinUCB, written C, uses $V_1 = \lambda I_d$, $b_1 = 0$. A *warm start* (M, c) uses $V_1 = \lambda I_d + M$ and $b_1 = c$ with $M = \sum_s x_s x_s^\top \succeq 0$ and $c = \sum_s r_s x_s$ the Gram matrix and response sum of some logged data. We say the warm start is *offline* if that data is not counted in the evaluated horizon and *online* if it is.

Frozen expert and handoff hybrid. A frozen expert E proposes $a_t^{\mathsf{E}} \in \mathcal{A}_t$ at each round by a rule that is never updated (for an LLM, the action whose embedding is closest to the LLM's output; see Section 4.1). For $\tau \in \{0, \dots, T\}$ the *handoff hybrid* H_τ plays a_t^{E} for $t \le \tau$, updates (V_t, b_t) with every played action and observed reward including those rounds, and plays LinUCB for $t > \tau$. Thus $\mathsf{H}_0 = \mathsf{C}$ and $\mathsf{H}_T = \mathsf{E}$, and the LinUCB phase of H_τ is LinUCB with the online warm start $(M_\tau, c_\tau) = \big(\sum_{t \le \tau} x_{t,A_t} x_{t,A_t}^\top, \sum_{t \le \tau} r_t x_{t,A_t}\big)$. The deterministic router used in our experiments (Appendix F) chooses τ adaptively and may interleave the two; the theory below is for a fixed τ, which is the right object for understanding why a multiplier appears, and we comment on adaptive τ after Proposition 14.

3 Main Results

3.1 The Decomposition

Everything in this paper is organized by one identity.

Setup for the decomposition. Recall the three policies: C is LinUCB initialized from λI_d (no prior data); E is a frozen expert that proposes actions by a fixed rule (e.g. an LLM) and is never updated; and H_τ is the handoff hybrid that follows E for the first τ rounds, feeds every observation into LinUCB's state (V_t, b_t), and then hands control to LinUCB for rounds $\tau+1, \dots, T$. For any policy π and interval I, we write $R_I(\pi) = \sum_{t \in I} \Delta_{t,\pi_t}$ for its pseudo-regret on I, and $R_T(\pi) = R_{[T]}(\pi)$.

Proposition 1 (Decomposition of the hybrid's advantage)**.** *For any instance, any expert* E*, any* τ*, and any realization of the noise,*

$$R_T(\mathsf{C}) - R_T(\mathsf{H}_\tau) = \underbrace{R_{[1,\tau]}(\mathsf{C}) - R_{[1,\tau]}(\mathsf{E})}_{A_\tau} + \underbrace{R_{[\tau+1,T]}(\mathsf{C}) - R_{[\tau+1,T]}(\mathsf{H}_\tau)}_{V_\tau},$$

$$R_T(\mathsf{E}) - R_T(\mathsf{H}_\tau) = \underbrace{R_{[\tau+1,T]}(\mathsf{E}) - R_{[\tau+1,T]}(\mathsf{H}_\tau)}_{B_\tau}.$$

A_τ is the expert's *direct* advantage: positive if the expert earns more than cold LinUCB would have on the routed rounds, negative if the expert's actions cost more regret than cold exploration. V_τ is the *information* value of the routed rounds: it is nonzero only through the warm start (M_τ, c_τ) that they leave in LinUCB's state, since apart from that state H_τ and C run the same algorithm on the same future. B_τ is the adaptive learner's advantage on the tail and is large only when the expert remains substantially suboptimal late while the warm-started learner does not. We write $R_{[1,\tau]}(\mathsf{E}) = \sum_{t \le \tau} \Delta_{t,a_t^{\mathsf{E}}}$ for the *acquisition regret* of the warm start: the price, counted in the online horizon, of the data in M_τ.

Remark 1. The identities of Proposition 1 hold pathwise in the reward-table model of Section 3.5, in which two policies can be run on the same noise, and therefore also in expectation. They also hold when the routed rounds are not an initial segment: for any set $\mathcal{T} \subseteq [T]$ of rounds on which the hybrid follows the expert, fixed in advance or chosen adaptively (as by the router of Appendix F), replacing

#	Instance	A_τ	V_τ	B_τ	Ratio to C	Ingredient isolated
1	canonical, offline (Prop. 2)	not charged	$V^{\mathrm{off}} \asymp T$	$B^{\mathrm{off}} \asymp T$	$\asymp T$	data is free
2	canonical, online (Prop. 3)	$\le \tau$	0	≤ 0	none	directions never recur
3	canonical, oracle (Cor. 1)	τ	0	$-(T-\tau)$	additive	expert too good
4	recurring, bad-arm expert (Prop. 8)	0	0	≥ 0	≤ 961	same price as cold
5	recurring, cheap probes (Thm. 2)	$\ge Km_0 - \varepsilon\tau_0$	≥ 0	$T-\tau_0$	$\asymp c^2/(\varepsilon L^2)$	price ratio, in the window
5′	as 5, cold kept busy in the window (Prop. 9)	$-\varepsilon\tau_0$	$\ge Km_0$	$T-\tau_0$	$\asymp c^2/(\varepsilon L^2)$	price ratio, on the tail
6	as 5, expert optimal on tail (Prop. 10)	as 5	as 5	0	as 5	expert optimal late
7	nuisance probes (Prop. 11)		$< R_{[\tau_0+1,T]}(\mathsf{C})$		≤ 65	wrong directions
8	as 5 with $\varepsilon \ge c^2/L^2$ (Thm. 2)				≤ 2	probe too dear
9	cold already cheap (Prop. 12)				$o(1)$ avg. reward	nothing to buy
10	bad arms with gap η (Prop. 13)				$\le 1 + c^2/(\varepsilon L^2)$	gap rescales prices

Table 1: The catalog: what the three terms of Proposition 1 can be. Row 1 is the offline identity, in which A is not charged; rows 2–10 are online handoffs H_τ. Rows 3 and 6 kill B_τ from the two sides. Rows 4, 7, 8 kill the multiplier over cold LinUCB in three different ways: the expert buys what cold buys at the same price, buys directions nobody will need, or buys the right directions too dearly. Row 9 is the case in which cold LinUCB's own bill is already negligible in average reward. Row 10 shows that the gap is a rescaling of both prices. Rows 5 and 5′ are the positive cases; they realize the same price ratio, collected as A_τ in row 5 and as V_τ in row 5′, and Theorem 1 shows no instance exceeds a price ratio. Blank cells are not claimed. Constants are those proved in Appendix C, not optimized.

$[1,\tau]$ by $\mathcal{T}$ and $[\tau+1,T]$ by its complement gives the same two identities, because each side is a difference of sums over a partition of $[T]$. In particular, $R_T(\mathsf{H}_\tau) \le \min\{R_T(\mathsf{C}), R_T(\mathsf{E})\}$ if and only if $A_\tau + V_\tau \ge 0$ and $B_\tau \ge 0$, and $\min\{R_T(\mathsf{C}), R_T(\mathsf{E})\}/R_T(\mathsf{H}_\tau) \to \infty$ along a sequence of instances—formally, a sequence $(d_T, \theta_T^\star, \{\mathcal{A}_t^{(T)}\}_{t\le T}, \mathsf{E}_T, \tau_T)_{T\to\infty}$, which is how every diverging ratio in this paper is to be read—if and only if both $A_\tau + V_\tau$ and B_τ exhaust the corresponding base regret up to lower-order terms. If instead the warm start (M, c) is built from data outside the horizon (an *offline* warm start $\mathsf{H}^{\mathrm{off}}$, LinUCB started from $(\lambda I + M, c)$), the routed rounds are not charged and the identities read $R_T(\mathsf{C}) - R_T(\mathsf{H}^{\mathrm{off}}) =: V^{\mathrm{off}}$ and $R_T(\mathsf{E}) - R_T(\mathsf{H}^{\mathrm{off}}) =: B^{\mathrm{off}}$, with no A term.

The proposition is bookkeeping and says nothing by itself. Its content is in what the three terms can be. Reward multiplication over both base policies means $A_\tau + V_\tau$ is almost all of $R_T(\mathsf{C})$ and B_τ is almost all of $R_T(\mathsf{E})$, so every question about when a handoff multiplies is a question about which of A_τ, V_τ, B_τ can be large at once. We answer it with eleven instances, each obtained from the canonical one (Proposition 2) by changing a single ingredient (Table 1); exactly two of them multiply over both base policies, they realize the same price ratio through the two channels A_τ and V_τ (Theorem 2, Proposition 9), and Theorem 1 says that no instance can do better than a price ratio. The pattern across the rows is the following.

> **Thesis.** A handoff multiplies regret savings over cold LinUCB if and only if the expert's routed actions buy coverage of the directions cold LinUCB would otherwise pay for, at a lower price than cold exploration pays, early enough for those directions to recur; and it multiplies over the expert if and only if, in addition, the expert is worse than the warm-started learner on the tail.

The catalog has eleven rows rather than the $2^3 = 8$ of a binary table because several terms admit qualitatively distinct failure modes—for instance, V_τ can vanish because the expert buys the same directions cold exploration buys at the same price (row 4), because it buys the wrong directions (row 7), or because it buys the right directions too dearly (row 8)—because the offline/online distinction (rows 1–2) and the two ways of killing the tail term (rows 3 and 6) each warrant separate entries, and because the positive case appears twice, with the saving collected in the routed window (row 5) and on the tail (row 5′).

3.2 Rows 1–3: The Canonical Instance

All proofs are in Appendices B and C.

Rows 1–2: the canonical instance, offline and online. The canonical instance, which we construct to model the separation reported by [13], uses horizon n, a single reward direction $g = e_1$, and n orthogonal "bad" arms $b_t = Le_{t+1}$ with $L > 1$.

Proposition 2 (Offline three-way separation)**.** *Fix $n \geq 1$, $L > 1$, $\lambda > 0$. There exist $d \geq n+1$, $\theta^\star$, and two-action sets $\{\mathcal{A}_t\}_{t=1}^n$ such that, for any sub-Gaussian noise: (1) a constant policy π_{const} has pseudo-regret exactly n; (2) cold LinUCB also incurs regret n, selecting the same action as π_{const} each round; (3) for m large enough, LinUCB warm-started with the design matrix that π_{const} generates from mn* offline *rollouts incurs $O(1)$ regret with probability $\geq 1-\delta$.*

The proof (Appendix B.2) is short and worth reading through the lens of Section 2. The true parameter is as simple as possible: one reward-bearing direction. Cold LinUCB fails anyway, because at each round the fresh arm b_t has width $\|b_t\|_{V_t^{-1}} = L/\sqrt{\lambda}$ regardless of what was played before, while g has width $1/\sqrt{\lambda}$ and estimated value 0, so optimism picks b_t every time. The offline design matrix $M = mL^2 \sum_{j\geq 2} e_j e_j^\top$ does not teach LinUCB that g is good; its response sum is zero in expectation. It teaches LinUCB that the *alternatives* are not uncertain: $\|b_t\|^2_{(\lambda I+M)^{-1}} = L^2/(\lambda + mL^2)$, which falls below the gap once m is moderate, and LinUCB then plays g. In the language of Proposition 1 this is the offline identity with $V^{\mathrm{off}} \asymp T$ and $B^{\mathrm{off}} \asymp T$; the acquisition of the data is not charged.

The same instance is hostile to online acquisition. Every round introduces a direction that no earlier observation touches, so the data any expert can collect during the first τ rounds says nothing about the arms that appear after round τ.

Proposition 3 (No online transfer on the canonical instance)**.** *Consider the instance of Proposition 2 with horizon T and confidence radius β, and let $\kappa := \sigma\sqrt{2\log(2T/\delta)}/\sqrt{\lambda}$. If $\beta(L-1)/\sqrt{\lambda} > 1+\kappa$, then with probability at least $1-\delta$, for* every *expert policy E and every $\tau \in \{0,\dots,T\}$, the handoff hybrid H_τ plays b_t at every round $t > \tau$. Consequently $R_{[\tau+1,T]}(\mathsf{H}_\tau) = R_{[\tau+1,T]}(\mathsf{C}) = T-\tau$ and $V_\tau = 0$.*

The condition on β is the same optimism condition that makes cold LinUCB fail in the first place (it only strengthens the inequality $\beta L/\sqrt{\lambda} > \beta/\sqrt{\lambda}$ by the largest value the estimate of g can take). The proposition says that the offline separation of Proposition 2 cannot be realized by any online router on this instance: not by a bad expert, whose data has $V_\tau = 0$, and not by a good one, which has $B_\tau \leq 0$ because it is already optimal. Online multiplication needs a third kind of expert, whose routed actions are cheap *and* cover directions that will matter later; row 5 constructs one.

Row 3: an oracle expert on the canonical instance.

Corollary 1 (Oracle expert on the canonical instance)**.** *On the instance of Proposition 3, under its hypothesis, let E play g at every round. With probability at least $1-\delta$, for every τ: $R_T(\mathsf{E}) = 0$, $R_T(\mathsf{H}_\tau) = T-\tau$, $A_\tau = \tau$, $V_\tau = 0$, $B_\tau = -(T-\tau)$. The hybrid improves on cold LinUCB by exactly the rounds it routes and on the expert by nothing.*

3.3 Prices and the Main Theorem

Rows 4–10 share a vocabulary. The word "price" has a precise meaning. For a direction $u \in \mathbb{R}^d$ and an action set $\mathcal{A}_t$, write

$$\pi_t(u) \;:=\; \min\Big\{ \frac{\Delta_{t,a}}{\langle x_{t,a}, u\rangle^2} :\; a \in \mathcal{A}_t,\; \langle x_{t,a}, u\rangle \neq 0 \Big\}, \qquad \pi(u) := \inf_t \pi_t(u), \tag{1}$$

the least pseudo-regret any policy must pay at round t to add one unit to the coverage $\sum_s \langle x_{s,A_s}, u\rangle^2$ of u. Coverage is what reduces widths (Lemma 3), so $\pi(u)$ is the exchange rate between regret and width reduction in direction u. The following lemma says that no policy, expert or otherwise, gets coverage for less.

Lemma 1 (No free coverage)**.** *For any policy π, any interval I of rounds, and any direction u, $\sum_{t\in I}\langle x_{t,\pi_t}, u\rangle^2 \leq R_I(\pi)/\pi(u)$. Consequently, if H_τ hands off with design matrix $V_{\tau+1} = \lambda I + M_\tau$, then for every unit vector u, $\|u\|^2_{V_{\tau+1}^{-1}} \;\geq\; 1/\big(\lambda + R_{[1,\tau]}(\mathsf{E})/\pi(u)\big)$.*

The lemma is one half of the main theorem. The other half is that cold LinUCB buys coverage of each direction at the price of the arm optimism forces it to pull, and stops once that arm's width falls below its gap. Together they bound the online multiplier by a ratio of prices on the directions the warm start certifies, plus cold LinUCB's regret on the directions it does not.

Setting for the theorem. Let $\mathcal{X} \subset \mathbb{R}^d$ be a finite set of arm vectors with $\|x\|_2 \le L$, let $\mathcal{A}_t \subseteq \mathcal{X}$ for every t, and suppose an arm $x^\star \in \mathcal{X}$ with $\mu(x^\star) = \max_{x\in\mathcal{X}} \mu(x)$ belongs to every $\mathcal{A}_t$. Then the gap of an arm is the same at every round, $\Delta(x) := \mu(x^\star) - \mu(x)$, and we write $S := \{x \in \mathcal{X} : \Delta(x) > 0\}$. For $x \in S$ put $u_x := x/\|x\|_2$ and

$$p(x) := \frac{\Delta(x)}{\|x\|_2^2}, \qquad c_x := \frac{4\beta^2\|x\|_2^2}{\Delta(x)^2}, \qquad N_x := \#\{t \in [T] : x \in \mathcal{A}_t\},$$

where $\beta := \beta_T$ is the largest confidence radius. Here $p(x)$ is the price of covering u_x by pulling x itself, and c_x is the coverage of u_x at which the width of x equals half its gap, so that the confidence bound alone excludes x once the coverage $\sum_s \langle x_s, u_x\rangle^2$ exceeds $c_x - \lambda$. For a set U of unit vectors write $\mathrm{cov}_U(x) := \sum_{u\in U}\langle x, u\rangle^2$. The theorem compares cold LinUCB with an arbitrary policy π at an arbitrary cutoff $t_0 \in \{0, \dots, T\}$; the handoff hybrid H_τ with $t_0 = \tau$ is the case of interest, and then rounds $1, \dots, t_0$ are the expert's, but nothing in the statement or proof uses this. Write x_t^π for the arm π plays at round t and define

$$\pi_U(t_0) := \min\Big\{\frac{\Delta(x_t^\pi)}{\mathrm{cov}_U(x_t^\pi)} : t \le t_0,\ \Delta(x_t^\pi) > 0,\ \mathrm{cov}_U(x_t^\pi) > 0\Big\}, \qquad C_U^{\mathrm{free}}(t_0) := \sum_{t\le t_0:\,\Delta(x_t^\pi)=0} \mathrm{cov}_U(x_t^\pi),$$

the price π pays for coverage of U up to t_0 (the minimum over an empty set is $+\infty$, and then $1/\pi_U(t_0) := 0$), and the coverage of U that π obtains for free by playing optimal arms. Finally let $U_{\mathrm{cert}}(t_0)$ be the set of directions u_x, $x \in S$, with $2\beta\|x\|_{(V_{t_0+1}^\pi)^{-1}} < \Delta(x)$, where $V_{t_0+1}^\pi = \lambda I + \sum_{t\le t_0} x_t^\pi (x_t^\pi)^\top$ is π's design matrix after round t_0: the directions π's first t_0 rounds certify. For $\pi = \mathsf{H}_\tau$ and $t_0 = \tau$ this is the hybrid's matrix at handoff, $\pi_U(\tau)$ is the expert's price, and $R_{[1,t_0]}(\pi)$ is the acquisition regret.

Theorem 1 (Price bound for the online multiplier)**.** *In the setting above, on the confidence event of cold LinUCB, the following hold for every policy π, every cutoff $t_0 \in \{0, \dots, T\}$, and every $U \subseteq U_{\mathrm{cert}}(t_0)$.*

(a) (Cold LinUCB pays its own price, once per direction.) $R_T(\mathsf{C}) \le \sum_{x\in S} \Delta(x)\min\{m_x, N_x\}$, *where $m_x := (c_x - \lambda)/\|x\|_2^2 + 1$ if $c_x \ge \lambda$ and $m_x := 0$ otherwise.*

(b) (Certification costs at least the price.)

$$\sum_{u_x\in U} (c_x - \lambda)_+ \;\le\; \sum_{u_x\in U}\sum_{t\le t_0} \langle x_t^\pi, u_x\rangle^2 \;\le\; \frac{R_{[1,t_0]}(\pi)}{\pi_U(t_0)} + C_U^{\mathrm{free}}(t_0).$$

(c) (Price bound.) *With $p_U := \max_{u_x\in U} p(x)$ and the price ratio $P_U := p_U/\pi_U(t_0)$,*

$$R_T(\mathsf{C}) \;\le\; P_U\, R_{[1,t_0]}(\pi) \;+\; p_U\, C_U^{\mathrm{free}}(t_0) \;+\; \sum_{u_x\in U} \Delta(x) \;+\; \sum_{x\in S:\,u_x\notin U} \Delta(x)\min\{m_x, N_x\}.$$

Since $R_T(\pi) \ge R_{[1,t_0]}(\pi)$, whenever $R_{[1,t_0]}(\pi) > 0$,

$$\frac{R_T(\mathsf{C})}{R_T(\pi)} \;\le\; P_U \;+\; \frac{p_U\, C_U^{\mathrm{free}}(t_0) + \sum_{u_x\in U}\Delta(x) + \sum_{x\in S:\,u_x\notin U}\Delta(x)\min\{m_x, N_x\}}{R_{[1,t_0]}(\pi)}. \tag{2}$$

For $\pi = \mathsf{H}_\tau$, $t_0 = \tau$, this bounds the multiplier of the handoff hybrid by the price ratio between cold exploration and the expert on the directions the warm start certifies.

The proof is in Appendix B.9; part (a) is the width-limited lemma of Section 3.6 combined with the Sherman–Morrison identity, part (b) is Lemma 1 summed over the certified directions, and (c) is arithmetic. Only the confidence event of cold LinUCB is used: certification is a property of the matrix $V_{t_0+1}^\pi$ and (c) does not require π to exploit it. Because π is arbitrary, the bound covers the fixed-τ hybrid, the adaptive router of Appendix F (take t_0 to be its last routed round), a CORRAL-style master, and cold LinUCB itself; every one of them beats cold LinUCB by at most the price ratio of the coverage it has certified by t_0, plus the uncertified residual. We stress that (2) bounds the ratio $R_T(\mathsf{C})/R_T(\pi)$ of the two policies' regrets on the same instance, rather than comparing an upper bound for one against an upper bound for the other: two policies sharing an asymptotic upper bound may still differ by a diverging factor, so a statement about the multiplier cannot be obtained from matching rates and must bound one policy's regret in terms of the other's, as (c) does.

Remark 2 (Two relaxations). (i) The factor 2 in the definition of U_{cert} is the crude one. On the confidence event with the true self-normalized radius $\beta_t^{\mathrm{true}} \le \beta$, the proof of Lemma 2 gives $\Delta_{t,A_t} \le (\beta_t + \beta_t^{\mathrm{true}})\|x_{t,A_t}\|_{V_t^{-1}}$ for the played arm, so the theorem holds with c_x replaced by $(\beta+\beta^{\mathrm{true}})^2\|x\|_2^2/\Delta(x)^2$ throughout and U_{cert} enlarged to the directions with $(\beta+\beta^{\mathrm{true}})\|x\| < \Delta(x)$; P_U is unchanged. (ii) If the optimal arm is not offered at every round, so that the gap of x varies with t, write $\Delta^-(x)$ and $\Delta^+(x)$ for its smallest and largest values over rounds at which x is offered. Part (a) holds with c_x computed from $\Delta^-(x)$ and each pull charged $\Delta^+(x)$, and (b) with $\Delta^-(x)$; the price ratio then carries a factor $\Delta^+(x)/\Delta^-(x)$, which is 1 under the standing assumption.

The residual in (2) has three parts: the coverage the expert obtains for free by playing optimal arms (zero when the optimal arm is orthogonal to the certified directions, as on every instance of this section), one pull of each certified arm, and cold LinUCB's regret on the arms whose widths the warm start does not push below their gaps. Only the last can be large, and it is exactly the part of the hybrid's advantage that is not bought at a price: on those arms the hybrid faces the same confidence bound as cold LinUCB, and no coverage argument lower-bounds its tail regret on them. Theorem 1 is therefore a statement about the certified saving, and it holds whether that saving is collected in the routed window (A_τ) or on the tail (V_τ): rows 5 and 5′ below realize the same price ratio through the two channels. The confidence radius cancels in P_U because it scales c_x, the coverage both policies must reach, identically.

Corollary 2 (Diverging online multipliers require vanishing prices)**.** *Consider a sequence of instances, policies π and cutoffs t_0 along which $R_T(\mathsf{C})/R_T(\pi) \to \infty$ and, for some choice of $U \subseteq U_{\mathrm{cert}}(t_0)$ on each instance, the residual in* (2) *is at most a fixed multiple of $R_{[1,t_0]}(\pi)$. Then $P_U \to \infty$: the price $\pi_U(t_0)$ that π pays for the certified coverage vanishes relative to the price p_U cold LinUCB pays for it, so π plays actions whose gap is negligible against the coverage they supply. For the handoff hybrid this means the expert has access to such actions. On the canonical instance of Proposition 2 no online expert certifies any direction that is offered after the handoff, so $U_{\mathrm{cert}}(\tau) = \emptyset$ for every $\tau < T$ and all of cold LinUCB's regret is in the residual; that is the price-theoretic content of Proposition 3. Offline, where the acquisition is not charged, the same instance multiplies by a factor of the horizon.*

Proof. From (2), $P_U \ge R_T(\mathsf{C})/R_T(\pi) - C$ for the fixed multiple C, so $P_U \to \infty$; since $p_U \le \max_{x\in S} p(x)$ is a property of the instance, this is $\pi_U(t_0) \to 0$ relative to p_U. On the canonical instance the arms offered after round τ are $b_t = Le_{t+1}$, $t > \tau$, and no arm played at or before round τ has a component along e_{t+1}, so $\|b_t\|_{V_{\tau+1}^{-1}} = L/\sqrt{\lambda}$ and $2\beta L/\sqrt{\lambda} > 1 = \Delta(b_t)$ under the hypothesis of Proposition 3; hence no such b_t is certified. □

When do cheap probes exist? The positive rows need an action of small gap whose feature vector carries a non-negligible component along a direction that cold LinUCB can otherwise cover only by pulling an arm of large gap. Two observations make this less special than it looks. First, the probe need not be a single designated arm: any near-optimal action whose embedding overlaps the bad direction serves, and in the embedding spaces used in practice (Section 4) near-optimal actions overlap many others. Second, the expert does not need to know that the probe is informative; it needs only to prefer it, which any policy with prior knowledge of the reward does when the probe is nearly as good as the best arm. What the expert supplies, and cold optimism cannot, is the *order* in which coverage is bought: optimism ranks a fresh bad arm above a cheap probe because the bad arm's width is larger, and pays the bad arm's price first (Lemma 10 in the appendix); a policy that already believes the bad arm is bad takes the probe at once. The price ratio $c^2/(\varepsilon L^2)$ of the rows below is the exchange rate between those two orders, and Theorem 1 says it is the only exchange rate available: an expert whose actions are not cheaper per unit of certified coverage than cold exploration's cannot multiply, however accurate it is.

3.4 The Two Channels of a Warm Start

A warm start (M, c) enters LinUCB through two objects: the design matrix $V_1 = \lambda I + M$, which fixes every confidence width, and the response sum $b_1 = c$, which fixes the estimate $\widehat{\theta}_1 = V_1^{-1}c$. Everything so far has bounded the value of a warm start through the first. We now separate the two. The conclusion is that in the model of Section 2 the first is the whole story, so Theorem 1 is a complete accounting of information value there; and that the second exists only when the reward is not linear in the features, in which case it has an explicit form.

Proposition 4 (Width-limited regret, two channels)**.** *At a round t at which LinUCB acts, let $\rho_t := \|\widehat{\theta}_t - \theta^\star\|_{V_t}$ be the realized estimation error, so $\rho_t \le \beta_t$ on $\mathcal{E}_\delta$. Then on $\mathcal{E}_\delta$*

$$\Delta_{t,A_t} \;\le\; \Phi_t(V_t, \rho_t) \;:=\; \max_{a \in \mathcal{A}_t} \min\big\{\Delta_{t,a},\; (\beta_t + \rho_t)\|x_{t,a}\|_{V_t^{-1}}\big\}, \tag{3}$$

and $\Phi_t(V, \rho)$ is non-increasing in V and in ρ. A warm start acts on the first argument through M and on the second through c. Since $0 \le \rho_t \le \beta_t$ we have $\Phi_t(V_t, \rho_t) \ge \frac{1}{2}\Phi_t(V_t)$: the response sum can tighten the width-limited bound by at most a factor of two, whereas the design matrix can reduce it by the price ratio, which is unbounded.

Proposition 5 (The response sum is determined by the design matrix)**.** *Let the warm start be built from pairs $(x_s, r_s)_{s \le \tau}$ with $r_s = \langle \theta^\star, x_s \rangle + \eta_s$ and η_s i.i.d. $\mathcal{N}(0, \sigma^2)$. Then*

$$c = M\theta^\star + \xi, \qquad \xi := \sum_{s \le \tau} \eta_s x_s \sim \mathcal{N}(0, \sigma^2 M), \qquad \widehat{\theta}_1 - \theta^\star = -\lambda V_1^{-1}\theta^\star + V_1^{-1}\xi.$$

Hence the law of (V_1, b_1) given M does not depend on which policy generated the logged actions: two experts, or two routed sets, with the same Gram matrix leave warm starts with the same law, and the tail trajectories of the corresponding hybrids—hence V_τ and B_τ—have the same law.

For σ-sub-Gaussian noise the distributional identity is replaced by the observation that the self-normalized bound of [1], the only property of ξ used in this paper, depends on the features only through M; every bound in Sections 3.6–3.7 is therefore unchanged when one expert is exchanged for another of equal coverage.

Corollary 3 (Coverage is complete under linear rewards)**.** *Under the model of Section 2, two experts whose routed rounds leave the same Gram matrix have the same V_τ in law. Any difference in V_τ between experts of equal coverage is a departure from the linear model.*

Proof. Immediate from Proposition 5: equal Gram matrices give warm starts (V_1, b_1) of the same law, and the LinUCB phase is a deterministic function of (V_1, b_1) and of the future contexts and noise, which are independent of the warm start. □

Misspecification opens the second channel. Suppose $r(x) = \langle \theta^\star, x \rangle + \varepsilon(x) + \eta$ for a fixed reference $\theta^\star$—for instance the best linear predictor—and a residual ε; the binary Precision@1 reward of Section 4 is of this form for every $\theta^\star$, and badly so, since no linear function of $\phi(x, a)$ reproduces an indicator. Regret is still measured against the true best action, with $\bar{r} := \langle \theta^\star, \cdot \rangle + \varepsilon(\cdot)$ and $\Delta_{t,a} = \bar{r}(x_{t,a_t^\star}) - \bar{r}(x_{t,a})$.

Proposition 6 (The transfer)**.** *Under the misspecified model a warm start built from $(x_s, r_s)_{s \le \tau}$ satisfies*

$$\widehat{\theta}_1 = \theta^\star - \lambda V_1^{-1}\theta^\star + m + V_1^{-1}\xi, \qquad m := V_1^{-1} \sum_{s \le \tau} \varepsilon(x_s)\, x_s.$$

Given M, the transfer *m is the only term that depends on which actions were logged: it is the ridge projection of the residuals the expert's own picks incur. Under linear rewards $\varepsilon \equiv 0$, $m = 0$, and Proposition 5 is recovered.*

Proposition 7 (What the transfer can buy)**.** *Under the misspecified model, at a round t at which LinUCB acts and for any reference $\widetilde{\theta} \in \mathbb{R}^d$, write $\widetilde{\rho}_t := \|\widehat{\theta}_t - \widetilde{\theta}\|_{V_t}$ and $\widetilde{\varepsilon}(x) := \bar{r}(x) - \langle \widetilde{\theta}, x \rangle$. Then*

$$\Delta_{t,A_t} \;\le\; (\beta_t + \widetilde{\rho}_t)\|x_{t,A_t}\|_{V_t^{-1}} \;+\; (\widetilde{\rho}_t - \beta_t)_+ \|x_{t,a_t^\star}\|_{V_t^{-1}} \;+\; \big[\widetilde{\varepsilon}(x_{t,a_t^\star}) - \widetilde{\varepsilon}(x_{t,A_t})\big],$$

simultaneously for every reference. The design matrix fixes the widths; the response sum fixes $\widetilde{\rho}_t$ for every reference at once. The transfer helps exactly when it moves $\widehat{\theta}$ toward a reference whose residual is smaller on the optimal arm than on the arms LinUCB would otherwise play, and the bound is width-limited whenever some reference has $\widetilde{\rho}_t \le \beta_t$ and a nonpositive residual difference on the played arm.

Proposition 7 is a statement about what the second channel *can* deliver, not a prediction of what a given expert will deliver: m depends on the residuals at the actions the expert chooses, which are not known before it is run. Section 4.3 measures m on real data, finds that it accounts for essentially the whole estimate at handoff, and shows that removing it collapses the differences between experts of equal accuracy; Appendix G reports what this costs us in predictive power.

3.5 The Template and Rows 4–10

The template. All instances below are built from the canonical one by three modifications: directions may recur, a cheap informative action may be available, and that action may disappear. Fix $K \geq 1$ and set $d = K + 1$ (or $d = 2K + 1$ when nuisance directions are present). The parameter is $\theta^\star = e_1$. Directions cycle: $j(t) := 2 + ((t-1) \bmod K)$. Three actions may be offered at round t:

$$g := e_1 \ (\mu = 1), \qquad b_t := L e_{j(t)} \ (\mu = 0), \qquad p_t := (1-\varepsilon)e_1 + c\, e_{j(t)} \ (\mu = 1-\varepsilon),$$

the *good* arm, the *bad* arm, and the *probe*. The instance $\mathcal{I}(K, L, c, \varepsilon, \tau_0)$ has $\mathcal{A}_t = \{g, b_t, p_t\}$ for $t \leq \tau_0$ and $\mathcal{A}_t = \{g, b_t\}$ for $t > \tau_0$. The price of e_j, $j \geq 2$, is $\min\{1/L^2, \varepsilon/c^2\}$ while probes are offered and $1/L^2$ afterwards. The expert is the frozen linear policy E_w that plays $\arg\max_{a \in \mathcal{A}_t} \langle \widehat{\theta}_{\mathsf{E}}, x_{t,a} \rangle$ with $\widehat{\theta}_{\mathsf{E}} = e_1 + w \sum_{j \geq 2} e_j$; ties do not occur for the values of w we use. Throughout, $\tau_0 = Kn$ for an integer n, so each direction is probed exactly n times by a policy that plays every probe.

Standing conventions. Noise is i.i.d. σ-sub-Gaussian and independent of the actions; we use the reward-table model in which the m-th pull of a given action vector receives the m-th entry of an i.i.d. sequence attached to that vector [31, §4.6], which has the same law as the sequential model and lets us compare two policies on the same noise. We fix

$$L \geq 8, \quad c \in (0, 1], \quad 0 < \varepsilon < c/L, \quad \lambda \ \geq \ \max\bigl\{\sigma^2 d\, \ell_0,\ 8\sigma^2 \ell_1,\ 60\bigr\}, \quad \beta \equiv 2\sqrt{\lambda}, \qquad (4)$$

with $\ell_0 := \log\bigl(1 + TL^2\bigr) + 2\log(1/\delta)$ and $\ell_1 := \log(2dT/\delta)$. The confidence radius is constant. The lower bound on λ makes $2\sqrt{\lambda}$ at least the radius of [1] at every round, so the confidence event of Section 3.6 holds for each policy with probability $\geq 1 - \delta$, and it makes the noise on any $\lambda/30$ pulls of one arm smaller than a quarter of that arm's confidence bonus. The constant $m_0 := \lfloor \lambda/30 \rfloor \geq 2$ is the number of pulls of a fresh bad arm that optimism forces before the good arm or the probe can be preferred to it. Conditions on T appear in each statement; they only say that the horizon is long enough for the claimed rounds to take place.

Row 4: an expert that mimics cold exploration.

Proposition 8 (Expert mimics cold exploration: same price, no gain)**.** *Let $\mathcal{I}(K, L, \cdot, \cdot, 0)$ be the instance with no probes ($\mathcal{A}_t = \{g, b_t\}$ for all t, $d = K + 1$), and let E be any policy that plays b_t at every round (for instance E_w with $w > 1/L$). With probability at least $1 - 2\delta$: (i) for every $\tau \leq Km_0$, H_τ and C play the same action at every round, so $A_\tau = V_\tau = 0$ and $R_T(\mathsf{H}_\tau) = R_T(\mathsf{C})$; (ii) for every $\tau > Km_0$, $R_T(\mathsf{H}_\tau) \geq \tau > Km_0$ while $R_T(\mathsf{C}) \leq K(4\beta^2 + 1)$, so $R_T(\mathsf{C})/R_T(\mathsf{H}_\tau) \leq (16\lambda + 1)/m_0 \leq 961$.*

Row 5: cheap recurring probes, the positive case. On $\mathcal{I}(K, L, c, \varepsilon, \tau_0)$ the expert E_w with $w \in (1/L, (1-\varepsilon)/(L-c))$ plays every probe while probes are offered and every bad arm afterwards (Lemma 12). Its routed actions cost ε each, cover $e_{j(t)}$ at rate c^2 per round, and recur; after τ_0 it is the worst possible policy. The hybrid that hands off exactly when the probes disappear therefore tests every clause of the thesis at once.

Theorem 2 (Online multiplication, and the price ratio as the multiplier)**.** *Let $\mathcal{I} = \mathcal{I}(K, L, c, \varepsilon, Kn)$ under* (4)*, with $\mathsf{E} = \mathsf{E}_w$ for some $w \in (1/L, (1-\varepsilon)/(L-c))$, and $\tau = \tau_0 = Kn$. With probability at least $1 - 3\delta$ the following hold simultaneously.*

(i) (Expert.) $R_T(\mathsf{E}) = \varepsilon Kn + (T - Kn)$.

(ii) (Cold, lower bound.) *If $n \geq m_0$ then $R_{[1,Kn]}(\mathsf{C}) \geq Km_0$; in particular $A_{\tau_0} \geq Km_0 - \varepsilon Kn$.*

(iii) (Cold, upper bound.) *For every $n \geq 1$, $R_T(\mathsf{C}) \leq K(8\beta^2 + 1) + \varepsilon Kn$.*

(iv) (Hybrid.) *If $n \geq 36\lambda L^2/c^2$ and $K \geq 36L^2$, then H_{τ_0} plays g at every round $t > \tau_0$, so $R_T(\mathsf{H}_{\tau_0}) = \varepsilon Kn$, $V_{\tau_0} = R_{[Kn+1,T]}(\mathsf{C}) \geq 0$, and $B_{\tau_0} = T - Kn$.*

Consequently, for $n \geq 36\lambda L^2/c^2$ and $K \geq 36L^2$,

$$\frac{m_0}{\varepsilon n} \ \leq \ \frac{R_T(\mathsf{C})}{R_T(\mathsf{H}_{\tau_0})} \ \leq \ 1 + \frac{8\beta^2 + 1}{\varepsilon n}, \qquad \frac{R_T(\mathsf{E})}{R_T(\mathsf{H}_{\tau_0})} = 1 + \frac{T - Kn}{\varepsilon Kn}, \qquad (5)$$

and at the smallest admissible $n = \lceil 36\lambda L^2/c^2 \rceil$ the first ratio is $\Theta(c^2/(\varepsilon L^2))$, with explicit constants $c^2/(2220\,\varepsilon L^2) \le R_T(\mathsf{C})/R_T(\mathsf{H}_{\tau_0}) \le 1 + c^2/(\varepsilon L^2)$ (the constants of this section are not optimized; see Appendix C, "Remarks on the constants").

The bracket (5) is the price ratio. Cold exploration buys coverage L^2 of a bad direction per unit of regret, so its price is $1/L^2$; the probe buys c^2 per ε, price ε/c^2; the ratio of the two prices is $c^2/(\varepsilon L^2)$, and the multiplier over cold LinUCB is that ratio up to constants, and is $O(1)$ when the ratio is. The confidence radius does not appear: both policies must push the same widths below the same level, and β scales the required coverage identically for both. The multiplier over the expert is $(T-\tau_0)/(\varepsilon\tau_0)$ and diverges as the tail lengthens, because the expert keeps paying for bad arms that the warm-started learner has learned to avoid. The lower bound in (ii) is attained inside the routed window, so on this instance $A_{\tau_0} > 0$ whenever $\varepsilon n < m_0$: the expert's probes are cheaper than the exploration cold LinUCB is forced into at the same rounds, and the data they leave is what prevents that exploration from resuming after τ_0. On this instance the saving is carried by A_{τ_0}, not by V_{τ_0}, and for a structural reason: the certificate in (iv) needs coverage of order $\beta^2 L^2$ of each bad direction, which the probe supplies at c^2 per round while cold LinUCB's forced pulls supply it at L^2 per round. The window long enough for the expert to certify ($n \ge 36\lambda L^2/c^2$ rounds per direction) is therefore longer by a factor of order L^2/c^2 than the exploration cold LinUCB is forced into ($m_0 \le$ pulls $\le 8\beta^2 + 1$ per direction), and cold LinUCB completes that exploration inside the window whenever the bad arm is offered there (exact-arithmetic simulations of the template give $R_{[Kn+1,T]}(\mathsf{C}) = 0$ across the admissible range). Theorem 1 does not distinguish the two channels: the certified saving is bounded by the same price ratio wherever it is collected. Row $5'$ shows that the tail channel is attainable with the same ratio when the routed window keeps cold LinUCB away from the directions it will be trapped on later.

Row $5'$: the same price ratio, collected on the tail. For the saving to appear as V_τ, the routed window must contain rounds at which cold LinUCB is *not* offered the bad arms it will later be trapped on, and is busy with something else; otherwise, as just explained, it completes its forced exploration inside the window. The following instance keeps cold LinUCB busy with *distractors*: arms of gap zero whose feature vectors carry a large component in a fresh direction, so that optimism prefers them to everything else while the expert probes. Fix $K, K', n \in \mathbb{N}$, put $d = 1 + K + K'$, $\theta^\star = e_1$, $L' := 12$, and $\tau_0 := Kn$. Directions $J := \{2, \dots, K+1\}$ are the bad directions and $J' := \{K+2, \dots, K+K'+1\}$ the distractor directions, cycled by $j(t) := 2 + ((t-1) \bmod K)$ and $j'(t) := K + 2 + ((t-1) \bmod K')$. The arms are

$$g = e_1\,(\mu = 1), \quad p_j = (1-\varepsilon)e_1 + c\,e_j\,(\mu = 1-\varepsilon), \quad b_j = Le_j\,(\mu = 0), \quad b'_{j'} = e_1 + L'e_{j'}\,(\mu = 1),$$

and the instance $\mathcal{J}(K, K', L, c, \varepsilon, n)$ offers $\mathcal{A}_t = \{g, p_{j(t)}, b'_{j'(t)}\}$ for $t \le \tau_0$ and $\mathcal{A}_t = \{g, b_{j(t)}\}$ for $t > \tau_0$. The expert plays the probe while probes are offered and the bad arm afterwards; the linear policy $\widehat{\theta}_{\mathsf{E}} = e_1 + w\sum_{j\in J} e_j - \sum_{j'\in J'} e_{j'}$ with $w \in (1/L, (1-\varepsilon)/(L-c))$ does this, by the computation of Lemma 12 together with $\langle \widehat{\theta}_{\mathsf{E}}, b'_{j'} \rangle = 1 - L' < 1$. The prices of e_j, $j \in J$, are ε/c^2 in the window and $1/L^2$ on the tail, as in Theorem 2; the distractor directions have price 0 and no bad arm uses them.

Proposition 9 (The price ratio collected as V_τ)**.** *Let $\mathcal{J} = \mathcal{J}(K, K', L, c, \varepsilon, n)$ under* (4) *strengthened to $L \ge 13$ and $\lambda \ge 16\sigma^2 d\,\ell_0$, with $n \ge 36\lambda L^2/c^2$, $K \ge 36L^2$, $K' \ge L'^2 Kn/\lambda$, and $T - \tau_0 \ge Km_0$. With probability at least $1 - 3\delta$ the following hold simultaneously.*

(i) (Expert.) $R_T(\mathsf{E}) = \varepsilon Kn + (T - Kn)$.

(ii) (Cold.) *Cold LinUCB plays the distractor at every round $t \le \tau_0$ and the bad arm at each of the first m_0 rounds at which every tail direction is offered, so $R_{[1,\tau_0]}(\mathsf{C}) = 0$ and $Km_0 \le R_{[\tau_0+1,T]}(\mathsf{C}) \le K(4\beta^2 + 1)$.*

(iii) (Hybrid.) H_{τ_0} *plays g at every round $t > \tau_0$, so $R_T(\mathsf{H}_{\tau_0}) = \varepsilon Kn$.*

Consequently $A_{\tau_0} = -\varepsilon Kn < 0$, $Km_0 \le V_{\tau_0} \le K(16\lambda + 1)$, $B_{\tau_0} = T - Kn$, and

$$\frac{m_0}{\varepsilon n} \;\le\; \frac{R_T(\mathsf{C})}{R_T(\mathsf{H}_{\tau_0})} \;\le\; \frac{16\lambda + 1}{\varepsilon n},$$

which at the smallest admissible n lies between $c^2/(2220\,\varepsilon L^2)$ and $c^2/(2\varepsilon L^2)$: the bracket of Theorem 2 up to its constant, now carried entirely by V_{τ_0}.

The proof (Appendix C.7) has one new ingredient: on the confidence event with the true radius, which the strengthened condition on λ makes at most $\frac{5}{8}\beta$, every index satisfies $U_t(x) \geq \mu(x) + \frac{3}{8}\beta\|x\|_{V_t^{-1}}$, and the distractor's width is at least $L'/\sqrt{2\lambda}$ as long as it has been pulled at most λ/L'^2 times, which the condition on K' guarantees. Its index then exceeds 7 while the indices of g and of the probe are below 4.3 and 5.6. Cold LinUCB never probes, reaches the tail with every bad direction untouched, and is trapped there exactly as in Theorem 2(ii); the hybrid's window is the same as in Theorem 2(iv). In the language of Theorem 1, rows 5 and 5′ have the same certified set $U = \{e_j : j \in J\}$, the same prices, and the same $P_U = c^2/(\varepsilon L^2)$; they differ only in which rounds cold LinUCB's counterfactual exploration falls on. Data streams with many more unresolved directions than any affordable window can certify (Section 4.3) are of the second kind.

Row 6: the same expert, optimal on the tail.

Proposition 10 (Expert optimal on the tail: large V_τ, no gain over the expert)**.** *Let $\mathcal{I} = \mathcal{I}(K, L, c, \varepsilon, Kn)$ with $n \geq 36\lambda L^2/c^2$ and $K \geq 36L^2$, and let $\mathsf{E} = \mathsf{E}_w$ with $w \in (\varepsilon/c, 1/L)$, which plays every probe and then g at every round. With probability at least $1 - 3\delta$, $R_T(\mathsf{E}) = R_T(\mathsf{H}_{\tau_0}) = \varepsilon Kn$ and $B_{\tau_0} = 0$, while A_{τ_0} and V_{τ_0} are exactly as in Theorem 2.*

Row 7: probes that cover the wrong directions.

Proposition 11 (Probes cover the wrong directions)**.** *Let $d = 2K + 1$ and replace the probe by $p'_t := (1 - \varepsilon)e_1 + c\, e_{K+j(t)}$, whose second coordinate is a nuisance direction that no bad arm uses, with $\mathcal{A}_t = \{g, b_t, p'_t\}$ for $t \leq \tau_0 = Kn$ and $\{g, b_t\}$ afterwards. Let E play p'_t for $t \leq \tau_0$ and b_t afterwards (for instance the linear policy with $\widehat{\theta}_{\mathsf{E}} = e_1 + w\sum_{j\geq 2} e_j$, $w \in (1/L, (1 - \varepsilon)/(L - c))$). Assume $n \geq 36\lambda L^2/c^2$, $K \geq 36L^2$, and $T - \tau_0 \geq K\lfloor\lambda/2\rfloor$. With probability at least $1 - 3\delta$,*

$$R_T(\mathsf{H}_{\tau_0}) \;\geq\; \varepsilon Kn + K\lfloor\lambda/2\rfloor, \qquad R_T(\mathsf{C}) \;\leq\; K(4\beta^2 + 1) + \varepsilon Kn, \qquad \frac{R_T(\mathsf{C})}{R_T(\mathsf{H}_{\tau_0})} \leq 65.$$

The principal angle between the expert's Gram matrix and the span of the bad directions is 90° and $V_{\tau_0} \leq R_{[\tau_0+1,T]}(\mathsf{C}) - K\lfloor\lambda/2\rfloor$.

Row 8: probes that are too dear.

Corollary 4 (Row 8: probes no cheaper than cold exploration)**.** *In the setting of Theorem 2, if $\varepsilon \geq c^2/L^2$ (the probe's price ε/c^2 is at least cold exploration's price $1/L^2$) then $R_T(\mathsf{C})/R_T(\mathsf{H}_{\tau_0}) \leq 2$ for every $n \geq 36\lambda L^2/c^2$.*

Row 9: cold LinUCB already cheap.

Proposition 12 (Cold LinUCB already cheap: no multiplier in average reward)**.** *On the no-probe instance $\mathcal{I}(K, L, \cdot, \cdot, 0)$, with probability at least $1-\delta$, $R_T(\mathsf{C}) \leq K(4\beta^2+1) = K(16\lambda+1)$. Hence for* every *policy π, $(R_T(\mathsf{C}) - R_T(\pi))/T \leq K(16\lambda+1)/T$, which with $\lambda = \sigma^2 d\ell_0$ is $O(\sigma^2 K^2 \ell_0/T)$.*

Remark 3 (What certification costs on the template). On the template, the certificate used in Theorem 2(iv), namely that the confidence bound alone excludes the bad arm in direction e_j, $2\beta L\|e_j\|_{V_{\tau+1}^{-1}} < 1$, requires by Lemma 1 that $R_{[1,\tau]}(\mathsf{E}) > \pi(e_j)\,(4\beta^2 L^2 - \lambda)$ for that direction, hence a total acquisition regret exceeding $K \min\{1/L^2, \varepsilon/c^2\}(4\beta^2 L^2 - \lambda)$ for all K directions. This is Theorem 1(b) written out for the template.

Row 10: the gap.

Proposition 13 (The gap enters only through the price)**.** *Fix $\eta \in (0, 1]$ and replace g by $g_\eta := \eta e_1$ ($\mu = \eta$) and p_t by $p_{t,\eta} := (1 - \varepsilon)\eta e_1 + c\, e_{j(t)}$ ($\mu = (1 - \varepsilon)\eta$), so that the gaps of b_t and p_t are η and $\varepsilon\eta$ and the prices of e_j are η/L^2 and $\varepsilon\eta/c^2$. Let $\mathsf{E} = \mathsf{E}_w$ with $w \in (\eta/L, (1 - \varepsilon)\eta/(L - c))$. Under (4), with $n \geq 36\lambda L^2/(\eta^2 c^2)$, $K \geq 36L^2/\eta^2$, and $\tau = \tau_0 = Kn$, with probability at least $1 - 3\delta$,*

$$\frac{\eta^2 c^2}{2220\,\varepsilon L^2} \;\leq\; \frac{\eta K m_0}{\varepsilon\eta Kn} \;\leq\; \frac{R_T(\mathsf{C})}{R_T(\mathsf{H}_{\tau_0})} \;\leq\; 1 + \frac{8\beta^2/\eta^2 + 1}{\varepsilon n} \;\leq\; 1 + \frac{c^2}{\varepsilon L^2}.$$

The upper bound is the price ratio $(\eta/L^2)/(\varepsilon\eta/c^2) = c^2/(\varepsilon L^2)$, in which η cancels; the lower bound loses a factor η^2 because our lower bound on cold LinUCB counts only the m_0 pulls per direction that optimism forces regardless of the gap.

Where the experiments sit. Each configuration of Section 4 has a row. WikiLinks with the Qwen expert is row 3: the expert is at the linear learner's ceiling, and the hybrid's gain is additive. WikiLinks with Flan-T5 is row 9: the routed data does reach useful directions ($V_\tau > 0$ on every seed), but cold LinUCB already resolves most of what matters and the multiplier is $1.01\times$; its principal angle of 82° alone would have suggested row 7, which the decomposition rules out (Section 4.2). AmazonCat with Flan-T5 is row $5'$ for all three experts—multipliers $1.71\times$, $2.72\times$ and $3.29\times$, routed rounds concentrated in the first few thousand, and 96–100% of the saving accruing after the last routed round (Figure 2), which is the tail channel. AmazonCat with BGE features lies between the two: cold LinUCB already reaches about a third of the correct answers, and once the gate is given a c at which it engages it adds a smaller tail gain (up to $1.23\times$ on held-out seeds). Section 3.8 turns the row assignment into quantities that can be estimated before any online round is played.

3.6 What Governs the Regret of LinUCB at a Single Round

Fix the confidence event $\mathcal{E}_\delta$ on which $\|\widehat{\theta}_t - \theta^\star\|_{V_t} \le \beta_t$ for all rounds t at which LinUCB acts, where (V_t, b_t) are built from all data so far including any warm start (M, c) formed from the same kind of data; by the self-normalized bound of [1] this event has probability at least $1-\delta$ with $\beta_t = \sigma\sqrt{2\log\Big(\det(V_t)^{1/2}\det(\lambda I)^{-1/2}/\delta\Big)} + \sqrt{\lambda}S$. Define, for a positive definite V,

$$\Phi_t(V) \;:=\; \max_{a\in\mathcal{A}_t} \min\Big\{\Delta_{t,a},\; 2\beta_t\|x_{t,a}\|_{V^{-1}}\Big\}. \tag{6}$$

Lemma 2 (Width-limited regret)**.** *On $\mathcal{E}_\delta$, at any round t at which LinUCB acts with design matrix V_t, the instantaneous pseudo-regret satisfies $\Delta_{t,A_t} \le \Phi_t(V_t)$. Moreover Φ_t is monotone: $V \preceq V'$ implies $\Phi_t(V') \le \Phi_t(V)$.*

The lemma is the familiar fact that an optimistic algorithm can only play an arm whose width exceeds half its gap, restated so that the gap and the width sit in one expression. It makes three facts visible at once. A competitor with a large gap but an even larger width is exactly as dangerous as its gap (Proposition 2: $\Delta = 1$, width $\beta L/\sqrt{\lambda} > 1/2$). A competitor with a tiny gap is cheap no matter how wide it is, so a heavy mass of near-ties does not by itself mean large regret. And the only way to reduce the regret LinUCB can incur on a given round is to reduce widths *on competitors whose widths exceed their gaps*; reducing the width of g in Proposition 2 would have achieved nothing.

There are exactly two ways widths get reduced: through data that arrives as a warm start, and through LinUCB's own observations. Both are exact linear algebra.

Lemma 3 (A warm start shrinks widths in the directions it covers)**.** *For any $M \succeq 0$ and any x, $\|x\|_{(\lambda I+M)^{-1}} \le \|x\|_{(\lambda I)^{-1}} = \|x\|_2/\sqrt{\lambda}$. If P is an orthogonal projector, $M \succeq mP$, and $x \in \mathrm{Im}(P)$, then $\|x\|^2_{(\lambda I+M)^{-1}} \le \|x\|_2^2/(\lambda+m)$. If instead $x \perp \mathrm{Im}(M)$, then $\|x\|_{(\lambda I+M)^{-1}} = \|x\|_2/\sqrt{\lambda}$: a warm start does nothing for directions it does not cover.*

Lemma 4 (One observation shrinks the width of another direction by an exact factor)**.** *For $V \succ 0$ and vectors x, z with $z \neq 0$,*

$$\|z\|^2_{(V+xx^\top)^{-1}} = \big(1-\rho_V(x,z)\big)\|z\|^2_{V^{-1}}, \qquad \rho_V(x,z) := \frac{(z^\top V^{-1}x)^2}{(1+\|x\|^2_{V^{-1}})\,\|z\|^2_{V^{-1}}} \in [0,1).$$

At a cold start, $\rho_{\lambda I}(x,z) = \cos^2(x,z)\cdot\|x\|_2^2/(\lambda+\|x\|_2^2)$.

Lemma 4 is where "feature coherence" enters, and it shows precisely how much of that notion survives. Observing x reduces the width of z by the squared cosine *in the V^{-1} geometry*, scaled by how novel x itself is. The Euclidean cosine is the right quantity only at $V = \lambda I$; as V becomes anisotropic the relevant angle is measured in V^{-1}, and two vectors with positive Euclidean inner product can have ρ_V near zero (or vice versa). In Proposition 2 the Euclidean cosine between any two bad arms is exactly zero, so $\rho = 0$ and no observation ever helps a future round; that is what makes the instance hard, not the ambient dimension as such. A large ambient dimension is necessary for many mutually V^{-1}-orthogonal directions to exist, but it is the lack of transfer, not the dimension, that cold LinUCB pays for.

How the earlier structural conditions arise. Conditions sometimes presented as independent structural axes (effective dimension, feature coherence, gap structure) are all readings of Lemma 2 together with Lemmas 3 and 4: a uniform gap $\Delta_{\min}$ caps Φ_t at $\Delta_{\max}$ but saves nothing while widths exceed $\Delta_{\min}/2$; a small rank r of the span of played directions caps the elliptical potential $\sum_t \|x_{t,A_t}\|^2_{V_t^{-1}}$ at $2r\log\big(1 + L^2T/(r\lambda)\big)$ and therefore the number of rounds at which any width can exceed a fixed level; and positive coherence lower-bounds $\rho_{\lambda I}$ at the start of learning only. None of the three is a condition under which a warm start cannot help by more than a constant factor, and we do not claim one here; what they control is how many rounds of cold exploration are needed before Φ_t falls below the level the warm start could have delivered immediately.

3.7 Regret of the Handoff Hybrid

Proposition 14 (Handoff regret in exact form)**.** *Fix τ and an expert E. On the event $\mathcal{E}_\delta$ (probability at least $1-\delta$), the handoff hybrid satisfies*

$$R_T(\mathsf{H}_\tau) \;\le\; \underbrace{\sum_{t\le\tau} \Delta_{t,a_t^{\mathsf{E}}}}_{\textit{acquisition regret}} \;+\; \min\left\{ \sum_{t=\tau+1}^{T} \Phi_t(V_t),\;\; 2\beta_T \max\Big\{1, \tfrac{L}{\sqrt{\lambda}}\Big\} \sqrt{2(T-\tau)\,\log \frac{\det V_{T+1}}{\det V_{\tau+1}}} \right\}, \tag{7}$$

where $V_{\tau+1} = \lambda I + M_\tau$ and the widths inside Φ_t are those of the hybrid's own trajectory. The determinant ratio satisfies $\log\det V_{T+1}/\det V_{\tau+1} \le \sum_{j=1}^{d} \log\big(1 + L^2(T-\tau)/\sigma_j(V_{\tau+1})\big)$ with σ_j the eigenvalues of $V_{\tau+1}$, and cold LinUCB satisfies the same bound with $\tau = 0$ and $V_1 = \lambda I$.

We state the bound with the exact determinant ratio and make no asymptotic simplification of it. Two remarks explain why.

The determinant term cannot certify a multiplier. Suppose $M_\tau \succeq \nu\tau P$ on an r-dimensional subspace containing all actions, the situation analyzed in earlier work. Then the second term in (7) is at most $2\beta_T\sqrt{2r(T-\tau)\log(1 + L^2(T-\tau)/(\lambda+\nu\tau))}$, and its ratio to the cold bound $2\beta_T\sqrt{2rT\log(1+L^2T/\lambda)}$ is

$$\Psi(\tau,\nu,T) = \sqrt{\frac{T-\tau}{T}} \cdot \sqrt{\frac{\log\big(1 + L^2(T-\tau)/(\lambda+\nu\tau)\big)}{\log\big(1+L^2T/\lambda\big)}}.$$

For $\tau = o(T)$ both logarithms are $\Theta(\log T)$ and $\Psi \to 1$; for $\tau = T^\alpha$ with $\nu\tau \gg \lambda$ it tends to $\sqrt{1-\alpha}$, bounded away from zero. It is tempting to read Ψ as $\sqrt{\lambda/(\lambda+\nu\tau)}$ by comparing the arguments of the two logarithms; that reading is wrong. The correct conclusion is the opposite one: a warm start that improves only the $\sqrt{T}$-rate term buys at most a constant factor, and any multiplier must come from the first term in the minimum, that is, from competitors whose widths the warm start pushes below their gaps so that Φ_t drops from the order of the gap to the order of the (now small) width or to zero. That is what happens in Proposition 2, where $\sum_t \Phi_t$ falls from T to $O(1)$ while the determinant bound barely moves.

What the bound says about multiplication. Combining (7) with Proposition 1, the hybrid beats cold LinUCB by more than a constant factor only if

$$\sum_{t>\tau} \big(\Delta_{t,A_t^{\mathsf{C}}} - \Phi_t(V_t^{\mathsf{H}})\big) \;\gg\; \sum_{t\le\tau} \Delta_{t,a_t^{\mathsf{E}}} - R_{[1,\tau]}(\mathsf{C}),$$

i.e., the width reduction that the routed data delivers on the competitors cold LinUCB would have played must exceed the net regret paid to acquire it; and it beats the expert only if the expert's tail regret exceeds the hybrid's. Neither condition is implied by low expert accuracy being "near-optimal early" or by an abstract coverage parameter. Both are properties of *which* actions the expert proposes, relative to the directions cold exploration leaves unresolved, and of what those actions cost. This is the quantity our diagnostics estimate.

Adaptive τ and interleaving. The router of Appendix F decides at each round whether to follow the expert or LinUCB by comparing confidence widths, so the expert rounds need not be contiguous. Proposition 14 applies verbatim with $\sum_{t\le\tau}$ replaced by the sum over expert-routed rounds and with the LinUCB-phase terms summed over LinUCB-routed rounds, since Lemma 2 is a round-level statement and the elliptical potential argument only uses that V_t is non-decreasing.

3.8 Pre-Runtime Estimates

Proposition 14 and Proposition 1 point to three quantities a practitioner would want before committing to LLM inference on a long trajectory, all estimable from a labeled held-out sample $\mathcal{D} = \{(\mathcal{A}_s, \{x_{s,a}\}, \{r_{s,a}\})\}_{s=1}^{n}$ and the expert's proposals a_s^{E} on it (formal definitions and consistency statements in Appendix G):

1. **Width-limited regret under cold and warm designs.** Fit a ridge estimate $\widehat{\theta}_{\mathrm{or}}$ on $\mathcal{D}$ to obtain gap estimates $\widehat{\Delta}_{s,a}$, and form the expert's held-out Gram matrix $\widehat{M}_{\mathsf{E}} = \sum_s x_{s,a_s^{\mathsf{E}}} x_{s,a_s^{\mathsf{E}}}^{\top}$. For a target number τ of routed rounds, compute $\widehat{\Phi}(\lambda I)$ and $\widehat{\Phi}(\lambda I + (\tau/n)\widehat{M}_{\mathsf{E}})$, each the average over s of $\max_a \min\{\widehat{\Delta}_{s,a}, 2\beta\|x_{s,a}\|_{V^{-1}}\}$. Their difference is the width reduction the expert's data would deliver on the competitors that matter, before any online round is played.
2. **Acquisition regret.** $\widehat{C}_{\mathsf{E}}(\tau) = \tau \cdot \frac{1}{n}\sum_s \widehat{\Delta}_{s,a_s^{\mathsf{E}}}$, which for binary reward is τ times the expert's held-out error rate.
3. **Tail advantage.** The expert's held-out accuracy compared with the accuracy of $\widehat{\theta}_{\mathrm{or}}$, the latter being an upper proxy for what the linear learner can reach; multiplication over the expert is impossible if the expert is already at that ceiling.

The prediction is that a handoff can multiply through the tail channel (V_τ) only if the width reduction in item 1, scaled by the number of future rounds on which it applies, exceeds the acquisition regret in item 2 by a wide margin, and item 3 leaves room; the concurrent channel (A_τ) is covered by the dry-run variant of Appendix G. These are estimates, not certificates; the thresholds we use are stated with the experiments and are provisional. Four cheaper statistics—effective rank, median minimum cosine, gap-distribution tail, and the principal angle between the expert's Gram matrix and the context covariance—are useful as proxies and we report them alongside: the rank bounds how many distinct unresolved directions can exist, the cosine is $\rho_{\lambda I}$ up to the norm factor of Lemma 4, the gap tail is the first argument of the minimum in (6), and the principal angle says whether $\widehat{M}_{\mathsf{E}}$ touches the directions that dominate the online context covariance at all.

4 Experiments

We measure the decomposition of Proposition 1 and the two channels of Section 3.4 on real data, running the router, the LLM alone, and cold LinUCB on the same data stream, and ask how far the quantities of Section 3.8 anticipate the outcome. We sweep the router's confidence parameter c over several orders of magnitude so that negative results are not hyperparameter artifacts. The setup mirrors [13].

Two scales for one effect. The literature reports *reward multiplication*, the ratio of a hybrid's average reward to that of a base policy [13]; with binary Precision@1 this is a ratio of accuracies, and we call it the *accuracy multiplier*. The theorems bound a different quantity, the ratio $R_T(\mathsf{C})/R_T(\pi)$ of pseudo-regrets. Every context has at least one correct action, so the best achievable reward is 1 per round and $R_T(\pi) = T(1 - \mathrm{acc}(\pi))$; alongside the accuracy multiplier we report the *regret ratio* $(1 - \max\{\mathrm{acc}_{\mathrm{LLM}}, \mathrm{acc}_{\mathrm{Cold}}\})/(1 - \mathrm{acc}_{\mathrm{Router}})$. The two agree when accuracies are high and differ sharply when they are low: on AmazonCat an accuracy multiplier of 3.29 from a base of 8.4% is a regret ratio of 1.27, that is 21% less regret. Statements about the theorems refer to the regret ratio. A_τ, V_τ and B_τ are differences of regret, counted in correct answers, and are the same on either scale.

4.1 Setup

Datasets. OneShotWikiLinks-311[1] ($T = 622{,}000$, $|\mathcal{A}| = 311$) pairs two model families (Flan-T5, Qwen2.5) with the same dataset, giving six configurations. **AmazonCat-13K** [11] ($T = 50{,}000$, $|\mathcal{A}| = 13{,}330$) pairs two embedding spaces (BERT-768d, BGE-1024d) with the same categories. Reward is binary Precision@1.

Policies. Each experiment tracks three policies on the same stream: (1) **Router** (Algorithm 1), the deterministic confidence-gated hybrid with expert design matrix $V^{(1)} = cI_d$, so the expert's

[1] `https://www.kaggle.com/generall/oneshotwikilinks`

width is $u_t^{(1)} = \|\phi(x_t, a_t^{(1)})\|_2/\sqrt{c}$ and LinUCB's is $u_t^{(0)} = \|\phi(x_t, a_t^{(0)})\|_{V_t^{-1}}$; the Router updates (V_t, b_t) on every round with the played action and observed reward, whichever policy chose it, so its LinUCB component is exactly the online-warm-started learner of Section 2; (2) **LLM only**; (3) **Cold LinUCB** from λI_d. Hyperparameters: $\lambda = 0.1$; the exploration scale is annealed per round as $\alpha_t = \max(0.1, 0.9985^t)$ on both datasets, reaching its floor at $t \approx 1.5 \times 10^3$. A seed fixes the order of the context stream and is shared by all policies in a run, so every comparison between policies or experts is paired on seed. Reported results use seeds 1–10. The confidence parameter c of every configuration is chosen on the disjoint block of seeds 11–20, by mean Router accuracy over the three experts, and the chosen value is then evaluated on seeds 1–10, so that tuning and evaluation never share a seed. On AmazonCat the tuning block selects $c = 100$ for BERT features (mean Router accuracy 17.2, 19.5, 22.2, 21.4 and 5.4% at $c = 1, 10, 10^2, 10^3, 10^4$), the value used throughout, and $c = 0.1$ for BGE features (Table 8). [**TODO:** WikiLinks sweep] On WikiLinks the Flan-T5 grid is $c \in \{1, 10, 10^2, 10^3, 10^4\}$ (Table 4) and the Qwen grid $c \in \{0.01, 0.1, 1, 10, 100\}$ (Table 8). The LLM expert proposes the action whose embedding is closest to the LLM's textual output, as in [13].

Pre-runtime statistics are computed on a held-out sample of 5,000 labeled contexts. Table 2 reports the four proxies of Appendix G.3, the held-out LLM accuracy (the acquisition-cost term of Section 3.8 up to the factor τ), and, for the configurations where we measured it, the price ratio P_U of Theorem 1. The proxies were designed for an earlier version of this work and are retained for comparison only: on the three AmazonCat Flan-T5 rows they agree to within 1.0° of subspace angle and are identical in the other three columns, while the observed multipliers span $1.71\times$ to $3.29\times$. The price ratio is different in kind, and is *not* a pre-runtime quantity. We measure it after the run from per-round logs, taking for U the leading directions of the features cold LinUCB played on rounds it lost after routing stopped, and the real streams lie outside the fixed-arm setting of Theorem 1; we report it as a post-hoc explanatory statistic. It bounds the regret ratio, and every observed regret ratio lies below it (Section 4.3); it is equal across the three experts, as the coverage channel is; and it is by construction blind to the transfer, which Section 3.4 shows is what distinguishes these experts. Appendix G reports the two estimators we built for the tail channel, $\widehat{N}$ and the dry run, and how far each gets.

Table 2: Pre-runtime statistics, with the observed multiplier for reference (Flan-T5 rows over ten seeds; AmazonCat Qwen rows over seeds 1–10, held out from the choice of c; WikiLinks Qwen rows pending the sweep). The four proxies are inherited from an earlier version of this work and are retained only for comparison; Appendix G.3 states what each does and does not bound. On the three AmazonCat Flan-T5 rows they are identical in three columns and span 1.0° of subspace angle while the multipliers span $1.71\times$ to $3.29\times$: they cannot separate the configurations. P_U is the price ratio of Theorem 1; unlike the other columns it is measured *after* the run, from per-round logs, and is a post-hoc statistic. It bounds the regret ratio and is blind to the transfer, which is what orders these three experts. The last column gives the accuracy multiplier and, after the slash, the regret ratio. Boldface marks values in the direction that leaves room for a warm start. LLM acc. is held-out top-1 accuracy; its complement, times τ, is the acquisition regret.

Dataset	Config	$\widehat{r}\log T/\sqrt{T}$	$\widehat{\alpha}$	Tail $\le \eta$	Subspace $\angle$	LLM acc.	P_U	Observed (acc. / regret)
WikiLinks	Flan-T5-small	0.05	0.19	**heavy**	**82.2°**	7.2%	—	1.01× / 1.00
WikiLinks	Flan-T5-base	0.05	0.19	**heavy**	**83.2°**	10.6%	—	1.01× / 1.00
WikiLinks	Flan-T5-large	0.05	0.19	**heavy**	**83.0°**	12.4%	—	0.88× / 0.98
WikiLinks	Qwen-0.5B	0.04	0.43	**heavy**	**81.9°**	18.7%	—	—
WikiLinks	Qwen-3B	0.04	0.43	**heavy**	**81.9°**	29.5%	—	—
WikiLinks	Qwen-7B	0.04	0.43	**heavy**	**81.9°**	38.7%	—	—
AmazonCat	Flan-T5-small (768d)	**0.16**	**0.04**	**heavy**	**85.1°**	5.2%	1.73	1.71× / 1.07
AmazonCat	Flan-T5-base (768d)	**0.16**	**0.04**	**heavy**	**84.1°**	3.6%	1.59	2.72× / 1.19
AmazonCat	Flan-T5-large (768d)	**0.16**	**0.04**	**heavy**	**84.2°**	5.3%	1.58	**3.29× / 1.27**
AmazonCat	Qwen-0.5B (1024d)	**0.15**	0.31	**heavy**	**83.1°**	21.4%	—	1.00× / 1.00
AmazonCat	Qwen-3B (1024d)	**0.15**	0.31	**heavy**	**83.1°**	22.4%	—	**1.23× / 1.13**
AmazonCat	Qwen-7B (1024d)	**0.15**	0.31	**heavy**	**83.1°**	27.2%	—	1.19× / 1.10

4.2 OneShotWikiLinks-311

Setup. 622,000 context–entity pairs over 311 entities. *Flan-T5 (BERT-768d features)*: embeddings from `bert-base-nli-mean-tokens`, joint feature $\phi(x, a) = x \odot a \in \mathbb{R}^{768}$, expert Flan-T5 [16] at 60M/250M/780M. *Qwen2.5 (BGE-1024d features)*: embeddings from `BAAI/bge-large-en-v1.5` [55], expert Qwen2.5-Instruct [27] at 0.5B/3B/7B.

Table 3: Final cumulative accuracy on OneShotWikiLinks-311, mean±std over seeds 1–10. Flan-T5 rows are at $c = 100$; the Qwen rows are filled at the c chosen on seeds 11–20. LLM accuracy and cold LinUCB do not depend on c. Mult. is the accuracy multiplier Router/$\max$(LLM, Cold); the regret ratio is within 0.02 of one on every Flan-T5 row. The lift column is the Router's gain over cold LinUCB in percentage points. **[TODO:** WikiLinks sweep]

Family	Size	LLM	Cold LinUCB	Router	Lift (pp)	Mult.
Flan-T5	small (60M)	7.16%	$10.71 \pm 0.04\%$	$10.84 \pm 0.04\%$	+0.13	1.01×
	base (250M)	10.56%	$10.71 \pm 0.04\%$	$10.87 \pm 0.05\%$	+0.16	1.01×
	large (780M)	12.43%	$10.71 \pm 0.04\%$	$10.88 \pm 0.04\%$	+0.17	0.88×
Qwen2.5	0.5B	18.72%	$18.88 \pm 0.04\%$	—	—	—
	3B	29.47%	$18.88 \pm 0.04\%$	—	—	—
	7B	38.74%	$18.88 \pm 0.04\%$	—	—	—

Results. Table 3 reports final cumulative accuracy at $c = 100$. The Router routes to the LLM on $< 0.3\%$ of rounds for Flan-T5. Routing is driven by feature geometry, not by reward: all three Flan-T5 sizes produce nearly identical routing curves despite an order-of-magnitude spread in accuracy.

Confidence sweep. To rule out a suboptimal c, the Flan-T5 configurations are swept over $c \in \{1, 10, 10^2, 10^3, 10^4\}$ on the tuning seeds 11–20 (Table 4), and the Qwen configurations over $c \in \{0.01, 0.1, 1, 10, 100\}$ (Table 8).

Table 4: Confidence sweep on the WikiLinks Flan-T5 (BERT-768d) configurations, tuning seeds 11–20: share of rounds routed to the LLM and final Router accuracy at each c. Cold LinUCB does not depend on c. **[TODO:** WikiLinks sweep]

	routed %			Router accuracy			
c	small	base	large	small	base	large	best mult.
1	—	—	—	—	—	—	—
10	—	—	—	—	—	—	—
10^2	—	—	—	—	—	—	—
10^3	—	—	—	—	—	—	—
10^4	—	—	—	—	—	—	—

[TODO: WikiLinks sweep] The decomposition explains the shape. For Flan-T5 the routed rounds do leave usable data—V_τ is 748, 832 and 886 correct answers and is positive on all ten seeds—but cold LinUCB reaches 10.71% on its own, so there is little left to buy: this is row 9 of Table 1, not the wrong-directions failure of row 7. Nor does WikiLinks have a dominant entity for a warm start to steer the learner toward, as AmazonCat does (Section 4.3): the most frequent of the 311 entities covers 0.36% of contexts. For Qwen the expert is far better than the learner (29–39% against 19%), so $B_\tau < 0$ for every τ and no handoff can beat the expert. **[TODO:** WikiLinks sweep]

4.3 AmazonCat-13K

AmazonCat-13K has 13,330 categories and $T = 50{,}000$ rounds. We run two feature spaces: *Flan-T5 (BERT-768d)* and *Qwen2.5 (BGE-1024d)*; hyperparameters as in Section 4.1.

The multiplying configuration. Flan-T5-large reaches $27.64 \pm 5.56\%$ against cold LinUCB's $8.39 \pm 2.46\%$ with only 3.2% of rounds routed to the LLM overall; routing is concentrated in the first $\sim 5{,}000$ rounds. Table 6 gives the shape of a positive case directly. $A_\tau = 53$ is negligible: it is positive, not negative, because the routed rounds fall early, while cold LinUCB is still near 2%, so even a 5% expert is marginally the better player there; but at 1,578 routed rounds the whole term is worth 53 correct answers against a total advantage of 9,624. $V_\tau = 9{,}571$ carries 99.4% of it, and is positive on all ten seeds. $B_\tau = 11{,}189$ is large because the expert stays at 5% while the warm-started learner passes 27%. This is the row-5′ signature of Proposition 9: the acquisition is nearly free, and the saving is collected on the tail, after routing has stopped. Figure 2 shows the same accounting in time: every routed round falls inside the first 5,000, and 96% of the 9,624 correct answers the Router gains over cold LinUCB are gained after its last routed round, while the expert itself stays frozen at 5.3%. The cumulative-accuracy curves part earlier than that only because an early lead weighs heavily in a short prefix.

Table 5: AmazonCat-13K, $T = 50{,}000$, mean $\pm$ standard deviation over ten seeds. The accuracy multiplier (Acc. mult.) is the ratio of the seed-averaged Router accuracy to the seed-averaged $\max\{\text{LLM, cold}\}$, which is less sensitive to a single unlucky denominator than the average of per-seed ratios; the regret ratio is the corresponding ratio of regrets, $(1-\max\{\text{LLM, cold}\})/(1-\text{Router})$, the quantity Theorem 1 bounds. Cold LinUCB does not depend on the expert, so its column is shared within each feature space. BERT rows are at $c = 100$, seeds 1–10. BGE rows are at $c = 0.1$, selected on the disjoint seed block 11–20 in Table 8 and evaluated here on seeds 1–10—at the $c = 100$ used for BERT the gate never engages and all six BGE rows are degenerate. On BERT the expert's held-out accuracy orders the experts differently from the multiplier; on BGE the two larger experts gain significantly (paired $t = 3.92$ and 2.59 against cold) and the smallest does not move ($t = 0.01$).

Features	LLM	LLM acc.	Cold LinUCB	Router	LLM%	Acc. mult.	Regret ratio
BERT-768d	Flan-T5-small	5.20%	$8.39 \pm 2.46\%$	$14.36 \pm 6.12\%$	2.8%	1.71×	1.07
	Flan-T5-base	3.61%	$8.39 \pm 2.46\%$	$22.80 \pm 9.00\%$	3.1%	2.72×	1.19
	Flan-T5-large	5.27%	$8.39 \pm 2.46\%$	$\mathbf{27.64 \pm 5.56}\%$	3.2%	**3.29×**	**1.27**
BGE-1024d	Qwen-0.5B	21.38%	$32.59 \pm 5.00\%$	$32.62 \pm 6.33\%$	0.04%	1.00×	1.00
	Qwen-3B	22.39%	$32.59 \pm 5.00\%$	$\mathbf{40.20 \pm 4.45}\%$	0.05%	**1.23×**	**1.13**
	Qwen-7B	27.17%	$32.59 \pm 5.00\%$	$38.71 \pm 6.38\%$	0.05%	1.19×	1.10

Table 6: The decomposition of Proposition 1 measured on the BERT-768d runs, in correct answers, mean over ten seeds. A_τ is summed over the routed rounds and V_τ, B_τ over the rest, using the routed set the Router actually selected (the identities hold for any routed set, by the remark after Proposition 1). Every configuration collects almost all of its advantage through V_τ: the expert's own answers are worth a few dozen correct rounds out of 1,578 routed, while the data they leave behind is worth thousands. V_τ is the only column that separates the three experts, and the multiplying configuration is also the only one whose V_τ is positive on every seed.

LLM	A_τ	V_τ	B_τ	seeds with $V_\tau > 0$
Flan-T5-small	47	$2{,}936 \pm 3{,}562$	4,582	9/10
Flan-T5-base	27	$7{,}178 \pm 4{,}772$	9,599	9/10
Flan-T5-large	53	$\mathbf{9{,}571 \pm 3{,}411}$	11,189	**10/10**

Equal accuracy, unequal information—and which channel carries it. The three Flan-T5 experts have held-out accuracies 5.20%, 3.61% and 5.27% and deliver accuracy multipliers of 1.71×, 2.72× and 3.29× (regret ratios 1.07, 1.19 and 1.27): accuracy orders them differently from the multiplier, and the least accurate expert beats a more accurate one by 60%. By Proposition 1 the only place they can differ is V_τ, and Table 6 confirms it: A_τ is 47, 27 and 53, indistinguishable as the equal accuracies require, while V_τ is 2,936, 7,178 and 9,571.

Section 3.4 says where that difference can live. The three experts cover the same directions to within 5%: summed over the routed rounds, the coverage of the directions cold LinUCB is trapped on is 1.40×10^5, 1.46×10^5 and 1.47×10^5, at prices 9.5×10^{-3}, 1.03×10^{-2} and 1.04×10^{-2}, so the price ratio of Theorem 1, measured post hoc as described in Section 4.1, is 1.73, 1.59 and 1.58 for the three—equal to within measurement error, and ordered opposite to the observed multipliers. It bounds the regret ratio, and the observed regret ratios lie below it for every expert; since the real stream is outside the theorem's setting we read this as consistency, not confirmation. By Corollary 3 the observed separation must then come from the transfer, and Table 7 and Figure 1 show that it does.

Removing m while leaving the binary problem otherwise intact collapses the spread of per-seed multipliers from 1.74 to 0.56 and turns every significant pairwise difference into a null result: large over small falls from $t = 5.19$ to $t = 0.62$. Suppressing the response sum entirely gives the same collapse, to a spread of 0.11; and the two transfer-free conditions agree with each other on every expert (paired $t = -1.18, +0.26, -0.60$), as they must if m is the only thing both lack. Removing the transfer costs each expert in proportion to what it gained, -2.22 for large ($t = 5.12$) against -0.63 for small ($t = 2.47$). A third route to the same conclusion replaces the reward by $\langle \theta_{\text{or}}, \phi \rangle + \mathcal{N}(0, \varsigma^2)$, making the model well specified so that Proposition 5 applies exactly: at $\varsigma \in \{0.001, 0.01, 0.499\}$ the three experts' regret ratios, measured against the oracle's best action, span 0.019, 0.010 and

Table 7: Isolating the two channels on AmazonCat/BERT-768d, ten seeds. In *full* a routed round contributes both the covariance and the response sum, as the Router does. In *transfer removed* the round contributes the covariance and the expert-independent part $M\theta_{\text{or}}$ of the response sum—the learner is fed $\langle \theta_{\text{or}}, \phi \rangle$ instead of the observed reward on routed rounds only, which deletes m of Proposition 6 and changes nothing else. In *no response sum* the round contributes the covariance alone. Multipliers are accuracy multipliers, Router over cold LinUCB, with the regret ratio in parentheses; t-statistics are paired over seeds against small, on each scale, $t^\star = 2.26$. With the transfer present the experts separate; without it, by either mechanism, no pairwise difference survives.

	Router accuracy		Multiplier over cold		
LLM	full	transfer removed	full	transfer removed	no response sum
Flan-T5-small	$14.36 \pm 6.12\%$	$9.71 \pm 2.08\%$	$1.71\times$ (1.07)	$1.16\times$ (1.02)	$1.53\times$ (1.05)
Flan-T5-base	$22.80 \pm 9.00\%$	$15.21 \pm 9.13\%$	$2.72\times$ (1.19)	$1.81\times$ (1.08)	$1.61\times$ (1.06)
Flan-T5-large	$27.64 \pm 5.56\%$	$11.62 \pm 5.97\%$	$3.29\times$ (1.27)	$1.38\times$ (1.04)	$1.53\times$ (1.05)
spread of per-seed multipliers			1.74	0.56	0.11
large − small, paired t			**+5.19**	+0.62	+0.24
base − small, paired t			**+3.03**	+1.25	+0.23
large − small, paired t, regret ratio			**+5.66**	+0.92	+0.15
base − small, paired t, regret ratio			**+3.33**	+1.74	+0.33

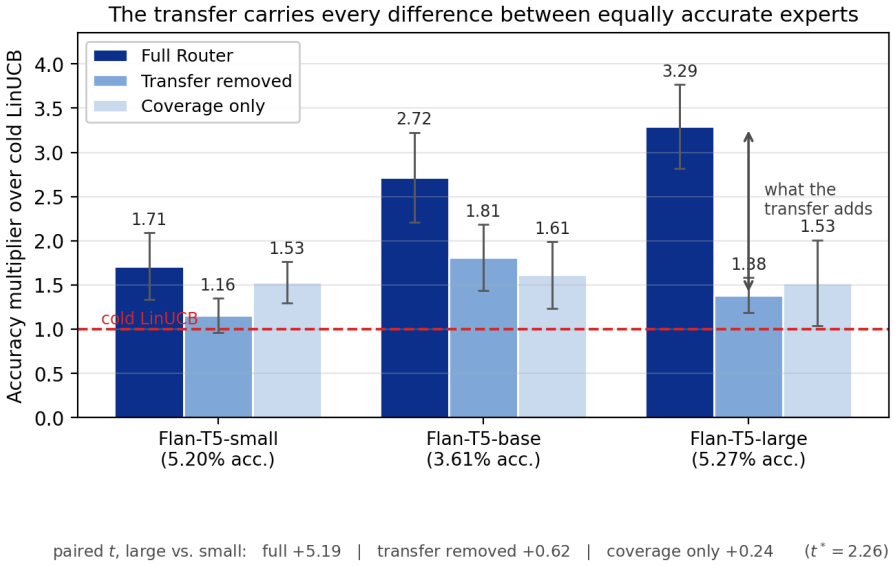


Figure 1: The two channels on AmazonCat/BERT-768d, ten seeds, with standard errors over seeds. The three experts have held-out accuracies of 5.20%, 3.61% and 5.27% and cover the same directions to within 5%, so Corollary 3 leaves the transfer as the only place they can differ. With it present (dark bars) they span $1.71\times$ to $3.29\times$; with it removed by either mechanism they collapse onto a common multiplier near $1.5\times$ and no pairwise difference survives. The two transfer-free conditions differ in what else they delete—*transfer removed* keeps the expert-independent part $M\theta_{\text{or}}$ of the response sum, *coverage only* discards the response sum entirely—yet they agree with each other on every expert, as they must if the transfer is the only thing both lack.

0.030, against 0.197 on the real reward; the ratios themselves lie between 0.97 and 1.09. Under linear rewards the experts are interchangeable, exactly as predicted.

Where the regret goes. On the regret scale the two channels have very different sizes. With the response sum suppressed, coverage alone gives regret ratios 1.051, 1.060 and 1.051: about 5% less regret than cold LinUCB, indistinguishable across experts, and below the price ratio for each, as it must be. The full Router gives 1.070, 1.187 and 1.266. For Flan-T5-base and Flan-T5-large, therefore, most of the regret saved is saved by the transfer, roughly 10 of 16 and 16 of 21 percentage

Table 8: Confidence sweep on the BGE-1024d configurations, tuning seeds 11–20, which select c. **[TODO:** WikiLinks sweep] At the $c = 100$ used for BERT the expert width $u^{(1)} = \|\phi\|/\sqrt{c}$ is permanently below LinUCB's, the gate never engages, and the Router equals the LLM exactly on every seed ($V_\tau = 0$, zero variance). Lowering c raises the expert's width and restores the gate. On AmazonCat, where cold LinUCB is strong, the Router at $c = 0.1$ beats it while routing 0.05% of rounds, and the same c reproduces the gain on the evaluation seeds (Table 5); at $c \geq 1$ it is dragged down toward a weaker expert. On WikiLinks the expert is genuinely better than the learner—row 3 of Table 1, where no combination can beat the better component.

		routed %			Router accuracy			
Dataset	c	0.5B	3B	7B	0.5B	3B	7B	best mult.
AmazonCat	0.01	0.0	0.0	0.0	34.77	34.77	34.77	1.00×
	0.1	0.04	0.05	0.05	33.84	39.68	**40.63**	**1.17**×
	1	75.2	71.9	73.5	28.09	29.26	36.50	1.05×
	10	100	100	100	21.38	22.39	27.17	0.78×
WikiLinks	0.01	—	—	—	—	—	—	—
	0.1	—	—	—	—	—	—	—
	1	—	—	—	—	—	—	—
	10	—	—	—	—	—	—	—
	100	—	—	—	—	—	—	—

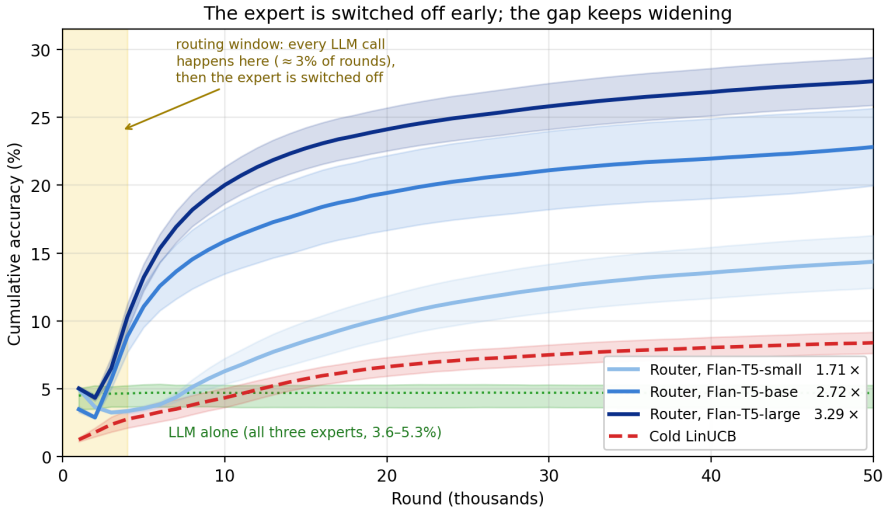


Figure 2: Learning dynamics on AmazonCat-13K (BERT-768d, $c = 100$), mean over ten seeds with shaded standard error. The shaded window contains every routed round: all calls to the LLM fall inside the first 5,000 rounds, after which the expert is switched off. The three Flan-T5 experts (held-out accuracies 5.20%, 3.61% and 5.27%, the band at the bottom) are consulted on equally few rounds, yet their Routers separate by nearly a factor of two; and 96–100% of each Router's advantage over cold LinUCB, counted in correct answers, accrues after its last routed round. That is the row-5′ signature: the saving is collected on the tail.

points. The separation between experts survives the change of scale intact: on per-seed regret ratios, large beats small with paired $t = +5.66$ and base beats small with $t = +3.33$ when the transfer is present, and neither difference is significant with the transfer removed ($t = +0.92, +1.74$) or the response sum suppressed ($t = +0.15, +0.33$); Figure 1 shows the same collapse on the accuracy scale. The price ratio correctly predicts that coverage makes the three experts interchangeable; it does not predict the size of the coverage gain, which it bounds loosely.

Finally, the transfer is not a small correction to the estimate at handoff; on this data it is essentially the whole of it. The ridge oracle assigns a correct action a score of about 2.5×10^{-3} where the reward is 1, so the residuals dominate: $\|m\|/\|\widehat{\theta}_{\tau+1}\| = 0.9998$, and the identity of Proposition 6 holds on the logged trajectories to 1.2×10^{-14}. Stripping m makes the three estimates agree to four decimals in held-out ranking quality ($0.2058, 0.2058, 0.2057$), while the transfers themselves separate the experts

with the same resolution the multipliers have: base and large each beat small ($t = 3.12$, $t = 2.56$) and are indistinguishable from one another ($t = 0.36$), which is the pair the multipliers also cannot separate.

What the transfer carries. On AmazonCat one category is present on 29.8% of contexts, so a policy that knew this in advance and always predicted it would score 29.8%. An online learner does not know it, and finding it is not easy: context-free bandits that try categories without using the context reach 0.04% (UCB1) and 2.31% (optimistic greedy) over the same $50{,}000$ rounds and $13{,}330$ categories, seeds 1–10. The Routers that multiply are the ones that discover this category. LinUCB settles on one frequent category and predicts it on a large share of rounds, and its accuracy then tracks that category's frequency: runs that settle on the dominant category finish near 28%, while cold LinUCB settles on a category present on 11% of contexts, or on none, in 18 of 20 seeds. Whether a Router settles on the dominant category is decided by a handful of the expert's routed answers. Across 177 Router runs on BERT features (every c and CORRAL, excluding configurations that route most rounds), a run whose expert gave no correct routed answer on this category reached it in 3 of 45 runs, and a run that received five or more reached it in all 54. In a controlled test, adding 15 correct answers on this category to Flan-T5-small ($+0.15$ points of held-out accuracy) doubled its multiplier ($1.33\times$ to $2.62\times$, paired $p = 0.002$), while 15 correct answers on random categories gave no significant gain ($1.65\times$, $p = 0.15$). This is the transfer of Proposition 6 made concrete: correct answers on a frequent category are the residuals that move the estimate, and they are why accuracy does not order the experts. Appendix E gives the details.

BGE-1024d. At $c = 100$ the expert width $u^{(1)} = \|\phi\|_2/\sqrt{c}$ is permanently below LinUCB's $u^{(0)}$, so the Router defers on every round and equals the LLM exactly: $V_\tau = 0$ with zero variance across seeds. This is a failure of the gate's hyperparameter, not evidence about V_τ, and it affected all six BGE configurations. Table 8 sweeps c down until the gate engages. The sweep is run on seeds 11–20 and selects $c = 0.1$; the same value is then evaluated on seeds 1–10. There cold LinUCB reaches 32.59%, while the Router reaches 40.20% with Qwen-3B and 38.71% with Qwen-7B, accuracy multipliers of $1.23\times$ and $1.19\times$ (regret ratios 1.13 and 1.10), while routing 0.05% of rounds. Both gains are significant (paired $t = 3.92$ and 2.59 against cold, $t^\star = 2.26$). With Qwen-0.5B the gate engages just as often and the Router does not move at all ($1.00\times$, $t = 0.01$), which is the expert-dependence the BERT rows also show. The accounting has the BERT signature where there is a gain: A_τ is 5 and 6 correct answers while V_τ is $3{,}801$ and $3{,}056$, positive on 9 and 8 of the ten seeds, against $V_\tau = 13$ for Qwen-0.5B. The multipliers are smaller than on BERT, which is what a stronger learner implies: when cold LinUCB already reaches a third of the correct answers, there is less left to buy. In the terms of the previous paragraph, cold LinUCB on BGE features settles on the dominant category by itself in 7 of 20 seeds, so the expert has less to add (Appendix E). Pushing c higher drags the Router toward a weaker expert and below cold LinUCB. On WikiLinks the Qwen experts are better than the learner (29.47% and 38.74% against 18.88%), which is row 3 of Table 1: an expert at or above the learner's ceiling, where $B_\tau \le 0$ and no combination can beat the better component. **[TODO:** WikiLinks sweep]

Summary across configurations. Twelve configurations span two datasets, two model families (60M to 7B), two embedding spaces and four orders of magnitude in c, with ten evaluation seeds per row and every c chosen on a disjoint block of ten tuning seeds. One configuration family multiplies clearly: AmazonCat with BERT-768d and Flan-T5 (accuracy multipliers $1.71\times$, $2.72\times$, $3.29\times$; regret ratios 1.07, 1.19, 1.27). AmazonCat with BGE-1024d and Qwen, evaluated on seeds held out from the choice of c, gives a smaller but significant gain (up to $1.23\times$ and 1.13, paired $t = 3.92$). The six WikiLinks configurations do not **[TODO:** WikiLinks sweep]; neither does BGE at the c that routes everything. In every configuration that multiplies, the accounting has the same shape: A_τ is a few dozen correct answers out of more than a thousand routed rounds, V_τ is in the thousands, and B_τ is large because the expert stays poor while the warm-started learner does not. Where multiplication fails, the failing term of Proposition 1 can be named: on WikiLinks with Flan-T5 cold LinUCB already resolves the directions that matter, so V_τ is positive but small; with Qwen the expert is above the learner's ceiling and $B_\tau < 0$; at large c on BGE the gate never hands back and $V_\tau = 0$ identically.

Across the three Flan-T5 experts the coverage channel is expert-independent, as Corollary 3 requires, and the transfer accounts for the whole of the observed separation and for most of the regret saved (Table 7, Figure 1). The deterministic Router differs from the randomized CORRAL master of [6] and the CORRAL-style routing of [13] only in how it chooses the routed rounds; both feed every

observation to the bandit learner, so the decomposition applies to both, and Proposition 5 adds that two routers with the same routed Gram matrix leave warm starts of the same law—any difference between routers must therefore appear in A_τ or in which residuals their routed sets happen to collect. Table 9 runs both alternatives.

Table 9: Alternative routing mechanisms on AmazonCat-13K (BERT-768d, $c = 100$), ten seeds, paired on seed with the gate. CORRAL is the randomized log-barrier master of [6] with importance-weighted updates; the offline arms spend τ expert pulls on contexts drawn from *outside* the online stream and then run plain LinUCB, so their acquisition is not charged as regret and $A_\tau = 0$ identically. $\tau = 1{,}600$ matches the gate's routed count. Cold LinUCB is bit-identical across all four arms on every seed, so each comparison isolates the router. The last column is the paired t statistic for gate minus that arm over the ten shared seeds ($t^\star = 2.26$). Mult. is the accuracy multiplier, with the regret ratio in parentheses.

LLM	Router	queries	Router acc.	Mult.	A_τ	V_τ	t vs. gate
Flan-T5-small	Gate (Alg. 1)	1,406	$14.36 \pm 6.12\%$	$1.71\times$ (1.07)	47	2,936	—
	CORRAL	22,860	$8.87 \pm 1.37\%$	$1.06\times$ (1.01)	-508	744	**2.64**
	Offline τ=1,600	1,600	$11.17 \pm 5.71\%$	$1.33\times$ (1.03)	0	1,386	1.16
	Offline τ=5,000	5,000	$15.95 \pm 7.11\%$	$1.90\times$ (1.09)	0	3,778	-0.59
Flan-T5-base	Gate (Alg. 1)	1,570	$22.80 \pm 9.00\%$	$2.72\times$ (1.19)	27	7,178	—
	CORRAL	6,939	$24.44 \pm 2.39\%$	$2.91\times$ (1.21)	-23	8,045	-0.63
	Offline τ=1,600	1,600	$20.97 \pm 8.89\%$	$2.50\times$ (1.16)	0	6,286	0.42
	Offline τ=5,000	5,000	$27.29 \pm 5.17\%$	$3.25\times$ (1.26)	0	9,449	-2.00
Flan-T5-large	Gate (Alg. 1)	1,597	$27.64 \pm 5.56\%$	$3.29\times$ (1.27)	53	9,571	—
	CORRAL	7,488	$25.42 \pm 2.31\%$	$3.03\times$ (1.23)	50	8,460	1.49
	Offline τ=1,600	1,600	$26.27 \pm 7.61\%$	$3.13\times$ (1.24)	0	8,937	0.42
	Offline τ=5,000	5,000	$25.76 \pm 6.53\%$	$3.07\times$ (1.23)	0	8,681	0.67

Three things follow. First, no baseline beats the gate significantly on any expert, and the gate beats CORRAL on Flan-T5-small ($t = 2.64$) while issuing $16\times$ fewer expert queries; on the other two experts CORRAL matches the gate at four to five times the query budget. Second, the offline arm at a matched budget of $1{,}600$ pulls is statistically indistinguishable from the gate on all three experts even though its acquisition is free, so the online routing decision costs essentially nothing—which is the content of $A_\tau \approx 0$, now visible from the other side. Tripling the free budget to $5{,}000$ still does not separate the two. Where an alternative does come out ahead of the gate in the mean, the margin is small, never significant, and paid for with a much larger budget: CORRAL on Flan-T5-base gains 1.6 points ($t = -0.63$) with $4.4\times$ the gate's expert queries, and the offline arm at $\tau = 5{,}000$ gains 1.6 and 4.5 points on Flan-T5-small and Flan-T5-base ($t = -0.59, -2.00$) only by drawing 3–$4\times$ as many expert answers, all collected before the online run begins. None of these trades buys a reliable improvement, so the gate is the cheapest router in the comparison and no alternative is significantly better on any expert.

Third, and most directly, the decomposition survives the change of master. In all twelve rows V_τ carries the value and A_τ is negligible beside it, and A_τ is exactly where the choice of router shows up: it is 0 by construction offline, a few dozen for the gate, and *negative* for CORRAL on the two weaker experts (-508 and -23), which is the price of routing $22{,}860$ rounds to a 5.2% expert while cold LinUCB is improving. The comparison also shows that coverage is not bought by volume: CORRAL routes $16\times$ as many rounds as the gate on Flan-T5-small and ends with $V_\tau = 744$ against the gate's $2{,}936$. What the gate supplies is not more expert pulls but pulls on the directions cold LinUCB is trapped on, which is what $\pi(u)$ in Theorem 1 measures and what Proposition 5 says is the whole of the coverage channel.

5 Conclusion

Reward multiplication in LLM–bandit hybrids is an accounting statement: the hybrid beats cold LinUCB by the expert's direct advantage plus the value of the data it leaves behind, and beats the expert by the learner's tail advantage. The data-value term is governed by whether the routed actions push the confidence widths of dangerous competitors below their gaps; the determinant-ratio term

that controls $\sqrt{T}$-rates cannot produce more than a constant factor. When the directions that trap cold LinUCB recur and the expert has a cheaper way of covering them, an online handoff multiplies, and the multiplier is the ratio of the two prices; when any one of recurrence, cheapness, correct targeting, or late suboptimality of the expert is removed, it does not.

A warm start, however, carries two objects, and the price accounting governs one of them. Under the linear model the second is redundant: the response sum is the Gram matrix applied to $\theta^\star$ plus noise whose law depends only on that matrix, so experts of equal coverage are interchangeable and the price ratio is a complete account of information value (Proposition 5). Real rewards are not linear, and there the estimate at handoff acquires a transfer—the ridge projection of the residuals onto the expert's own picks—which we derive in closed form and bound (Propositions 6 and 7). On AmazonCat it is the larger channel. A linear function of the joint features assigns a correct action a score of 2.5×10^{-3} against a reward of 1, so the transfer is essentially the entire estimate at handoff. Coverage alone saves about 5% of cold LinUCB's regret, the same for every expert, while the full router saves up to 21%; and every significant difference between three experts of equal accuracy disappears when the transfer is removed, whether by suppressing the response sum, by replacing the routed-round signal with its expert-independent part, or by linearizing the reward. The practical consequence is that a warm-start expert cannot be chosen by its accuracy, nor by the coverage its actions provide: what decides its value is the residuals its choices carry—on AmazonCat, a handful of correct answers on the most frequent category. Information value is not policy value; and on real data it is not coverage alone either.

Limitations. The theory assumes linear reward, and on our main dataset the channel it leaves out is the larger one: the transfer has a closed form and a regret bound (Proposition 7), but no price-style quantity that predicts its size before the expert is run. Reward models suited to binary feedback, such as generalized linear bandits, may shrink the residuals the transfer is built from; we have not tested whether the two-channel picture changes under them. The price ratio we report on real data is measured after the run, from the trajectories it is compared against, on streams outside the fixed-arm setting of Theorem 1; it is an explanatory statistic, not a prediction. Theorem 1 bounds the online multiplier on any finite-arm instance with the optimal arm always available; real streams satisfy this only approximately, and the bound has a residual—cold LinUCB's regret on arms the warm start does not certify—that is not itself governed by a price. We have no lower bound showing that the residual can be large without a large price ratio, and no lower bound on $R_T(\mathsf{H}_\tau)$ beyond the acquisition regret, so the theorem characterizes the certified saving rather than the multiplier outright. The positive instances (Theorem 2, Proposition 9) use a fixed confidence radius and a regularization scaled to the dimension, and their constants are not optimized. The constants in the catalog are not optimized and the lower bound on cold LinUCB's regret there counts only the rounds that optimism forces. The pre-runtime estimates require a labeled held-out sample and the expert's proposals on it; their thresholds are provisional and were not tuned across datasets. The extension of the width-based analysis to other confidence-based learners (OFUL, SupLinUCB, Thompson sampling) is immediate for Lemmas 3 and 4, which are statements about design matrices, but Lemma 2 uses the optimistic selection rule and would need an analogue for posterior sampling. The large AmazonCat multipliers depend on a dominant category that the learner can be steered to (Appendix E); where no category dominates, as on WikiLinks, we see no multiplication, and we have not tested other datasets with a skewed label distribution. Only two real datasets are used; the effect is large in one feature space and modest in the other, where cold LinUCB is already strong; multi-expert behaviour is not stress-tested; LLM-guided recommendation can inherit biases from pretrained models and logged data.

Broader impact. The work is foundational theory and pre-runtime estimation for contextual bandits. A positive impact is reducing unnecessary LLM inference in sequential decision systems by identifying when cold-start LinUCB suffices, which lowers compute cost and energy use. A potential negative impact is that LLM-guided contextual-bandit systems deployed in recommendation or ranking may inherit biases from pretrained models or logged data without auditing; concrete harms depend on follow-on applications not studied here.

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A Related Work

This appendix expands on the related-work discussion deferred from the introduction. We cover (i) linear contextual bandits and confidence geometry, (ii) model selection and expert aggregation, (iii) warm starts and prior information, (iv) the geometry of information gain in linear bandits, and (v) large language models for sequential decision making.

Sequential decision making with contextual information has been studied extensively in the contextual bandit literature, where the learner must balance exploration and exploitation under partial feedback [29, 5, 22, 31]. In the linear setting, algorithms such as LinUCB and OFUL achieve regret bounds that depend explicitly on the geometry of the accumulated design matrix through self-normalized confidence ellipsoids [15, 1]. Subsequent work has developed statistically efficient contextual bandit algorithms under general function approximation assumptions [4, 20, 21, 48, 63, 45, 59]. A central insight of this line of work is that the conditioning of the covariance matrix determines how quickly uncertainty shrinks and therefore how rapidly regret decreases. Our work builds on this perspective by showing that actions selected by a frozen predictor can reshape this geometry early in learning and accelerate subsequent adaptation.

Model selection and expert aggregation in bandits. A closely related direction studies how to combine multiple candidate policies or base algorithms online. Classical expert-learning methods provide regret guarantees relative to the best fixed expert in hindsight [9]. In the bandit setting, CORRAL [6] uses Online Mirror Descent with a log-barrier regularizer to achieve regret competitive with the best base algorithm; subsequent refinements tighten the rates and extend the framework to contextual and stochastic settings [39, 36, 64, 62, 19]. These approaches are inherently *randomized* and target robustness: the master performs nearly as well as the best base algorithm regardless of which one that is. Our objective is different—we want a *deterministic* rule that exploits the complementarity between a good early frozen policy and a strong asymptotic adaptive learner to *strictly outperform* both. Correspondingly, our analysis does not bound a relative regret against the best base, but instead tracks how early actions selected by one policy reshape the design matrix and accelerate the statistical efficiency of another.

Warm starts and prior information in contextual bandits. Several works have investigated how prior information can be incorporated into contextual bandit algorithms through initialization, offline datasets, or pretrained predictors. Such approaches can substantially improve short-horizon performance by reducing uncertainty before online interaction begins [10, 58]. Related directions in

transfer learning and structured bandits further show that leveraging information across tasks or user populations can accelerate adaptation and reduce exploration requirements [32, 24]. Meta-learning approaches similarly learn priors that enable faster exploration in new environments [47]. Empirical work on neural-linear bandit architectures also demonstrates that representation pretraining can significantly improve early decision quality [44]. Our analysis formalizes these intuitions in a linear bandit framework: informative early actions contract the confidence ellipsoid on specific competitors (Lemmas 3 and 4), and the resulting saving is a difference of width-limited regrets rather than an improvement of the determinant-ratio rate, which Proposition 14 shows cannot by itself exceed a constant factor.

Geometry of information gain in linear bandits. A central theme in the analysis of linear bandit algorithms is that learning efficiency is governed by the geometry of the accumulated design matrix. Early work on stochastic linear optimization under bandit feedback showed that regret depends explicitly on the growth of the covariance matrix associated with the chosen actions and the resulting confidence ellipsoid [18]. Subsequent analyses of linearly parameterized bandits further clarified how the eigenvalue structure of this matrix determines parameter identifiability and sample complexity [46]. Later work established sharp self-normalized confidence bounds and finite-time regret guarantees expressed directly in terms of the design matrix and its inverse [1], while related results connected spectral growth of the covariance matrix to optimal arm identification in linear settings [49]. From an information-theoretic perspective, regret can also be related to determinant growth and effective dimension of the covariance operator [50, 54]. The price (1) is the single-direction shadow of the allocation programs behind asymptotic instance-dependent lower bounds, where regret is minimized subject to coverage constraints on every direction [25, 30]; our use of it is non-asymptotic and comparative: two policies on the same instance pay different prices for the same coverage. Prior side information has also been analyzed as hints or predictable sequences in online learning [40, 17]. The comparison with [17] is the closest: there the learner receives a hint about $\theta^\star$ and the regret improves as a function of the hint's accuracy, with the standard rate recovered when the hint is poor. Our warm start is not a hint about $\theta^\star$ but data in the design matrix, and its value is not a function of accuracy: two experts of equal accuracy can leave design matrices of very different value (Section 4.3), and an expert of zero accuracy can leave one of enormous value (Proposition 2). What governs the value is the price ratio of Theorem 1. The warm-start analyses of [10, 58] bound the regret of a learner started from logged data through the quality of that data; they are offline in our sense (row 1 of Table 1), and our contribution relative to them is the online accounting, in which the acquisition of the data is charged and can exceed its value. Our work extends this geometric viewpoint by showing that a fixed policy can shape the spectrum of the design matrix during early rounds of interaction, thereby improving conditioning of the confidence ellipsoid and accelerating the learning rate of subsequent adaptive exploration.

Large language models for sequential decision making. Recent work has explored the use of large language models (LLMs) as policies in contextual bandit environments, demonstrating that frozen pretrained models can achieve strong short-horizon performance even without online updates [13]. More broadly, LLMs exhibit strong in-context learning capabilities [12, 56, 23, 7], and instruction tuning further improves their effectiveness in downstream decision tasks [38, 16]. Several recent approaches interpret sequential decision making itself as a sequence-modeling problem and adapt transformer architectures to interactive settings [14, 60, 42, 51, 41, 33]. Closest to our work is the approach of [13], who combine an LLM policy with an adaptive contextual bandit learner using an online model-selection master and empirically document the reward-multiplication phenomenon on large-scale recommendation datasets. Our contribution relative to that line is theoretical: we give a regret-based explanation for why such combinations can beat both components, and replace the randomized CORRAL-style master with a deterministic confidence-gated rule whose routing decision directly exposes the geometric warm-start mechanism. A concurrent line of work [26] also uses LLM uncertainty to gate exploration, but does so by calibrating an embedding-based uncertainty estimate to adjust UCB-style exploration bonuses, rather than by comparing self-normalized confidence radii of a frozen predictor and an online learner; their analysis bounds the regret of a single calibrated agent through the accuracy of the LLM's reward predictions, whereas we give an exact decomposition of the hybrid's advantage, bound the multiplier over the cold learner on any finite-arm instance by a ratio of coverage prices, and show the bound is attained; in their setting the LLM's value enters through prediction error, in ours through the design matrix, and the two can disagree by a factor of the horizon (Proposition 2). Two concurrent works are closer still. [**?**] warm-start a linear

bandit from LLM-generated preferences and bound the warm-started estimator's error through a single prior-error term measured in the geometry of the pretraining design matrix, which treats the design matrix and the response vector as one object; our analysis separates the two, shows that under linear rewards the second is determined by the first, and identifies the expert-dependent remainder in closed form, which is the part that differs across equally accurate experts on our data. [**?**] inject LLM pseudo-observations into LinUCB with a calibration-dependent weight on both the design matrix and the response vector, a routing rule close in spirit to ours, and analyze it through a bias–variance argument without a regret bound; the decomposition of Proposition 1 and the price bound of Theorem 1 apply to any such rule, since both hold for an arbitrary routed set. Additional evidence that LLMs can exhibit in-context reinforcement learning behavior [37, 28] and that LLM-derived priors can jump-start exploration [8, 57] further motivates the study of principled mechanisms for combining frozen predictors with online learners.

B Proofs

Section B.1 proves the decomposition identity. Section B.2 reproduces the proof of Proposition 2 from the original construction. Section B.3 proves Proposition 3. Sections B.4–B.6 prove the three lemmas of Section 3.6, and Section B.7 proves Proposition 14.

B.1 Proof of Proposition 1

Proof. Split $[T]$ into $[1,\tau]$ and $[\tau+1,T]$. On $[1,\tau]$ the hybrid plays a_t^{E}, so $R_{[1,\tau]}(\mathsf{H}_\tau) = R_{[1,\tau]}(\mathsf{E})$. Hence

$$R_T(\mathsf{C}) - R_T(\mathsf{H}_\tau) = \big(R_{[1,\tau]}(\mathsf{C}) - R_{[1,\tau]}(\mathsf{E})\big) + \big(R_{[\tau+1,T]}(\mathsf{C}) - R_{[\tau+1,T]}(\mathsf{H}_\tau)\big) = A_\tau + V_\tau,$$

and

$$R_T(\mathsf{E}) - R_T(\mathsf{H}_\tau) = \big(R_{[1,\tau]}(\mathsf{E}) - R_{[1,\tau]}(\mathsf{E})\big) + \big(R_{[\tau+1,T]}(\mathsf{E}) - R_{[\tau+1,T]}(\mathsf{H}_\tau)\big) = B_\tau.$$

Both are identities between sums of gaps, so they hold for every noise realization and therefore in expectation. The "if and only if" statements are restatements of the identities. □

B.2 Proof of Proposition 2

We first record the standard subgaussian concentration lemma used in the high-probability part of the argument, and then construct the linear bandit instance and verify the three claims of Proposition 2.

Lemma 5 (Subgaussian concentration, Corollary 5.5 of [31])**.** *Assume that $X_1 - \mu, \dots, X_n - \mu$ are independent, σ-subgaussian random variables. Then for any $\varepsilon \geq 0$,*

$$\mathbb{P}(\hat{\mu} \geq \mu + \varepsilon) \leq \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right), \qquad \mathbb{P}(\hat{\mu} \leq \mu - \varepsilon) \leq \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right),$$

where

$$\hat{\mu} = \frac{1}{n}\sum_{t=1}^{n} X_t.$$

Problem setup. Let the dimension satisfy $d \geq n+1$, where n is the horizon. The unknown parameter is

$$\theta^\star = e_1 \in \mathbb{R}^d,$$

where $\{e_i\}_{i=1}^d$ denotes the standard basis. Rewards follow the stochastic linear model

$$X_t = \langle \theta^\star, A_t \rangle + \eta_t,$$

where η_t are independent mean-zero σ-subgaussian random variables.

At each round $t = 1, \dots, n$, the learner is offered two actions,

$$g := e_1, \qquad b_t := L e_{t+1},$$

where $L > 1$ is a fixed constant. Their expected rewards are

$$\langle \theta^\star, g \rangle = 1, \qquad \langle \theta^\star, b_t \rangle = 0.$$

Both LinUCB variants below use the rule

$$\mathrm{UCB}_t(a) = a^\top \hat{\theta}_{t-1} + \beta \|a\|_{V_{t-1}^{-1}},$$

with

$$V_{t-1} = V_0 + \sum_{s<t} A_s A_s^\top, \qquad \hat{\theta}_{t-1} = V_{t-1}^{-1} \sum_{s<t} A_s X_s.$$

Algorithm 1: a constant policy. Consider a policy π_{const} that selects an action by maximizing $\langle \hat{\theta}_{\mathrm{const}}, a \rangle$ for some fixed parameter estimate $\hat{\theta}_{\mathrm{const}} \in \mathbb{R}^d$ that is never updated. For concreteness, take

$$\hat{\theta}_{\mathrm{const}} = \frac{1}{L} \sum_{j=2}^{n+1} e_j,$$

so that

$$\langle \hat{\theta}_{\mathrm{const}}, g \rangle = 0, \qquad \langle \hat{\theta}_{\mathrm{const}}, b_t \rangle = 1 \quad \text{for every } t \le n.$$

Then π_{const} selects b_t at every round t. Since each pull of b_t yields expected reward 0 while g has expected reward 1, the cumulative pseudo-regret of π_{const} on the horizon n is exactly n. This proves part (1) of Proposition 2.

The failure mode is structural rather than statistical: π_{const} never queries direction e_1, so no amount of data can refine $\hat{\theta}_{\mathrm{const}}$ in the reward-relevant direction. The policy has no learning mechanism. We will see, however, that the trajectory it produces is precisely what suppresses the misplaced optimism of LinUCB in the bad directions.

Algorithm 2: LinUCB with isotropic initialization. The first LinUCB variant uses

$$V_0 = \lambda I_d, \qquad \hat{\theta}_0 = 0.$$

At $t = 1$,

$$\|g\|^2_{V_0^{-1}} = \frac{1}{\lambda}, \qquad \|b_1\|^2_{V_0^{-1}} = \frac{L^2}{\lambda}.$$

Since $\hat{\theta}_0 = 0$, we have

$$\mathrm{UCB}_1(b_1) > \mathrm{UCB}_1(g),$$

so the algorithm selects b_1.

Now suppose inductively that the algorithm has selected $b_1, \dots, b_{t-1}$ in the first $t-1$ rounds. Then

$$V_{t-1} = \lambda I_d + \sum_{s=1}^{t-1} b_s b_s^\top = \lambda I_d + L^2 \sum_{j=2}^{t} e_j e_j^\top.$$

Hence V_{t-1} is diagonal in the standard basis. In particular, for the fresh direction e_{t+1},

$$e_{t+1}^\top V_{t-1}^{-1} e_{t+1} = \frac{1}{\lambda}, \qquad \|b_t\|^2_{V_{t-1}^{-1}} = \frac{L^2}{\lambda}, \qquad \|g\|^2_{V_{t-1}^{-1}} = \frac{1}{\lambda}.$$

The key point is that although the observed rewards on $b_1, \dots, b_{t-1}$ contain noise, the estimate

$$\hat{\theta}_{t-1} = V_{t-1}^{-1} \sum_{s=1}^{t-1} b_s X_s$$

lies entirely in $\mathrm{span}\{e_2, \dots, e_t\}$. Therefore,

$$g^\top \hat{\theta}_{t-1} = 0, \qquad b_t^\top \hat{\theta}_{t-1} = 0,$$

because both $g = e_1$ and the fresh arm $b_t = L e_{t+1}$ are orthogonal to $\mathrm{span}\{e_2, \dots, e_t\}$.

Thus, even in the stochastic-reward setting, the comparison between g and b_t is determined purely by the confidence terms:

$$\mathrm{UCB}_t(g) = \frac{\beta}{\sqrt{\lambda}}, \qquad \mathrm{UCB}_t(b_t) = \frac{\beta L}{\sqrt{\lambda}}.$$

Since $L > 1$, we have $\mathrm{UCB}_t(b_t) > \mathrm{UCB}_t(g)$ for all $t \le n$. Hence the isotropically initialized algorithm selects b_t at every round and, like π_{const}, incurs cumulative pseudo-regret n. This proves part (2) of Proposition 2.

Algorithm 3: LinUCB with an anisotropic warm start. We now define an initialization that is realized by exactly the trajectory π_{const} would produce. Fix an integer $m \geq 1$, and let

$$V_0' = \lambda I_d + mL^2 \sum_{j=2}^{n+1} e_j e_j^\top, \qquad \hat{\theta}_0' = 0.$$

This corresponds to having observed each bad direction Le_j, $j = 2, \ldots, n+1$, exactly m times before the online interaction begins — precisely the kind of data π_{const} generates if rolled out for mn rounds on a sequence of action sets containing each bad direction. Thus V_0' is a genuine design matrix obtainable from a collection of outer products produced by a constant logging policy.

At $t = 1$,

$$\|g\|^2_{(V_0')^{-1}} = \frac{1}{\lambda}, \qquad \|b_1\|^2_{(V_0')^{-1}} = L^2 \cdot \frac{1}{\lambda + mL^2} = \frac{L^2}{\lambda + mL^2}.$$

Hence, if

$$m > \frac{\lambda(L^2 - 1)}{L^2},$$

then $\frac{1}{\lambda} > \frac{L^2}{\lambda + mL^2}$, and therefore

$$\mathrm{UCB}_1'(g) > \mathrm{UCB}_1'(b_1).$$

So the warm-started algorithm selects g at round 1.

Suppose now that the algorithm continues to select g for the first s rounds. Then

$$V_s' = \lambda I_d + mL^2 \sum_{j=2}^{n+1} e_j e_j^\top + s\, e_1 e_1^\top.$$

Since each reward observed on g has mean 1,

$$\sum_{i=1}^{s} X_i = s + \sum_{i=1}^{s} \eta_i,$$

and therefore

$$g^\top \hat{\theta}_s' = e_1^\top (V_s')^{-1} \sum_{i=1}^{s} g X_i = \frac{1}{\lambda + s} \sum_{i=1}^{s} X_i.$$

Applying Lemma 5 with $\mu = 1$, we obtain that with probability at least $1 - \delta$, simultaneously for all $1 \leq s \leq n$,

$$\frac{1}{s} \sum_{i=1}^{s} X_i \geq 1 - \sqrt{\frac{2\sigma^2 \log(2n/\delta)}{s}}.$$

On this event,

$$g^\top \hat{\theta}_s' \geq \frac{s}{\lambda + s} \left(1 - \sqrt{\frac{2\sigma^2 \log(2n/\delta)}{s}}\right).$$

For every fresh bad arm $b_{s+1} = Le_{s+2}$,

$$b_{s+1}^\top \hat{\theta}_s' = 0,$$

since the online trajectory has only sampled the direction e_1, and the warm-start contributes only to the design matrix, not to the response vector. Moreover,

$$\|b_{s+1}\|^2_{(V_s')^{-1}} = \frac{L^2}{\lambda + mL^2}, \qquad \|g\|^2_{(V_s')^{-1}} = \frac{1}{\lambda + s}.$$

On the same high-probability event,

$$\mathrm{UCB}_{s+1}'(g) \geq \frac{s}{\lambda + s} \left(1 - \sqrt{\frac{2\sigma^2 \log(2n/\delta)}{s}}\right) + \frac{\beta}{\sqrt{\lambda + s}},$$

while

$$\mathrm{UCB}'_{s+1}(b_{s+1}) = \frac{\beta L}{\sqrt{\lambda + mL^2}}.$$

Consequently, if m is chosen large enough so that for every $1 \le s \le n$,

$$\frac{s}{\lambda + s}\left(1 - \sqrt{\frac{2\sigma^2 \log(2n/\delta)}{s}}\right) + \frac{\beta}{\sqrt{\lambda + s}} > \frac{\beta L}{\sqrt{\lambda + mL^2}}, \tag{8}$$

then, with probability at least $1 - \delta$, the warm-started algorithm selects g at every round, and the cumulative pseudo-regret over the n online rounds is 0 on this event. This proves part (3) of Proposition 2; condition (8) is the explicit "m sufficiently large" requirement referenced in the proposition statement. The condition is satisfiable by a finite m whenever the left-hand side is positive for every s, which holds for any $\beta \ge \sigma\sqrt{2\log(2n/\delta)}$ (in particular for the radius of [1]), since then $\beta/\sqrt{\lambda + s} \ge \sigma\sqrt{2\log(2n/\delta)}\,\sqrt{s}/(\lambda + s)$. □

Remark 4. The warm-started algorithm in the proof uses $\widehat{\theta}'_0 = 0$, i.e., it carries the offline Gram matrix but not the offline response sum. Because every offline observation is of an arm with expected reward 0, the offline response sum has mean zero in each bad direction and carrying it would only add a mean-zero perturbation to $\widehat{\theta}$ in directions orthogonal to g; the comparison between g and b_{s+1} is unchanged up to the noise term already absorbed in the choice of m. We keep the convention of the original construction.

B.3 Proof of Proposition 3

Proof. Work on the instance of Section B.2 with horizon T: $\theta^\star = e_1$, $\mathcal{A}_t = \{g, b_t\}$ with $g = e_1$ and $b_t = Le_{t+1}$. Fix an arbitrary expert E and a handoff time τ. Whatever the expert plays during rounds $1, \dots, \tau$, and whatever LinUCB plays afterwards, the set of arms played before round t is contained in $\{g\} \cup \{b_s : s < t\}$, all of which are orthogonal to e_{t+1}. Hence for every t, the design matrix $V_t = \lambda I + \sum_{s<t} x_{s,A_s} x_{s,A_s}^\top$ is block diagonal with the block on e_{t+1} equal to λ, and $\widehat{\theta}_t = V_t^{-1} b_t$ lies in $\mathrm{span}\{e_1, \dots, e_t\}$. Therefore

$$\|b_t\|^2_{V_t^{-1}} = \frac{L^2}{\lambda}, \qquad \langle \widehat{\theta}_t, b_t \rangle = 0, \qquad \text{for every } t \text{ and every trajectory.}$$

So $\mathrm{UCB}_t(b_t) = \beta L/\sqrt{\lambda}$ exactly, for all t.

Now bound $\mathrm{UCB}_t(g)$. Let s_t be the number of pulls of g before round t and $X_1, \dots, X_{s_t}$ the rewards observed on those pulls; since V_t is diagonal with entry $\lambda + s_t$ on e_1 and the rewards on bad arms enter b_t only in directions orthogonal to e_1,

$$\langle \widehat{\theta}_t, g \rangle = \frac{1}{\lambda + s_t} \sum_{i=1}^{s_t} X_i, \qquad \|g\|^2_{V_t^{-1}} = \frac{1}{\lambda + s_t} \le \frac{1}{\lambda}.$$

By Lemma 5 and a union bound over $s \in \{1, \dots, T\}$, with probability at least $1 - \delta$ we have $\sum_{i \le s} X_i \le s + \sqrt{2\sigma^2 s \log(2T/\delta)}$ for all $s \le T$ simultaneously. On that event,

$$\langle \widehat{\theta}_t, g \rangle \le \frac{s_t}{\lambda + s_t} + \frac{\sqrt{2\sigma^2 s_t \log(2T/\delta)}}{\lambda + s_t} \le 1 + \frac{\sigma\sqrt{2\log(2T/\delta)}}{\sqrt{\lambda + s_t}} \le 1 + \kappa,$$

using $\sqrt{s_t}/(\lambda + s_t) \le 1/\sqrt{\lambda + s_t} \le 1/\sqrt{\lambda}$. Hence $\mathrm{UCB}_t(g) \le 1 + \kappa + \beta/\sqrt{\lambda}$. Under the hypothesis $\beta(L-1)/\sqrt{\lambda} > 1 + \kappa$ this is strictly less than $\beta L/\sqrt{\lambda} = \mathrm{UCB}_t(b_t)$, so LinUCB, whenever it acts, plays b_t. The argument did not use anything about rounds $1, \dots, \tau$ beyond the fact that they cannot produce coverage of e_{t+1} for $t > \tau$, so it holds for every expert and every τ. Since $\Delta_{t,b_t} = 1$, $R_{[\tau+1,T]}(\mathsf{H}_\tau) = T - \tau$. Cold LinUCB plays b_t on every round by part (2) of Proposition 2 (no condition on β is needed there because $\widehat{\theta}_t \perp g$ on its own trajectory), so $R_{[\tau+1,T]}(\mathsf{C}) = T - \tau$ and $V_\tau = 0$. □

B.4 Proof of Lemma 2

Proof. Fix a round t at which LinUCB acts, and write $x^\star = x_{t,a_t^\star}$, $x = x_{t,A_t}$, $w(u) = \|u\|_{V_t^{-1}}$. On $\mathcal{E}_\delta$, Cauchy–Schwarz gives $|\langle \widehat{\theta}_t - \theta^\star, u\rangle| \le \beta_t w(u)$ for every u. By the selection rule, $\langle \widehat{\theta}_t, x\rangle + \beta_t w(x) \ge \langle \widehat{\theta}_t, x^\star\rangle + \beta_t w(x^\star) \ge \langle \theta^\star, x^\star\rangle$, and $\langle \widehat{\theta}_t, x\rangle \le \langle \theta^\star, x\rangle + \beta_t w(x)$. Combining, $\Delta_{t,A_t} = \langle \theta^\star, x^\star - x\rangle \le 2\beta_t w(x)$. Thus $\Delta_{t,A_t} = \min\{\Delta_{t,A_t}, 2\beta_t w(x_{t,A_t})\} \le \max_a \min\{\Delta_{t,a}, 2\beta_t w(x_{t,a})\} = \Phi_t(V_t)$. Monotonicity: $V \preceq V'$ implies $V'^{-1} \preceq V^{-1}$, so every width is non-increasing and so is each term in the maximum. □

B.5 Proof of Lemma 3

Proof. $\lambda I + M \succeq \lambda I$ gives $(\lambda I + M)^{-1} \preceq (\lambda I)^{-1}$, which is the first claim. For the second, $\lambda I + M \succeq \lambda I + mP$, so $x^\top(\lambda I + M)^{-1}x \le x^\top(\lambda I + mP)^{-1}x$, and for $x \in \mathrm{Im}(P)$ the right-hand side is $\|x\|_2^2/(\lambda+m)$ because $(\lambda I+mP)^{-1}$ acts as $(\lambda+m)^{-1}$ on $\mathrm{Im}(P)$. For the third, if $x \perp \mathrm{Im}(M)$ then x is an eigenvector of $\lambda I + M$ with eigenvalue λ, so $x^\top(\lambda I + M)^{-1}x = \|x\|_2^2/\lambda$. □

B.6 Proof of Lemma 4

Proof. By the Sherman–Morrison formula, $(V + xx^\top)^{-1} = V^{-1} - \dfrac{V^{-1}xx^\top V^{-1}}{1 + x^\top V^{-1}x}$. Multiplying on both sides by z,

$$z^\top(V + xx^\top)^{-1}z = z^\top V^{-1}z - \frac{(z^\top V^{-1}x)^2}{1 + \|x\|_{V^{-1}}^2} = \|z\|_{V^{-1}}^2\Big(1 - \frac{(z^\top V^{-1}x)^2}{(1 + \|x\|_{V^{-1}}^2)\|z\|_{V^{-1}}^2}\Big).$$

The bracket is at least $1 - \dfrac{\|x\|_{V^{-1}}^2\|z\|_{V^{-1}}^2}{(1 + \|x\|_{V^{-1}}^2)\|z\|_{V^{-1}}^2} > 0$ by Cauchy–Schwarz in the V^{-1} inner product, so $\rho_V \in [0, 1)$. At $V = \lambda I$, $z^\top V^{-1}x = \langle z, x\rangle/\lambda$, $\|x\|_{V^{-1}}^2 = \|x\|_2^2/\lambda$, $\|z\|_{V^{-1}}^2 = \|z\|_2^2/\lambda$, and substituting gives $\rho_{\lambda I}(x, z) = \dfrac{\langle x, z\rangle^2}{\|x\|_2^2\|z\|_2^2} \cdot \dfrac{\|x\|_2^2}{\lambda + \|x\|_2^2}$. □

B.7 Proof of Proposition 14

Proof. The acquisition term is the definition of $R_{[1,\tau]}(\mathsf{H}_\tau) = R_{[1,\tau]}(\mathsf{E})$. For the LinUCB phase we give two bounds and take the minimum.

First bound. By Lemma 2 applied at each $t > \tau$, $\sum_{t>\tau} \Delta_{t,A_t} \le \sum_{t>\tau} \Phi_t(V_t)$.

Second bound. From the proof of Lemma 2, $\Delta_{t,A_t} \le 2\beta_t\|x_{t,A_t}\|_{V_t^{-1}} \le 2\beta_T\|x_{t,A_t}\|_{V_t^{-1}}$ for $t > \tau$ (the radius β_t is non-decreasing in t). Cauchy–Schwarz over $t = \tau + 1, \dots, T$ gives

$$\sum_{t>\tau} \Delta_{t,A_t} \le 2\beta_T\sqrt{(T - \tau)\sum_{t>\tau}\|x_{t,A_t}\|_{V_t^{-1}}^2}.$$

By the elliptical potential lemma [1, Lemma 11], $\sum_{t>\tau}\min\{1, \|x_{t,A_t}\|_{V_t^{-1}}^2\} \le 2\log\det V_{T+1}/\det V_{\tau+1}$; since $\|x_{t,A_t}\|_{V_t^{-1}}^2 \le L^2/\lambda$, the truncation at 1 costs at most a factor $\max\{1, L^2/\lambda\}$ on the sum of squares, which is the factor $\max\{1, L/\sqrt{\lambda}\}$ in (7) after the square root; with $L^2 \le \lambda$ nothing is lost. This gives the second term of (7).

Determinant ratio. Let $V_{\tau+1} = \sum_j \sigma_j u_j u_j^\top$ be an eigendecomposition and $G = \sum_{t>\tau} x_{t,A_t}x_{t,A_t}^\top$. Then $\det(V_{\tau+1} + G)/\det V_{\tau+1} = \det\Big(I + V_{\tau+1}^{-1/2}GV_{\tau+1}^{-1/2}\Big) \le \prod_j(1 + u_j^\top G u_j/\sigma_j)$ by Hadamard's inequality for positive definite matrices, and $u_j^\top G u_j = \sum_{t>\tau}\langle u_j, x_{t,A_t}\rangle^2 \le L^2(T - \tau)$. Taking logarithms gives the stated bound. If all played actions lie in an r-dimensional subspace $\mathrm{Im}(P)$ then G is supported there and only the r eigenvalues of $V_{\tau+1}$ restricted to $\mathrm{Im}(P)$ appear, which is how the factor r in the remark after the theorem arises; if in addition $V_{\tau+1} \succeq (\lambda + \nu\tau)P$ on that subspace, each $\sigma_j \ge \lambda + \nu\tau$.

The cold bound is the case $\tau = 0$, $V_1 = \lambda I$, $\sigma_j = \lambda$. The ratio Ψ in the text is the quotient of the two isotropic bounds and is stated only to show that it does not tend to zero. □

Remark 5 (Validity of the confidence event with an online warm start). The hybrid's LinUCB phase uses $(V_{\tau+1}, b_{\tau+1})$ built from the expert-routed rounds together with the observed rewards on those rounds. The self-normalized bound of [1] applies to any sequence of (feature, reward) pairs in which the feature at round s is measurable with respect to the history before s, whoever chose it, so the event $\mathcal{E}_\delta$ covers the hybrid's trajectory. If instead the warm start carried M_τ but dropped c_τ ("passive" updates), the estimate $\widehat{\theta}_t$ would be biased toward zero in the directions of M_τ and the event would not be guaranteed; the Router in our experiments updates both, and Algorithm 1 reflects this.

B.8 Proofs for Section 3.4

Proof of Proposition 4. Write $x^\star = x_{t,a_t^\star}$, $x = x_{t,A_t}$, $w(u) = \|u\|_{V_t^{-1}}$. Cauchy–Schwarz in the V_t inner product gives $|\langle \widehat{\theta}_t - \theta^\star, u\rangle| \le \rho_t w(u)$ for every u; this is the inequality in the proof of Lemma 2 with the realized error in place of its bound. The selection rule gives

$$\langle \widehat{\theta}_t, x\rangle + \beta_t w(x) \;\ge\; \langle \widehat{\theta}_t, x^\star\rangle + \beta_t w(x^\star) \;\ge\; \langle \theta^\star, x^\star\rangle + (\beta_t - \rho_t) w(x^\star) \;\ge\; \langle \theta^\star, x^\star\rangle,$$

using $\rho_t \le \beta_t$, and $\langle \widehat{\theta}_t, x\rangle \le \langle \theta^\star, x\rangle + \rho_t w(x)$. Subtracting, $\Delta_{t,A_t} \le (\beta_t + \rho_t) w(x)$, and taking the maximum over a of the minimum with the gap gives (3). Monotonicity in V is Lemma 2; in ρ it is immediate. Finally $\min\{\Delta, (\beta+\rho)w\} \ge \min\{\Delta, \beta w\} \ge \frac{1}{2}\min\{\Delta, 2\beta w\}$. □

Proof of Proposition 5. $c = \sum_s r_s x_s = \sum_s (\langle \theta^\star, x_s\rangle + \eta_s) x_s = \big(\sum_s x_s x_s^\top\big)\theta^\star + \sum_s \eta_s x_s = M\theta^\star + \xi$. Conditionally on the features, ξ is Gaussian with mean zero and covariance $\sigma^2 \sum_s x_s x_s^\top = \sigma^2 M$. Then $\widehat{\theta}_1 - \theta^\star = V_1^{-1}(M\theta^\star + \xi) - \theta^\star = V_1^{-1}(M - V_1)\theta^\star + V_1^{-1}\xi = -\lambda V_1^{-1}\theta^\star + V_1^{-1}\xi$. Both $V_1 = \lambda I + M$ and the law of ξ are functions of M alone, so the conditional law of (V_1, b_1) given M is the same whatever policy produced the actions. The LinUCB phase is a deterministic function of (V_1, b_1) and of the future contexts and noise, which are independent of the warm start; hence the tail trajectories agree in law, and so do V_τ and B_τ. □

Proof of Proposition 6. $c = \sum_s (\langle \theta^\star, x_s\rangle + \varepsilon(x_s) + \eta_s) x_s = M\theta^\star + \sum_s \varepsilon(x_s) x_s + \xi$, and $V_1^{-1} M\theta^\star = \theta^\star - \lambda V_1^{-1}\theta^\star$. □

Proof of Proposition 7. Fix $\widetilde{\theta}$ and write $w(u) = \|u\|_{V_t^{-1}}$, $x^\star = x_{t,a_t^\star}$, $x = x_{t,A_t}$. Cauchy–Schwarz gives $|\langle \widehat{\theta}_t - \widetilde{\theta}, u\rangle| \le \widetilde{\rho}_t w(u)$. By the definition of $\widetilde{\varepsilon}$,

$$\bar{r}(x^\star) = \langle \widetilde{\theta}, x^\star\rangle + \widetilde{\varepsilon}(x^\star) \le \langle \widehat{\theta}_t, x^\star\rangle + \widetilde{\rho}_t w(x^\star) + \widetilde{\varepsilon}(x^\star).$$

The selection rule gives $\langle \widehat{\theta}_t, x^\star\rangle \le \langle \widehat{\theta}_t, x\rangle + \beta_t w(x) - \beta_t w(x^\star)$, and $\langle \widehat{\theta}_t, x\rangle \le \langle \widetilde{\theta}, x\rangle + \widetilde{\rho}_t w(x) = \bar{r}(x) - \widetilde{\varepsilon}(x) + \widetilde{\rho}_t w(x)$. Chaining the three displays,

$$\bar{r}(x^\star) - \bar{r}(x) \le (\beta_t + \widetilde{\rho}_t) w(x) + (\widetilde{\rho}_t - \beta_t) w(x^\star) + \widetilde{\varepsilon}(x^\star) - \widetilde{\varepsilon}(x),$$

and $(\widetilde{\rho}_t - \beta_t) w(x^\star) \le (\widetilde{\rho}_t - \beta_t)_+ w(x^\star)$. □

B.9 Proof of Theorem 1

Throughout, work on the confidence event $\mathcal{E}_\delta$ of cold LinUCB (Section 3.6), so that Lemma 2 applies at every round, and write $\beta = \beta_T \ge \beta_t$.

(a) Fix $x \in S$ and suppose cold LinUCB plays x at round t after m earlier pulls of x. From the proof of Lemma 2, the played arm satisfies $\Delta(x) = \Delta_{t,A_t} \le 2\beta_t \|x\|_{V_t^{-1}} \le 2\beta \|x\|_{V_t^{-1}}$. Since $V_t \succeq \lambda I + m\, xx^\top$ (the m earlier pulls are among the summands of V_t), Lemma 3 and the Sherman–Morrison identity give $\|x\|^2_{V_t^{-1}} \le \|x\|^2_{(\lambda I + m xx^\top)^{-1}} = \|x\|_2^2/(\lambda + m\|x\|_2^2)$. Hence $\Delta(x)^2(\lambda + m\|x\|_2^2) \le 4\beta^2\|x\|_2^2$, i.e. $\lambda + m\|x\|_2^2 \le c_x$. If $c_x < \lambda$ this is impossible already for $m = 0$, so x is never played. If $c_x \ge \lambda$, every pull of x has $m \le (c_x - \lambda)/\|x\|_2^2$, so the number of pulls is at most $(c_x - \lambda)/\|x\|_2^2 + 1 = m_x$. The number of pulls is also at most N_x, the number of rounds at which x is offered. Each pull costs exactly $\Delta(x)$ because $x^\star \in \mathcal{A}_t$, and pulls of optimal arms cost nothing. Summing over $x \in S$ gives (a).

(b) Fix $u_x \in U \subseteq U_{\mathrm{cert}}(t_0)$ and write $C(u_x) := \sum_{t \le t_0} \langle x_t^\pi, u_x \rangle^2$ for the coverage of u_x after round t_0. Since $x^\top V_{t_0+1}^\pi x = \lambda \|x\|_2^2 + \sum_{t \le t_0} \langle x_t^\pi, x \rangle^2 = \|x\|_2^2 (\lambda + C(u_x))$, Cauchy–Schwarz in the $V_{t_0+1}^\pi$ inner product (as in the proof of Lemma 1) gives $\|x\|^2_{(V_{t_0+1}^\pi)^{-1}} \ge \|x\|_2^4 / (x^\top V_{t_0+1}^\pi x) = \|x\|_2^2 / (\lambda + C(u_x))$. Certification, $4\beta^2 \|x\|^2_{(V_{t_0+1}^\pi)^{-1}} < \Delta(x)^2$, therefore forces $\lambda + C(u_x) > 4\beta^2 \|x\|_2^2 / \Delta(x)^2 = c_x$, i.e. $C(u_x) > c_x - \lambda$; and trivially $C(u_x) \ge 0 \ge (c_x - \lambda)_+$ when $c_x < \lambda$. This is the first inequality. For the second, exchange the order of summation: $\sum_{u_x \in U} C(u_x) = \sum_{t \le t_0} \mathrm{cov}_U(x_t^\pi)$. Split the rounds $t \le t_0$ into those with $\Delta(x_t^\pi) = 0$, which contribute $C_U^{\mathrm{free}}(t_0)$ by definition, and those with $\Delta(x_t^\pi) > 0$. At a round of the second kind with $\mathrm{cov}_U(x_t^\pi) > 0$, the definition of $\pi_U(t_0)$ as a minimum gives $\mathrm{cov}_U(x_t^\pi) \le \Delta(x_t^\pi)/\pi_U(t_0)$; at a round with $\mathrm{cov}_U(x_t^\pi) = 0$ the same inequality holds trivially. Summing, the rounds of the second kind contribute at most $\sum_{t \le t_0} \Delta(x_t^\pi)/\pi_U(t_0) = R_{[1,t_0]}(\pi)/\pi_U(t_0)$ (if there are no rounds of the second kind with positive coverage, the contribution is 0, which matches the convention $1/\pi_U(t_0) = 0$).

(c) Split the sum in (a) into $u_x \in U$ and $u_x \notin U$. For $u_x \in U$, use $\min\{m_x, N_x\} \le m_x \le (c_x - \lambda)_+ / \|x\|_2^2 + 1$ to get $\Delta(x) \min\{m_x, N_x\} \le p(x)(c_x - \lambda)_+ + \Delta(x) \le p_U (c_x - \lambda)_+ + \Delta(x)$. Summing over $u_x \in U$ and applying (b),

$$\sum_{u_x \in U} \Delta(x) \min\{m_x, N_x\} \;\le\; p_U \Big(\frac{R_{[1,t_0]}(\pi)}{\pi_U(t_0)} + C_U^{\mathrm{free}}(t_0) \Big) + \sum_{u_x \in U} \Delta(x),$$

and $p_U / \pi_U(t_0) = P_U$. The terms with $u_x \notin U$ are carried unchanged. Finally $R_T(\pi) \ge R_{[1,t_0]}(\pi)$ because pseudo-regret is nonnegative on every round; dividing gives (2). For $\pi = \mathsf{H}_\tau$ and $t_0 = \tau$, $R_{[1,\tau]}(\mathsf{H}_\tau) = R_{[1,\tau]}(\mathsf{E})$ because the hybrid plays the expert's actions on $[1, \tau]$. □

The template as a check. On the instance of Theorem 2 take $U = \{e_j : j \in J\}$, all of which are certified at τ_0 by (iv). Then $p(b_j) = 1/L^2$, so $p_U = 1/L^2$; the expert plays only probes in the window, $\mathrm{cov}_U(p_j) = c^2$ and $\Delta(p_j) = \varepsilon$, so $\pi_U(\tau_0) = \varepsilon / c^2$ and $P_U = c^2/(\varepsilon L^2)$; $C_U^{\mathrm{free}} = 0$ because $g \perp U$; $\sum_U \Delta = K$; and the uncertified arms are the probes, for which $\Delta(p_j) \min\{m_{p_j}, N_{p_j}\} \le \varepsilon n$. The theorem gives $R_T(\mathsf{C}) / R_T(\mathsf{H}_{\tau_0}) \le c^2/(\varepsilon L^2) + (K + \varepsilon K n)/(\varepsilon K n) = c^2/(\varepsilon L^2) + 1 + 1/(\varepsilon n)$, consistent with the upper bound $1 + (8\beta^2 + 1)/(\varepsilon n) \le 1 + c^2/(\varepsilon L^2)$ proved directly. The same computation applies verbatim to Proposition 9, where the distractors have gap 0 and contribute nothing.

C Proofs for the Catalog of Section 3.1

Throughout this appendix the conventions (4) are in force, $\alpha := 1 - \varepsilon$, $m_0 := \lfloor \lambda / 30 \rfloor$, and "LinUCB acts at round t" means that the action at t is chosen by the LinUCB rule (every round for C; rounds $t > \tau$ for H_τ). Ties in the LinUCB rule are broken arbitrarily; every comparison we use is strict. For a policy π and a round t we write $s(t)$, $m_j(t)$, $n_j(t)$ for the numbers of pulls of g, b in direction e_j, and p in direction e_j before round t (all of them counted from the start, including expert-routed rounds), and we drop the argument t when no confusion arises. In the variant of Proposition 13 the roles of g and p are played by g_η and $p_{t,\eta}$ and $\alpha := (1 - \varepsilon)\eta$; we prove everything for general $\eta \in (0, 1]$ and set $\eta = 1$ outside that proposition.

C.1 Events and their probabilities

Lemma 6 (Good events)**.** *Fix a policy π on any instance of the template. Let $\mathcal{E}_{\mathrm{conf}}(\pi)$ be the event that $\|\widehat{\theta}_t - \theta^\star\|_{V_t} \le \beta$ for every $t \in [T]$, where (V_t, b_t) are formed from all of π's data before t. Let $\mathcal{E}_{\mathrm{noise}}(\pi)$ be the event that for every direction $j \in \{2, \dots, d\}$ and every $m \in [T]$, the sum $S_j^{(m)}$ of the noises on the first m pulls of b in direction e_j by π satisfies $|S_j^{(m)}| \le \sigma \sqrt{2 m \ell_1}$. Then $\mathbb{P}(\mathcal{E}_{\mathrm{conf}}(\pi)) \ge 1 - \delta$ and $\mathbb{P}(\mathcal{E}_{\mathrm{noise}}(\pi)) \ge 1 - \delta$. Moreover, under* (4)*, $\beta \ge 4\sigma\sqrt{2\ell_1}$.*

Proof. All arm vectors of the template have Euclidean norm at most L: $\|g_\eta\| = \eta \le 1$, $\|b_t\| = L$, $\|p_{t,\eta}\|^2 = \alpha^2 + c^2 \le 2 \le L^2$. Also $\|\theta^\star\| = 1$. By Theorem 2 of [1], with probability at least $1 - \delta$, simultaneously for all t, $\|\widehat{\theta}_t - \theta^\star\|_{V_t} \le \sigma \sqrt{2 \log\big(\det(V_t)^{1/2} \det(\lambda I)^{-1/2} / \delta \big)} + \sqrt{\lambda}$.

Since $V_t \preceq (\lambda + tL^2)I$, $\det(V_t)^{1/2}\det(\lambda I)^{-1/2} \le (1 + tL^2/\lambda)^{d/2}$, so the radius is at most $\sigma\sqrt{d\log(1+TL^2/\lambda) + 2\log(1/\delta)} + \sqrt{\lambda} \le \sigma\sqrt{d\,\ell_0} + \sqrt{\lambda} \le 2\sqrt{\lambda} = \beta$, using $\lambda \ge 1$ and $\lambda \ge \sigma^2 d\ell_0$. This proves the first claim. The theorem of [1] only requires the noise to be conditionally sub-Gaussian given the past and the current action, which holds in the reward-table model, and it applies to the data of any policy, not only to LinUCB's own.

For the second claim, in the reward-table model the noises on successive pulls of b in direction e_j form an i.i.d. σ-sub-Gaussian sequence $(\eta_{j,i})_{i\ge1}$ regardless of the policy, and $S_j^{(m)} = \sum_{i\le m}\eta_{j,i}$. By Hoeffding's inequality for sub-Gaussian sums, $\mathbb{P}(|S_j^{(m)}| > \sigma\sqrt{2m\ell_1}) \le 2e^{-\ell_1} = \delta/(dT)$. A union bound over $d-1 < d$ directions and T values of m gives probability at most δ for the complement. Finally $\beta = 2\sqrt{\lambda} \ge 2\sqrt{8\sigma^2\ell_1} = 4\sigma\sqrt{2\ell_1}$. □

Lemma 7 (Confidence consequences). *On $\mathcal{E}_{\mathrm{conf}}(\pi)$, for every t and every $x \in \mathbb{R}^d$, $|\langle\widehat{\theta}_t - \theta^\star, x\rangle| \le \beta\|x\|_{V_t^{-1}}$. Hence the index $U_t(x) := \langle\widehat{\theta}_t, x\rangle + \beta\|x\|_{V_t^{-1}}$ satisfies $\mu(x) \le U_t(x) \le \mu(x) + 2\beta\|x\|_{V_t^{-1}}$, where $\mu(x) = \langle\theta^\star, x\rangle$.*

Proof. Cauchy–Schwarz in the V_t inner product: $|\langle\widehat{\theta}_t - \theta^\star, x\rangle| \le \|\widehat{\theta}_t - \theta^\star\|_{V_t}\|x\|_{V_t^{-1}} \le \beta\|x\|_{V_t^{-1}}$. □

C.2 The state of LinUCB on the template

Every action vector of the template is g_η, some b in direction e_j, or some p_η in direction e_j, and the probe is the only one that couples two coordinates. The design matrix therefore has an arrowhead structure whose inverse is explicit.

Lemma 8 (Arrowhead inverse). *Let $V = \lambda I + Se_1e_1^\top + \sum_{j=2}^{d}\big(M_je_je_j^\top + n_jq_jq_j^\top\big)$ with $S, M_j, n_j \ge 0$ and $q_j = \alpha e_1 + ce_j$. Put*

$$D_j := \lambda + M_j + n_jc^2, \qquad u_j := \alpha c n_j,$$

$$Q := \lambda + S + \alpha^2\sum_{j\ge2} n_j - \sum_{j\ge2}\frac{u_j^2}{D_j} = \lambda + S + \alpha^2\sum_{j\ge2}\frac{n_j(\lambda + M_j)}{D_j}.$$

Then $Q \ge \lambda$ and

$$(V^{-1})_{11} = \frac{1}{Q}, \qquad (V^{-1})_{1j} = -\frac{u_j}{D_jQ},$$

$$(V^{-1})_{jj} = \frac{1}{D_j} + \frac{u_j^2}{D_j^2Q}, \qquad (V^{-1})_{jj'} = \frac{u_ju_{j'}}{D_jD_{j'}Q}\ \ (j \ne j'),$$

for $j, j' \ge 2$. In particular, if $n_j = 0$ then $(V^{-1})_{jj} = 1/D_j$ and $(V^{-1})_{1j} = (V^{-1})_{jj'} = 0$ for all $j' \ne j$; and if $n_j \ge 1$ and $M_j \ge 1$ then $(V^{-1})_{jj} \le 1/D_j + 1/M_j$.

Proof. Write $V = \begin{pmatrix} a & u^\top \\ u & D \end{pmatrix}$ with $a = \lambda + S + \alpha^2\sum_j n_j$, $u = (u_2, \dots, u_d)^\top$, and $D = \mathrm{diag}(D_2, \dots, D_d)$; this is the matrix in the statement because $q_jq_j^\top = \alpha^2e_1e_1^\top + \alpha c(e_1e_j^\top + e_je_1^\top) + c^2e_je_j^\top$. The Schur complement of D is $a - u^\top D^{-1}u = Q$, and the block inverse formula gives $(V^{-1})_{11} = Q^{-1}$, $(V^{-1})_{1,\cdot} = -Q^{-1}u^\top D^{-1}$, and $(V^{-1})_{\cdot,\cdot} = D^{-1} + D^{-1}uu^\top D^{-1}Q^{-1}$ on the lower block, which are the displayed entries. The second expression for Q follows from $\alpha^2n_j - u_j^2/D_j = \alpha^2n_j(D_j - c^2n_j)/D_j = \alpha^2n_j(\lambda + M_j)/D_j$, and it shows $Q \ge \lambda$. If $n_j = 0$ then $u_j = 0$ and the special case follows. For the last claim, $Q \ge \alpha^2n_j(\lambda + M_j)/D_j \ge \alpha^2n_jM_j/D_j$, so $u_j^2/(D_j^2Q) \le \alpha^2c^2n_j^2D_j/(D_j^2\alpha^2n_jM_j) = c^2n_j/(D_jM_j) \le 1/M_j$, using $c^2n_j \le D_j$. □

Lemma 9 (State of a policy on the template). *Let π be any policy on an instance of the template and t a round. Then V_t is the matrix of Lemma 8 with $S = \eta^2 s$, $M_j = m_jL^2$, n_j the probe counts, and $q_j = p_\eta$'s direction vector. For a direction $j \ge 2$ with $n_j = 0$:*

(a) $\|e_j\|^2_{V_t^{-1}} = 1/(\lambda + m_j L^2)$ *and* $\widehat{\theta}_{t,j} = LS_j^{(m_j)}/(\lambda + m_j L^2)$*, where* $S_j^{(m_j)}$ *is the sum of the noises on the* m_j *pulls of* b *in direction* e_j*;*

(b) $\|g_\eta\|^2_{V_t^{-1}} = \eta^2/Q \le 1/\lambda$ *and* $\|p_\eta\|^2_{V_t^{-1}} = \alpha^2/Q + c^2/(\lambda + m_j L^2) \le 2/\lambda$ *for the probe in direction* e_j*.*

For any direction with $m_j \ge 1$*,* $\|e_j\|^2_{V_t^{-1}} \le 2/(m_j L^2)$*.*

Proof. The form of V_t is immediate from the definitions. For (a), Lemma 8 with $n_j = 0$ gives $(V_t^{-1})_{jj} = 1/D_j = 1/(\lambda + m_j L^2)$ and that the j-th row of V_t^{-1} vanishes off the diagonal. The j-th coordinate of the response vector b_t is $L\sum r$ over the pulls of b in direction e_j plus $c\sum r$ over the pulls of p in that direction; the second sum is empty, and since $\mu(b) = 0$ the first is $LS_j^{(m_j)}$. Hence $\widehat{\theta}_{t,j} = (V_t^{-1})_{jj} LS_j^{(m_j)}$. For (b), $\|g_\eta\|^2_{V_t^{-1}} = \eta^2 (V_t^{-1})_{11} = \eta^2/Q$ and $\|p_\eta\|^2_{V_t^{-1}} = \alpha^2 (V_t^{-1})_{11} + 2\alpha c (V_t^{-1})_{1j} + c^2 (V_t^{-1})_{jj} = \alpha^2/Q + c^2/D_j$, using $(V_t^{-1})_{1j} = 0$; both bounds use $Q \ge \lambda$, $D_j \ge \lambda$, $\alpha \le 1$, $c \le 1$, $\eta \le 1$. The last claim is the last claim of Lemma 8 with $M_j = m_j L^2 \ge L^2 \ge 1$ together with $1/D_j \le 1/(m_j L^2)$, and it holds trivially when $n_j = 0$. □

C.3 Three reusable facts

Lemma 10 (Optimism trap)**.** *Let* π *be a policy on an instance of the template and suppose LinUCB acts for* π *at round* t*. On* $\mathcal{E}_{\mathrm{conf}}(\pi) \cap \mathcal{E}_{\mathrm{noise}}(\pi)$*, if the direction* $j = j(t)$ *satisfies* $n_j(t) = 0$ *and* $m_j(t) < m_0$*, then*

$$U_t(b_t) \;\ge\; \tfrac{3}{4}\,\frac{\beta L}{\sqrt{\lambda + m_j(t) L^2}} \;>\; \max\{U_t(g_\eta), U_t(p_{t,\eta})\},$$

where the probe term is omitted if the probe is not offered at t*. In particular LinUCB plays* b_t*.*

Proof. Write $m = m_j(t)$ and $D = \lambda + mL^2$. By Lemma 9(a), $U_t(b_t) = L\widehat{\theta}_{t,j} + \beta L \|e_j\|_{V_t^{-1}} = L^2 S_j^{(m)}/D + \beta L/\sqrt{D}$. On $\mathcal{E}_{\mathrm{noise}}$, $|L^2 S_j^{(m)}/D| \le L^2 \sigma \sqrt{2m\ell_1}/D \le L\sigma\sqrt{2\ell_1}/\sqrt{D}$, since $L\sqrt{m} \le \sqrt{D}$; by Lemma 6 this is at most $(\beta/4) L/\sqrt{D}$. This gives the first inequality. For the second, by Lemma 7 and Lemma 9(b), $U_t(g_\eta) \le \eta + 2\beta\eta/\sqrt{Q} \le 1 + 2\beta/\sqrt{\lambda}$ and $U_t(p_{t,\eta}) \le (1 - \varepsilon)\eta + 2\beta\sqrt{2/\lambda} \le 1 + 2\sqrt{2}\,\beta/\sqrt{\lambda}$. Since $\beta = 2\sqrt{\lambda}$, $1 \le \beta/(2\sqrt{\lambda})$, so both indices are at most $(\tfrac{1}{2} + 2\sqrt{2})\beta/\sqrt{\lambda}$. It remains to check $\tfrac{3}{4} L/\sqrt{D} > (\tfrac{1}{2} + 2\sqrt{2})/\sqrt{\lambda}$, i.e. $D < \lambda L^2 \,(3/4)^2/(\tfrac{1}{2} + 2\sqrt{2})^2$. Numerically $(3/4)^2/(\tfrac{1}{2} + 2\sqrt{2})^2 = 0.5625/11.0784 > 0.0507$. Since $m \le m_0 - 1 < \lambda/30$, $D < \lambda + \lambda L^2/30 = \lambda L^2 (1/L^2 + 1/30) \le \lambda L^2 (1/64 + 1/30) < 0.0491\,\lambda L^2$, using $L \ge 8$. The inequality holds with room to spare. □

Lemma 11 (Bad-pull cap)**.** *Let* π *be a policy on an instance of the template. On* $\mathcal{E}_{\mathrm{conf}}(\pi)$*, at any round* t *at which LinUCB acts for* π *and plays* b_t *with direction* j*,* $m_j(t) \le 8\beta^2/\eta^2$*; if moreover* $n_j(t) = 0$*, then* $m_j(t) \le 4\beta^2/\eta^2$*. Consequently the number of rounds at which LinUCB acting for* π *plays* b *in direction* e_j *is at most* $8\beta^2/\eta^2 + 1$*, and at most* $4\beta^2/\eta^2 + 1$ *if no probe in direction* e_j *is ever played by* π*.*

Proof. By Lemma 2, a bad arm with gap η is played only if $\eta \le 2\beta L \|e_j\|_{V_t^{-1}}$, i.e. $\|e_j\|^2_{V_t^{-1}} \ge \eta^2/(4\beta^2 L^2)$. If $m_j(t) \ge 1$, Lemma 9 gives $\|e_j\|^2_{V_t^{-1}} \le 2/(m_j(t) L^2)$, so $m_j(t) \le 8\beta^2/\eta^2$; with $n_j(t) = 0$ the sharper bound $\|e_j\|^2_{V_t^{-1}} = 1/(\lambda + m_j(t) L^2) \le 1/(m_j(t) L^2)$ gives $m_j(t) \le 4\beta^2/\eta^2$. Since $m_j(\cdot)$ increases by one at each such round, the number of rounds at which b in direction e_j is played with $m_j(t) \le M$ is at most $M + 1$. □

Lemma 12 (Expert routing)**.** *Let* $\varepsilon L < c$ *and let* E_w *be the frozen linear policy with* $\widehat{\theta}_{\mathsf{E}} = e_1 + w\sum_{j \ge 2} e_j$ *on the instance with* $g_\eta, b_t, p_{t,\eta}$*.*

(a) If $w \in (\eta/L, (1 - \varepsilon)\eta/(L - c))$*, a nonempty interval, then* E_w *plays* $p_{t,\eta}$ *whenever the probe is offered and* b_t *otherwise.*

(b) If $w \in (\varepsilon\eta/c, \eta/L)$, a nonempty interval, then E_w plays $p_{t,\eta}$ whenever the probe is offered and g_η otherwise.

All three comparisons are strict.

Proof. The scores are $\langle\widehat{\theta}_{\mathsf{E}}, g_\eta\rangle = \eta$, $\langle\widehat{\theta}_{\mathsf{E}}, b_t\rangle = wL$, and $\langle\widehat{\theta}_{\mathsf{E}}, p_{t,\eta}\rangle = (1-\varepsilon)\eta + wc$. The probe beats g_η iff $wc > \varepsilon\eta$ and beats b_t iff $w < (1-\varepsilon)\eta/(L-c)$; b_t beats g_η iff $w > \eta/L$. The interval in (a) is nonempty iff $(1-\varepsilon)\eta L > \eta(L-c)$, i.e. $c > \varepsilon L$, and every w in it satisfies $w > \eta/L > \varepsilon\eta/c$ because $c > \varepsilon L$. The interval in (b) is nonempty iff $\varepsilon\eta/c < \eta/L$, again $c > \varepsilon L$; for w in it, $w < \eta/L < (1-\varepsilon)\eta/(L-c)$. □

Lemma 13 (Widths at handoff after n probes in each direction)**.** *Let $V = \lambda I + n\sum_{j=2}^{K+1} q_j q_j^\top$ with $q_j = \alpha e_1 + c e_j$, $\alpha > 0$, $n \geq 1$. Then for every $j \in \{2, \dots, K+1\}$,*

$$\|e_j\|_{V^{-1}}^2 \;\leq\; \frac{1}{nc^2} + \frac{1}{K\lambda}, \qquad \textit{and} \qquad Q = \lambda + \frac{\alpha^2 K n\lambda}{\lambda + nc^2} \;\geq\; \lambda\Big(1 + \frac{\alpha^2 K}{2c^2}\Big) \;\textit{if } nc^2 \geq \lambda.$$

Proof. Apply Lemma 8 with $S = 0$, $M_j = 0$, $n_j = n$: $D_j = D := \lambda + nc^2$, $u_j = \alpha c n$, and $Q = \lambda + \alpha^2 K n\lambda/D$. Then $(V^{-1})_{jj} = 1/D + (\alpha c n)^2/(D^2 Q)$. The first term is at most $1/(nc^2)$. For the second, $Q \geq \alpha^2 K n\lambda/D$, so $(\alpha c n)^2/(D^2 Q) \leq (\alpha c n)^2 D/(D^2 \alpha^2 K n\lambda) = c^2 n/(DK\lambda) \leq 1/(K\lambda)$, using $c^2 n \leq D$. The bound on Q follows from $D \leq 2nc^2$ when $nc^2 \geq \lambda$. □

C.4 Proof of Lemma 1

Fix u. At a round $t \in I$ with $\langle x_{t,\pi_t}, u\rangle = 0$ the contribution to the coverage is zero. At a round with $\langle x_{t,\pi_t}, u\rangle \neq 0$, the definition (1) gives $\Delta_{t,\pi_t}/\langle x_{t,\pi_t}, u\rangle^2 \geq \pi_t(u) \geq \pi(u)$, i.e. $\langle x_{t,\pi_t}, u\rangle^2 \leq \Delta_{t,\pi_t}/\pi(u)$ (if $\pi(u) = 0$ the bound is vacuous). Summing over I gives the first claim. For the second, let u be a unit vector and $V = V_{\tau+1} = \lambda I + M_\tau$. By Cauchy–Schwarz in the V inner product, $1 = (u^\top u)^2 = (u^\top V^{1/2} V^{-1/2} u)^2 \leq (u^\top V u)(u^\top V^{-1} u)$, so $\|u\|_{V^{-1}}^2 \geq 1/(\lambda + u^\top M_\tau u)$, and $u^\top M_\tau u = \sum_{t \leq \tau} \langle x_{t,A_t}, u\rangle^2 \leq R_{[1,\tau]}(\mathsf{E})/\pi(u)$ by the first claim applied to the expert on $[1, \tau]$. □

C.5 Proof of Theorem 2

We work on $\mathcal{E}_{\mathrm{conf}}(\mathsf{C}) \cap \mathcal{E}_{\mathrm{noise}}(\mathsf{C}) \cap \mathcal{E}_{\mathrm{conf}}(\mathsf{H}_{\tau_0})$, which has probability at least $1 - 3\delta$ by Lemma 6. Set $\eta = 1$.

(i) The expert. By Lemma 12(a), E_w plays p_t at every $t \leq \tau_0$ (gap ε) and b_t at every $t > \tau_0$ (gap 1), so $R_T(\mathsf{E}) = \varepsilon K n + (T - Kn)$.

(ii) Cold LinUCB, lower bound. We claim that at every round $t \in [T]$ with $m_{j(t)}(t) < m_0$, cold LinUCB plays b_t. We argue by induction on t. Suppose the claim holds for all rounds before t and let $j = j(t)$ with $m_j(t) < m_0$. If cold LinUCB had played the probe in direction e_j at some earlier round t' (so $j(t') = j$), then $m_j(t') \leq m_j(t) < m_0$ and the claim at t' says it played $b_{t'}$, a contradiction; hence $n_j(t) = 0$. Lemma 10 now gives that LinUCB plays b_t. (The base case $t = 1$ is the same argument with no earlier rounds.) Direction e_j is offered at exactly n rounds in $[1, Kn]$, and at each of the first $\min\{n, m_0\}$ of them $m_j < m_0$, so cold LinUCB plays the bad arm, paying 1 each time. With $n \geq m_0$ this gives $R_{[1,Kn]}(\mathsf{C}) \geq K m_0$, and $A_{\tau_0} = R_{[1,Kn]}(\mathsf{C}) - R_{[1,Kn]}(\mathsf{E}) \geq K m_0 - \varepsilon K n$ by (i).

(iii) Cold LinUCB, upper bound. Cold LinUCB acts at every round. By Lemma 11 it plays b in each of the K directions at most $8\beta^2 + 1$ times over the horizon, each costing 1. Probes exist only at rounds $t \leq Kn$ and cost ε each, so probe pulls cost at most $\varepsilon K n$. Pulls of g cost nothing. Hence $R_T(\mathsf{C}) \leq K(8\beta^2 + 1) + \varepsilon K n$.

(iv) The hybrid. During $t \leq \tau_0$ the hybrid plays the expert's actions, which by (i) are the probes, each direction exactly n times. Hence $V_{\tau_0+1} = \lambda I + n\sum_{j=2}^{K+1} p^{(j)} p^{(j)\top}$ with $p^{(j)} = \alpha e_1 + c e_j$, and Lemma 13 gives, for every bad direction, $\|e_j\|_{V_{\tau_0+1}^{-1}}^2 \leq 1/(nc^2) + 1/(K\lambda)$. With $nc^2 \geq 36\lambda L^2 =$

$9\beta^2L^2$ and $K\lambda \geq 36\lambda L^2 = 9\beta^2L^2$ this is at most $2/(9\beta^2L^2)$, so $2\beta L\|e_j\|_{V_{\tau_0+1}^{-1}} \leq 2\sqrt{2}/3 < 1$. Now fix $t > \tau_0$. The hybrid's design matrix satisfies $V_t \succeq V_{\tau_0+1}$, so by Lemma 3 $\|e_{j(t)}\|_{V_t^{-1}} \leq \|e_{j(t)}\|_{V_{\tau_0+1}^{-1}}$, and by Lemma 7, $U_t(b_t) \leq 0 + 2\beta L\|e_{j(t)}\|_{V_t^{-1}} < 1 \leq U_t(g)$. Since $\mathcal{A}_t = \{g, b_t\}$, LinUCB plays g, with gap 0. Therefore $R_{[\tau_0+1,T]}(\mathsf{H}_{\tau_0}) = 0$, $R_T(\mathsf{H}_{\tau_0}) = R_{[1,\tau_0]}(\mathsf{E}) = \varepsilon Kn$, $V_{\tau_0} = R_{[Kn+1,T]}(\mathsf{C}) - 0 \geq 0$, and $B_{\tau_0} = (T - Kn) - 0$.

The ratios. From (ii), (iii), (iv): $R_T(\mathsf{C})/R_T(\mathsf{H}_{\tau_0}) \geq Km_0/(\varepsilon Kn) = m_0/(\varepsilon n)$ and $\leq (K(8\beta^2 + 1) + \varepsilon Kn)/(\varepsilon Kn) = 1 + (8\beta^2+1)/(\varepsilon n)$; the expert ratio is (i) over (iv). At $n = \lceil 36\lambda L^2/c^2 \rceil$, since $36\lambda L^2/c^2 \geq 36 \cdot 60 \cdot 64 > 1$ we have $n \leq 37\lambda L^2/c^2$, and $m_0 \geq \lambda/30 - 1 \geq \lambda/60$ because $\lambda \geq 60$; hence $m_0/(\varepsilon n) \geq (\lambda/60)c^2/(37\varepsilon\lambda L^2) = c^2/(2220\,\varepsilon L^2)$. For the upper bound, $8\beta^2 + 1 = 32\lambda + 1$ and $n \geq 36\lambda L^2/c^2$ give $(8\beta^2 + 1)/(\varepsilon n) \leq (32\lambda + 1)c^2/(36\varepsilon\lambda L^2) \leq c^2/(\varepsilon L^2)$ since $\lambda \geq 1$. □

C.6 Proof of Corollary 4

Let $\varepsilon \geq c^2/L^2$ and $n \geq 36\lambda L^2/c^2$, so that $\varepsilon n \geq (c^2/L^2)(36\lambda L^2/c^2) = 36\lambda$. With $\beta = 2\sqrt{\lambda}$ from (4) we have $8\beta^2 + 1 = 32\lambda + 1$, hence

$$\frac{8\beta^2+1}{\varepsilon n} \;\leq\; \frac{32\lambda+1}{36\lambda} \;=\; \frac{8}{9} + \frac{1}{36\lambda} \;\leq\; \frac{8}{9} + \frac{1}{2160} \;<\; 1,$$

using $\lambda \geq 60$. The upper bound of (5) then gives $R_T(\mathsf{C})/R_T(\mathsf{H}_{\tau_0}) \leq 1 + (8\beta^2+1)/(\varepsilon n) < 2$. □

C.7 Proof of Proposition 9

Events. All arm vectors of $\mathcal{J}$ have norm at most L: $\|b'_{j'}\|_2^2 = 1 + L'^2 = 145 \leq L^2$ since $L \geq 13$, and the others are as in Lemma 6. Let $\mathcal{E}'_{\mathrm{conf}}(\mathsf{C})$ be the event that $\|\widehat{\theta}_t - \theta^\star\|_{V_t} \leq \frac{5}{4}\sqrt{\lambda}$ for every t on cold LinUCB's trajectory. By Theorem 2 of [1] and the computation in the proof of Lemma 6, the self-normalized radius is at most $\sigma\sqrt{d\ell_0} + \sqrt{\lambda} \leq \frac{1}{4}\sqrt{\lambda} + \sqrt{\lambda}$ under $\lambda \geq 16\sigma^2 d\ell_0$, so $\mathbb{P}(\mathcal{E}'_{\mathrm{conf}}(\mathsf{C})) \geq 1-\delta$, and $\mathcal{E}'_{\mathrm{conf}}(\mathsf{C}) \subseteq \mathcal{E}_{\mathrm{conf}}(\mathsf{C})$ since $\frac{5}{4}\sqrt{\lambda} \leq \beta = 2\sqrt{\lambda}$. We work on $\mathcal{E}'_{\mathrm{conf}}(\mathsf{C}) \cap \mathcal{E}_{\mathrm{noise}}(\mathsf{C}) \cap \mathcal{E}_{\mathrm{conf}}(\mathsf{H}_{\tau_0})$, of probability at least $1-3\delta$ (Lemma 6; the noise event concerns the pulls of b in the directions e_j, $j \in J$, and its proof is unchanged). On $\mathcal{E}'_{\mathrm{conf}}(\mathsf{C})$, Cauchy–Schwarz gives $|\langle \widehat{\theta}_t - \theta^\star, x\rangle| \leq \frac{5}{4}\sqrt{\lambda}\|x\|_{V_t^{-1}} = \frac{5}{8}\beta\|x\|_{V_t^{-1}}$ for every x, hence for the index $U_t(x) = \langle \widehat{\theta}_t, x\rangle + \beta\|x\|_{V_t^{-1}}$,

$$\mu(x) + \tfrac{3}{8}\beta\|x\|_{V_t^{-1}} \;\leq\; U_t(x) \;\leq\; \mu(x) + \tfrac{13}{8}\beta\|x\|_{V_t^{-1}}. \tag{9}$$

(i) The expert. The expert plays $p_{j(t)}$ (gap ε) at every $t \leq \tau_0$ and $b_{j(t)}$ (gap 1) at every $t > \tau_0$, so $R_T(\mathsf{E}) = \varepsilon Kn + (T - Kn)$.

(ii) Cold LinUCB in the window. We show by induction on $t \leq \tau_0$ that cold LinUCB plays $b'_{j'(t)}$ at round t. Suppose this holds before t; then no probe and no bad arm has been played, so for every $j \in J$ no played vector has a component along e_j, $V_t e_j = \lambda e_j$, and $V_t^{-1} e_j = e_j/\lambda$. Three width bounds follow. First, $\|g\|_{V_t^{-1}}^2 = (V_t^{-1})_{11} \leq 1/\lambda$ because $V_t \succeq \lambda I$. Second, with $j = j(t)$, $\|p_j\|_{V_t^{-1}}^2 = (1-\varepsilon)^2(V_t^{-1})_{11} + 2(1-\varepsilon)c\, e_1^\top V_t^{-1} e_j + c^2(V_t^{-1})_{jj} = (1-\varepsilon)^2(V_t^{-1})_{11} + c^2/\lambda \leq (1+c^2)/\lambda \leq 2/\lambda$, using $e_1^\top V_t^{-1} e_j = e_1^\top e_j/\lambda = 0$. Third, with $j' = j'(t)$ and m the number of earlier pulls of $b'_{j'}$: the only played vector with a component along $e_{j'}$ is $b'_{j'}$ itself, so $e_{j'}^\top V_t e_{j'} = \lambda + mL'^2$, and Cauchy–Schwarz in the V_t inner product gives $\|b'_{j'}\|_{V_t^{-1}}^2 \geq \langle b'_{j'}, e_{j'}\rangle^2/(e_{j'}^\top V_t e_{j'}) = L'^2/(\lambda + mL'^2)$. Direction j' is offered at $\lceil \tau_0/K' \rceil$ rounds in the window, so $m \leq \tau_0/K' \leq \lambda/L'^2$ by the condition on K', and $\|b'_{j'}\|_{V_t^{-1}}^2 \geq L'^2/(2\lambda)$. Now apply (9) with $\beta = 2\sqrt{\lambda}$:

$$U_t(b'_{j'}) \;\geq\; 1 + \tfrac{3}{8}\cdot 2\sqrt{\lambda}\cdot\frac{L'}{\sqrt{2\lambda}} = 1 + \frac{3L'}{4\sqrt{2}} > 7.3, \qquad U_t(g) \;\leq\; 1 + \tfrac{13}{8}\cdot 2\sqrt{\lambda}\cdot\frac{1}{\sqrt{\lambda}} = 4.25, \qquad U_t(p_j) \;\leq\; 1 - \varepsilon + \tfrac{13}{8}\cdot 2\sqrt{\lambda}\cdot\frac{\sqrt{2}}{\sqrt{\lambda}} < 5$$

using $L' = 12$. So cold LinUCB plays $b'_{j'(t)}$, whose gap is 0. The base case $t = 1$ is the same computation with $m = 0$. Hence $R_{[1,\tau_0]}(\mathsf{C}) = 0$, and at round $\tau_0 + 1$ every direction e_j, $j \in J$, is untouched: $m_j = n_j = 0$.

(ii) Cold LinUCB on the tail. For $t > \tau_0$ the action set is $\{g, b_{j(t)}\}$ and the directions e_j, $j \in J$, are touched only by the bad arms, so for each of them e_j is an eigenvector of V_t with eigenvalue $\lambda + m_j(t)L^2$, the j-th coordinate of the response vector is $LS_j^{(m_j(t))}$, and the conclusions of Lemma 9(a) hold verbatim; Lemma 9(b)'s bound $U_t(g) \le 1 + 2\beta/\sqrt{\lambda}$ holds on $\mathcal{E}_{\mathrm{conf}}(\mathsf{C})$ by Lemma 7 and $\|g\|_{V_t^{-1}} \le 1/\sqrt{\lambda}$. These are the only properties of the template that the proofs of Lemmas 10 and 11 use for a direction with $n_j = 0$, so both lemmas apply to cold LinUCB on $\mathcal{J}$ at every tail round (with no probe term). The induction of Theorem 2(ii) now shows that cold LinUCB plays b_t at every tail round with $m_{j(t)}(t) < m_0$; since $T - \tau_0 \ge Km_0$ and the tail directions cycle, each direction is offered at least m_0 times on the tail, giving $R_{[\tau_0+1,T]}(\mathsf{C}) \ge Km_0$. Lemma 11 with $n_j = 0$ bounds the pulls of b in each direction by $4\beta^2 + 1$, and g costs nothing, so $R_{[\tau_0+1,T]}(\mathsf{C}) \le K(4\beta^2 + 1) = K(16\lambda + 1)$.

(iii) The hybrid. During $t \le \tau_0$ the hybrid plays the expert's actions, which by (i) are the probes, each direction $j \in J$ exactly n times; no distractor and no bad arm is played. Hence $V_{\tau_0+1} = \lambda I + n\sum_{j\in J} p_j p_j^\top$, which is the matrix of Theorem 2(iv), and the argument there (Lemma 13, Lemma 3, Lemma 7 on $\mathcal{E}_{\mathrm{conf}}(\mathsf{H}_{\tau_0})$) shows that $U_t(b_t) < 1 \le U_t(g)$ at every $t > \tau_0$, so the hybrid plays g and $R_{[\tau_0+1,T]}(\mathsf{H}_{\tau_0}) = 0$. Therefore $R_T(\mathsf{H}_{\tau_0}) = \varepsilon Kn$.

The terms and the ratios. $A_{\tau_0} = R_{[1,\tau_0]}(\mathsf{C}) - R_{[1,\tau_0]}(\mathsf{E}) = 0 - \varepsilon Kn$; $V_{\tau_0} = R_{[\tau_0+1,T]}(\mathsf{C}) - 0 \in [Km_0, K(16\lambda + 1)]$; $B_{\tau_0} = (T - Kn) - 0$. Since $R_T(\mathsf{C}) = R_{[\tau_0+1,T]}(\mathsf{C})$ and $R_T(\mathsf{H}_{\tau_0}) = \varepsilon Kn$, the ratio lies between $m_0/(\varepsilon n)$ and $(16\lambda + 1)/(\varepsilon n)$. At $n = \lceil 36\lambda L^2/c^2 \rceil$ the lower bound is at least $c^2/(2220\,\varepsilon L^2)$ exactly as in the proof of Theorem 2, and $(16\lambda+1)/(\varepsilon n) \le (16\lambda+1)c^2/(36\varepsilon\lambda L^2) \le c^2/(2\varepsilon L^2)$ since $\lambda \ge 1$. □

C.8 Proof of Proposition 8

Work on $\mathcal{E}_{\mathrm{conf}}(\mathsf{C}) \cap \mathcal{E}_{\mathrm{noise}}(\mathsf{C})$ (probability $\ge 1 - 2\delta$). There are no probes, so $n_j \equiv 0$ for every policy.

(i) Let $t \le Km_0$ and $j = j(t)$. Direction e_j was offered at $\lceil t/K \rceil - 1 \le m_0 - 1$ earlier rounds, so $m_j(t) \le m_0 - 1 < m_0$ for cold LinUCB, and Lemma 10 (with the probe term absent) says cold LinUCB plays b_t. Hence cold LinUCB plays b_t at every $t \le Km_0$, and so does E. Fix $\tau \le Km_0$. The two policies C and H_τ play the same action vectors in the same order during $[1, \tau]$; in the reward-table model they therefore observe the same rewards, so their states $(V_{\tau+1}, b_{\tau+1})$ coincide. From round $\tau + 1$ on both run LinUCB from that state on the same action sets and the same reward table, so they play the same actions at every round. Thus $R_I(\mathsf{C}) = R_I(\mathsf{H}_\tau)$ for every interval I, $A_\tau = R_{[1,\tau]}(\mathsf{C}) - R_{[1,\tau]}(\mathsf{E}) = \tau - \tau = 0$, and $V_\tau = 0$.

(ii) For any τ, $R_T(\mathsf{H}_\tau) \ge R_{[1,\tau]}(\mathsf{E}) = \tau$. By Lemma 11 with $n_j \equiv 0$, cold LinUCB plays each bad direction at most $4\beta^2 + 1$ times, so $R_T(\mathsf{C}) \le K(4\beta^2 + 1) = K(16\lambda + 1)$. For $\tau > Km_0$, $R_T(\mathsf{C})/R_T(\mathsf{H}_\tau) \le K(16\lambda + 1)/(Km_0) \le (16\lambda + 1)/(\lambda/60) = 960 + 60/\lambda \le 961$. □

C.9 Proof of Proposition 10

Work on the event of Theorem 2. By Lemma 12(b), E_w plays every probe during $[1, \tau_0]$ and g afterwards, so $R_T(\mathsf{E}) = \varepsilon Kn$. The hybrid's routed rounds are the same probes as in Theorem 2(iv), so its state at handoff and its subsequent play are identical to those established there: $R_T(\mathsf{H}_{\tau_0}) = \varepsilon Kn$ and $R_{[\tau_0+1,T]}(\mathsf{H}_{\tau_0}) = 0$. Hence $B_{\tau_0} = 0 - 0 = 0$. Cold LinUCB does not depend on the expert, so A_{τ_0} and V_{τ_0} are as in Theorem 2. □

C.10 Proof of Proposition 11

Work on $\mathcal{E}_{\mathrm{conf}}(\mathsf{C}) \cap \mathcal{E}_{\mathrm{conf}}(\mathsf{H}_{\tau_0}) \cap \mathcal{E}_{\mathrm{noise}}(\mathsf{H}_{\tau_0})$ (probability $\ge 1 - 3\delta$). Here $d = 2K + 1$; coordinates $2, \ldots, K + 1$ are the bad directions and $K + 2, \ldots, 2K + 1$ the nuisance directions. The probe p'_t couples e_1 with a nuisance coordinate only, so Lemma 8 applies with the nuisance coordinates playing the role of the probed j's and the bad coordinates having $n_j = 0$ for every policy at every round.

Cold LinUCB. Since $n_j \equiv 0$ on bad coordinates, Lemma 11 gives at most $4\beta^2 + 1$ bad pulls in each of the K directions, costing 1 each; probe pulls occur only at $t \le Kn$ and cost ε each; so $R_T(\mathsf{C}) \le K(4\beta^2 + 1) + \varepsilon Kn$.

The hybrid. During $[1, \tau_0]$ it plays the Kn probes, paying εKn. At handoff, $V_{\tau_0+1} = \lambda I + n\sum_i q_i q_i^\top$ over the K nuisance directions with $q_i = \alpha e_1 + c e_{K+1+i}$, and the bad coordinates are untouched: by Lemma 8 (with $M_j = n_j = 0$ on them), $\|e_j\|^2_{V^{-1}_{\tau_0+1}} = 1/\lambda$ for every bad direction and the estimate has zero j-th coordinate. By Lemma 13 applied to the nuisance block, $Q \ge \lambda(1 + \alpha^2 K/(2c^2))$ once $nc^2 \ge \lambda$, which holds since $n \ge 36\lambda L^2/c^2$. As $\alpha = 1 - \varepsilon \ge 1 - 1/L \ge 7/8$ and $c \le 1$, $Q \ge \lambda(1 + 49K/128) \ge 16\lambda = 4\beta^2$ because $K \ge 36L^2 \ge 2304$. Hence at every $t > \tau_0$, since Q only grows, $U_t(g) \le 1 + 2\beta/\sqrt{Q} \le 2$ by Lemma 7. On the other hand, for $t > \tau_0$ and $j = j(t)$ with $m_j(t) = m$, Lemma 9(a) and $\mathcal{E}_{\mathrm{noise}}(\mathsf{H}_{\tau_0})$ give, exactly as in the proof of Lemma 10, $U_t(b_t) \ge \frac{3}{4}\beta L/\sqrt{\lambda + mL^2}$, which exceeds 2 as soon as $\lambda + mL^2 < \frac{9}{64}\beta^2 L^2 = \frac{9}{16}\lambda L^2$, i.e. $m < \frac{9}{16}\lambda - \lambda/L^2$. With $L \ge 8$, $\frac{9}{16}\lambda - \lambda/L^2 \ge \frac{35}{64}\lambda > \lambda/2$, so this holds whenever $m \le \lfloor \lambda/2 \rfloor - 1$. Thus at every round $t > \tau_0$ at which $m_{j(t)}(t) < \lfloor \lambda/2 \rfloor$ the hybrid plays the bad arm. Each direction is offered at least $\lfloor (T - \tau_0)/K \rfloor \ge \lfloor \lambda/2 \rfloor$ times after τ_0, so each is pulled at least $\lfloor \lambda/2 \rfloor$ times, and $R_{[\tau_0+1,T]}(\mathsf{H}_{\tau_0}) \ge K\lfloor \lambda/2 \rfloor$. Hence $R_T(\mathsf{H}_{\tau_0}) \ge \varepsilon Kn + K\lfloor \lambda/2 \rfloor$.

The ratio and V_{τ_0}. Since $\lambda \ge 60$, $\lfloor \lambda/2 \rfloor \ge \lambda/4$, and for $x \ge 0$, $(a + x)/(b + x) \le \max\{a/b, 1\}$; with $a = K(16\lambda + 1)$, $b = K\lambda/4$, $x = \varepsilon Kn$ this gives $R_T(\mathsf{C})/R_T(\mathsf{H}_{\tau_0}) \le 64 + 4/\lambda \le 65$. Finally $V_{\tau_0} = R_{[\tau_0+1,T]}(\mathsf{C}) - R_{[\tau_0+1,T]}(\mathsf{H}_{\tau_0}) \le R_{[\tau_0+1,T]}(\mathsf{C}) - K\lfloor \lambda/2 \rfloor$. The expert's Gram matrix $n\sum_i q_i q_i^\top$ has range in $\mathrm{span}\{e_1, e_{K+2}, \ldots, e_{2K+1}\}$, which is orthogonal to the span of the bad directions, so the principal angle is 90°. □

C.11 Proof of Proposition 12

On $\mathcal{E}_{\mathrm{conf}}(\mathsf{C})$, Lemma 11 with $n_j \equiv 0$ gives $R_T(\mathsf{C}) \le K(4\beta^2 + 1) = K(16\lambda + 1)$; since $R_T(\pi) \ge 0$ for every π, $(R_T(\mathsf{C}) - R_T(\pi))/T \le K(16\lambda + 1)/T$, and with $\lambda = \sigma^2 d\ell_0$ and $d = K + 1$ this is $O(\sigma^2 K^2 \ell_0/T)$. For the second claim, the certificate $2\beta L\|e_j\|_{V^{-1}_{\tau+1}} < 1$ means $\|e_j\|^2_{V^{-1}_{\tau+1}} < 1/(4\beta^2 L^2)$, while Lemma 1 gives $\|e_j\|^2_{V^{-1}_{\tau+1}} \ge 1/(\lambda + R_{[1,\tau]}(\mathsf{E})/\pi(e_j))$; combining, $R_{[1,\tau]}(\mathsf{E})/\pi(e_j) > 4\beta^2 L^2 - \lambda$. On the template, $\pi(e_j) = \min\{1/L^2, \varepsilon/c^2\}$ while probes are offered; the acquisition regret must exceed $\pi(e_j)(4\beta^2 L^2 - \lambda)$ for each of the K directions that has to be certified, and the contributions of different directions come from disjoint sets of rounds because no action vector of the template has nonzero components on two bad directions. □

C.12 Proof of Proposition 13

The proof of Theorem 2 goes through with η carried along; we indicate the changes. Work on the same three events. (i) By Lemma 12(a), $R_T(\mathsf{E}) = \varepsilon\eta Kn + \eta(T - Kn)$. (ii) Lemma 10 was proved for general η, and the induction is unchanged; each forced bad pull now costs η, so $R_{[1,Kn]}(\mathsf{C}) \ge \eta K m_0$. (iii) Lemma 11 gives at most $8\beta^2/\eta^2 + 1$ bad pulls in each direction at cost η, and probe pulls cost $\varepsilon\eta$, so $R_T(\mathsf{C}) \le \eta K(8\beta^2/\eta^2 + 1) + \varepsilon\eta Kn$. (iv) At handoff, Lemma 13 with $\alpha = (1 - \varepsilon)\eta > 0$ gives $\|e_j\|^2_{V^{-1}_{\tau_0+1}} \le 1/(nc^2) + 1/(K\lambda) \le 2\eta^2/(9\beta^2 L^2)$ under $nc^2 \ge 9\beta^2 L^2/\eta^2$ and $K\lambda \ge 9\beta^2 L^2/\eta^2$, i.e. $n \ge 36\lambda L^2/(\eta^2 c^2)$ and $K \ge 36L^2/\eta^2$. Then $U_t(b_t) \le 2\beta L\|e_j\|_{V_t^{-1}} \le (2\sqrt{2}/3)\eta < \eta \le U_t(g_\eta)$ for all $t > \tau_0$ by Lemmas 3 and 7, so the hybrid plays g_η and $R_T(\mathsf{H}_{\tau_0}) = \varepsilon\eta Kn$. The ratio bounds follow: the lower bound is $\eta K m_0/(\varepsilon\eta Kn) = m_0/(\varepsilon n)$, and at $n = \lceil 36\lambda L^2/(\eta^2 c^2) \rceil \le 37\lambda L^2/(\eta^2 c^2)$ this is at least $(\lambda/60)\eta^2 c^2/(37\varepsilon\lambda L^2) = \eta^2 c^2/(2220\,\varepsilon L^2)$; the upper bound is $1 + (8\beta^2/\eta^2 + 1)/(\varepsilon n) \le 1 + (32\lambda + \eta^2)c^2/(36\varepsilon\lambda L^2) \le 1 + c^2/(\varepsilon L^2)$. □

C.13 Proof of Corollary 1

On the event of Proposition 3, the hybrid plays b_t at every $t > \tau$, so $R_{[\tau+1,T]}(\mathsf{H}_\tau) = T - \tau$, while the routed rounds cost nothing because the expert plays the optimal arm; hence $R_T(\mathsf{H}_\tau) = T - \tau$ and $R_T(\mathsf{E}) = 0$. Cold LinUCB also plays b_t at every round by Proposition 3 with $\tau = 0$, so $R_{[1,\tau]}(\mathsf{C}) = \tau$ and $R_{[\tau+1,T]}(\mathsf{C}) = T - \tau$. Therefore $A_\tau = \tau - 0$, $V_\tau = (T - \tau) - (T - \tau) = 0$, and $B_\tau = 0 - (T - \tau)$. □

C.14 Remarks on the constants

The constants 36, 30, 65, 961, 2220 come from three places: the factor $\frac{3}{4}$ by which noise can reduce a bad arm's index below its bonus, the factor $\frac{1}{2} + 2\sqrt{2}$ by which the good arm's or the probe's index can exceed its mean at a cold start, and the choice $\beta = 2\sqrt{\lambda}$. None of them is tight. What matters in Theorem 2 is the structure of (5): the lower bound is (forced cold pulls per direction) divided by (probe cost per direction), the upper bound is (maximal cold pulls per direction) divided by the same, and both numerators are independent of ε, c, and K. The gap between $m_0 \asymp \lambda$ and $8\beta^2 \asymp \lambda$ is a constant factor, and that is why the multiplier is pinned to the price ratio up to constants. The same structure with η gives the bounds in Proposition 13, where the looseness is a factor η^2 rather than a constant because the forced pulls in Lemma 10 are counted before the good arm's index has come down to its mean; a sharper count of cold LinUCB's pulls after the good arm is well estimated, as in the proof of Proposition 11, would remove it, but with probes available to cold LinUCB the required control of the estimate in a coupled coordinate is more delicate and we have not carried it out.

D Numerical Validation of the Catalog

Every claim in Section 3.1 is a statement about an explicitly constructed instance, so each can be checked by simulating that instance. We do so for the arrowhead identity and for each catalog row whose proof asserts a numerical bound. The simulations use the parameters named in the corresponding proof, 10 seeds, and no tuning; they are reported in Table 10. Their purpose is to catch an error in a constant or an inequality direction, not to estimate anything: a proved claim that fails here is a bug in the proof.

Table 10: Simulated instances against their proved claims. "Claim" is what the cited result asserts; "simulated" is what the instance produces. Row numbers refer to Table 1. Each block uses the instance and parameters named in the corresponding proof, so the absolute magnitudes are not comparable across blocks; the first four row-5 entries are the instance at $\varepsilon = 0.002$ and the last two the ε-sweep instance.

Result	Quantity	Claim	Simulated
Lemma 8	max entrywise error, 200 trials	0	$< 10^{-10}$
Prop. 2	cold regret	$T = 2{,}000$	2,000
	warm regret at $m = 20$	0	0 (all seeds)
Prop. 3	tail regret, good expert	$T - \tau$	$T - \tau$ exactly
	tail regret, bad expert	$T - \tau$	$T - \tau$ exactly
Prop. 8 (row 4)	hybrid vs. cold, $\tau \le Km_0$	identical	identical
	multiplier	$\le 4\beta^2 + 1 = 961$	0.343
Thm. 2 (row 5)	expert regret, part (i)	$\varepsilon Kn + (T - Kn) = 201{,}244$	201,244
	cold regret, part (iii)	$\le K(8\beta^2{+}1) + \varepsilon Kn = 62{,}716$	3,820
	hybrid regret, part (iv)	$\varepsilon Kn = 1{,}244$	1,244
	non-good plays after handoff	0	0
	handoff certificate $2\beta L\|e_j\|_{V^{-1}}$	< 1	0.941
	(ratio $-$ 1) $\cdot\, \varepsilon$, 64$\times$ sweep in ε	constant	0.0044–0.0039
Prop. 9 (row 5′)	regret inside window	0	0.000
	probes played	0	0
	tail bad pulls per direction	$[m_0,\, 4\beta^2{+}1] = [2, 961]$	93–94
	tail regret	$\ge Km_0 = 8$	374
Prop. 11 (row 7)	multiplier	≤ 65	0.99
	bad pulls per direction	$\Theta(\beta^2)$	214
Cor. 4 (row 8)	multiplier at $\varepsilon \ge c^2/L^2$	≤ 2	0.63, 0.39, 0.06

Two entries carry more weight than the rest. The certificate row is the only place where the construction could fail silently: Theorem 2 requires a K for which the coverage condition holds, and the simulation confirms that the K the proof constructs does satisfy it ($0.941 < 1$) rather than merely being claimed to. The ε-sweep row tests the mechanism rather than the conclusion. Theorem 2

predicts that the multiplier exceeds one by an amount of order $1/\varepsilon$; over a 64-fold range of ε the product $(\text{ratio} - 1) \cdot \varepsilon$ moves only between 0.0044 and 0.0039, so the predicted scaling is what the instance exhibits, not just the predicted sign.

The gaps between claim and simulation are all in the direction the proofs allow, and are the constants discussed in Appendix C, "Remarks on the constants": cold regret on the row-5 template comes in at $3{,}820$ against a bound of $62{,}716$, and the row-4 and row-7 multipliers at 0.343 and 0.99 against caps of 961 and 65. In every case the instance does what the row says it does—rows 4, 7 and 8 kill the multiplier, row $5'$ collects its saving only on the tail with an empty window—while the numerical bound is loose by the stated constant factors.

E The Dominant Category on AmazonCat-13K

This appendix gives the evidence behind the paragraph "What the transfer carries" of Section 4.3. All numbers are on BERT-768d features unless stated.

The label distribution. On the $50{,}000$ contexts of the online stream the four most frequent of the $13{,}330$ categories are present on 29.8%, 17.0%, 11.3% and 7.8% of contexts (category identifiers 1471, 7961, 7892, 7891). A policy that always predicts the first scores 29.77%; it uses the label distribution, which an online learner is not given. Context-free bandits that must discover it do not: over the same stream, $50{,}000$ rounds and seeds 1–10, UCB1 (one pull of every category first) scores 0.04% and an optimistic empirical-mean greedy rule 2.31%.

Where a run settles. We say a run *settles* on a category if that category accounts for more than 20% of the learner's plays in the last fifth of the horizon. For the dominant category the classification does not depend on the threshold: on the runs of Table 11, moving it anywhere between 10% and 30% changes the number of runs that settle on 1471 by at most one on BERT features and not at all on BGE features. The counts for the less frequent categories are more sensitive on BERT, where runs that settle on 7892 predict it on about a fifth of late rounds. Table 11 reports where runs settle and how accurate the runs that settle on each category are. On BERT features cold LinUCB settles on the dominant category in 1 of 20 seeds; the two multiplying experts move it there in 17 of 20. On BGE features cold LinUCB already settles there in 7 of 20 seeds, which is why the expert has less to add; there the runs that settle on the dominant category reach 41.2%, above the 29.8% of always predicting it, so the learner also uses the context. Knowing only where each run settles predicts mean Router accuracy well: predicting a configuration's accuracy as the frequency of each settled category times the mean accuracy of runs settling there in the *other* configurations gives $R^2 = 0.93$ over 25 BERT configurations and $R^2 = 0.93$ over 7 BGE configurations. These are every configuration that routes fewer than half the rounds: on BERT, the gate at $c \in \{1, 10, 10^2, 10^3\}$, CORRAL, the two offline arms of Table 9 and the transfer-removed ablation of Table 7, for each of the three experts, plus cold LinUCB; on BGE, the gate at $c \in \{0.01, 0.1\}$ for each expert, plus cold LinUCB.

Table 11: Where runs settle on AmazonCat-13K (seeds 1–20; BERT at $c = 100$, BGE at $c = 0.1$). Each cell counts runs that settle on that category; the percentage under each category is its frequency in the stream. The bottom block gives the mean final accuracy of the runs, pooled over the four rows, that settle on each category.

Features	Policy	1471 (29.8%)	7961 (17.0%)	7892 (11.3%)	7891 (7.8%)	none	Accuracy
BERT-768d	Cold LinUCB	1	0	9	1	9	8.34%
	Flan-T5-small	3	0	14	0	3	13.71%
	Flan-T5-base	17	0	3	0	0	25.10%
	Flan-T5-large	17	0	3	0	0	26.90%
BGE-1024d	Cold LinUCB	7	12	0	0	1	33.68%
	Qwen-0.5B	8	9	0	2	1	33.23%
	Qwen-3B	17	2	0	1	0	39.94%
	Qwen-7B	16	3	0	1	0	39.67%
Accuracy of settled runs, BERT		28.08%	—	11.61%	6.24%	5.93%	
Accuracy of settled runs, BGE		41.21%	30.84%	—	24.66%	25.96%	

A handful of routed answers decides it. Table 12 groups every Router run that routes fewer than half the rounds by the number k of routed rounds on which the expert proposed the dominant category and was correct, and reports the fraction that settle on it. On BERT features the fraction rises from 7% at $k = 0$ to 80% at $k = 4$, and every one of the 54 runs with $k \geq 5$ settles there. On BGE features the fraction at $k = 0$ is already 43%, the rate at which the learner finds the category by itself, and again every run with $k \geq 5$ settles there. The pooled runs span experts, values of c and, on BERT, the CORRAL master, so the curve is a property of the learner and the data rather than of one expert. The three BERT experts differ in exactly this count: in the $c = 100$ runs the mean k is 0.20 for Flan-T5-small, 3.70 for Flan-T5-base and 4.00 for Flan-T5-large.

Table 12: Runs that settle on the dominant category, by the number k of correct routed answers on it. BERT: 177 runs ($c \in \{1, 10, 10^2, 10^3\}$ and CORRAL). BGE: 99 runs ($c \in \{0.01, 0.1\}$).

k	0	1	2	3	4	≥ 5
BERT-768d	3/45 (7%)	11/29 (38%)	9/17 (53%)	13/17 (76%)	12/15 (80%)	54/54 (100%)
BGE-1024d	25/58 (43%)	3/3	5/9	6/8	8/9	12/12 (100%)

Controlled test. Table 13 changes the expert's answers on a fixed set of 15 contexts and nothing else. The runs use an exact re-implementation of the Router and of cold LinUCB, checked to make the same choice as the experimental runner on every round of a 3,000-round comparison, on 10,000 held-out contexts that do not appear in the online stream, with a horizon of 10,000 rounds and 20 seeds; the seed fixes the order of the contexts and is shared across conditions, so every comparison is paired. Multipliers are relative to cold LinUCB on the same runs (4.63%). $+15$ *on* 1471 replaces 15 wrong answers of Flan-T5-small by the dominant category on contexts that contain it. $+15$ *random* gives the same number of correct answers, on the same contexts where possible, on a uniformly chosen true category other than 1471. $+15$ *on* 7961 gives 15 correct answers on the second most frequent category. -15 *on* 1471 replaces the 15 correct dominant-category answers of Flan-T5-large by wrong answers. The first three conditions raise held-out accuracy by the same 0.15 points; only the category differs.

Table 13: Changing 15 of the expert's 10,000 held-out answers. Tests are paired over 20 seeds against the unmodified expert: a paired t-test on Router accuracy and an exact McNemar test on whether the run settles on 1471.

Expert	Change	Expert acc.	Router acc.	Mult.	Settle on 1471	p (acc. / settle)
Flan-T5-small	none	5.34%	6.14%	1.33×	1/20	—
	+15 random	5.49%	7.63%	1.65×	2/20	0.15 / 1.00
	+15 on 7961	5.49%	9.39%	2.03×	3/20	0.010 / 0.50
	+15 on 1471	5.49%	**12.10%**	**2.62×**	**8/20**	**0.002 / 0.016**
Flan-T5-large	none	5.32%	13.79%	2.98×	10/20	—
	−15 on 1471	5.17%	10.09%	2.18×	6/20	0.067 / 0.39

At equal accuracy, the gain grows with the frequency of the category the added answers point to: none for random categories, a significant gain for the second most frequent category (where 5 of 20 runs settle on that category instead), and the largest gain for the dominant one. Removing the dominant-category answers from Flan-T5-large lowers its multiplier but not significantly; its other answers still lead 6 of 20 runs to the dominant category. Individual seeds are sensitive to any change in the routed data, which can tip a run toward or away from a category, so we compare rates over seeds rather than reading individual pairs.

F Confidence-Gated Meta-Algorithm: Full Description

This appendix expands the brief description of Section 3.7 into the full round-by-round walkthrough, states the deterministic confidence-gated routing rule, and provides the pseudocode used in our experiments. The Router is the adaptive-τ version of the handoff hybrid H_τ of Section 2: it chooses, round by round, whether the expert or LinUCB acts, and feeds every observation to LinUCB. Proposition 14 applies to it with the acquisition term summed over expert-routed rounds.

Rather than encoding prior knowledge through a single improved initialization, we assume access to *multiple frozen policy estimators* $\{(\widehat{\theta}_i, V^{(i)})\}_{i=1}^m$. Each pair should be thought of as a constant policy obtained offline: $\widehat{\theta}_i$ is a fixed linear predictor and $V^{(i)}$ is the corresponding design matrix that captures the geometry of the data on which that predictor was trained. Our goal is not to compete with the single best expert in hindsight, an objective closer in spirit to online model-selection methods such as CORRAL [6, 13], which provide robustness guarantees of the form "perform nearly as well as the best base algorithm." Here we want to collect the data-value term V_τ of Proposition 1 when it is available while paying as little acquisition regret as possible, and to fall back to LinUCB otherwise.

Setting. At each round $t = 1, \dots, T$, the learner observes a finite action set $\mathcal{A}_t$ with feature vectors $\{x_{t,a}\}_{a \in \mathcal{A}_t} \subseteq \mathbb{R}^d$, selects an action $A_t \in \mathcal{A}_t$, and observes reward $r_t = \langle \theta^\star, x_{t,A_t} \rangle + \eta_t$, with η_t conditionally mean-zero sub-Gaussian. We are given m frozen estimators $\{(\widehat{\theta}_i, V^{(i)})\}_{i=1}^m$ and an online LinUCB learner with state $(V_t, b_t, \widehat{\theta}_t)$, where $\widehat{\theta}_t = V_t^{-1} b_t$. The frozen estimators are never updated; only LinUCB is updated online.

Expert proposals and geometry-aware confidence. Each frozen expert $i \in [m]$ proposes the greedy action $a_t^{(i)} \in \arg\max_{a \in \mathcal{A}_t} \langle \widehat{\theta}_i, x_{t,a} \rangle$. To measure how trustworthy expert i is on the current round, we evaluate the uncertainty of its proposed action under its own design geometry,

$$u_t^{(i)} \;:=\; \big\| x_{t,a_t^{(i)}} \big\|_{(V^{(i)})^{-1}}, \qquad \|x\|_{M^{-1}} \;:=\; \sqrt{x^\top M^{-1} x}.$$

The interpretation is simple. If the proposed action lies in directions that are well covered by the offline data underlying expert i, then $u_t^{(i)}$ is small and the expert is more trustworthy; if the current context points in directions poorly represented in $V^{(i)}$, then $u_t^{(i)}$ is large and the expert should be trusted less. We treat LinUCB itself as an additional candidate, indexed by 0, with proposal $a_t^{(0)} \in \arg\max_a \langle \widehat{\theta}_t, x_{t,a} \rangle + \alpha_t \|x_{t,a}\|_{V_t^{-1}}$ and uncertainty score $u_t^{(0)} := \|x_{t,a_t^{(0)}}\|_{V_t^{-1}}$.

Deterministic confidence-gated selection. The natural rule is to prefer the frozen expert whose proposed action has the smallest uncertainty score, unless the online learner is already at least as confident. Concretely, with $i_t^\star \in \arg\min_{i \in [m]} u_t^{(i)}$,

$$I_t \;=\; \begin{cases} i_t^\star, & \text{if } u_t^{(i_t^\star)} < u_t^{(0)}, \\ 0, & \text{otherwise,} \end{cases} \qquad A_t \;:=\; a_t^{(I_t)}.$$

The master is entirely deterministic. Early in learning, LinUCB typically has a large confidence radius, so a frozen expert may be preferred; as data accumulate, V_t grows and the LinUCB uncertainty shrinks, and eventually the adaptive learner becomes the most confident option and takes over automatically.

Optional permanent handover to LinUCB. To formalize the idea that LinUCB should eventually dominate, we include a one-way switching criterion. With upper-confidence scores $U_t^{(i)} = \langle \widehat{\theta}_i, x_{t,a_t^{(i)}} \rangle + \alpha_i \|x_{t,a_t^{(i)}}\|_{(V^{(i)})^{-1}}$ for each frozen expert and lower-confidence score $L_t^{(0)} = \langle \widehat{\theta}_t, x_{t,a_t^{(0)}} \rangle - \alpha_t \|x_{t,a_t^{(0)}}\|_{V_t^{-1}}$ for LinUCB, if $L_t^{(0)} \geq \max_{i \in [m]} U_t^{(i)}$ for K consecutive rounds we permanently hand control to LinUCB.

Distinction from randomized model selection. The deterministic, sample-by-sample geometric routing distinguishes Algorithm 1 from CORRAL-style randomized masters in two ways. First, it can exploit context-level information about which estimator is reliable on *this particular* action set; second, because it is deterministic, the choice of which rounds to route is a function of the current widths rather than of a sampled index. Both rules feed the played action and the observed reward into (V_t, b_t), so both produce the online warm start (M_τ, c_τ) whose value is V_τ in Proposition 1; the decomposition itself does not distinguish them, and any difference between them shows up in which rounds are routed, i.e., in A_τ and in the composition of M_τ.

Algorithm 1 Confidence-Gated Master over Frozen Policies and LinUCB

Require: Frozen estimators $\{(\widehat{\theta}_i, V^{(i)})\}_{i=1}^m$, regularization $\lambda > 0$, confidence parameters $\{\alpha_t\}_{t=1}^T$

Initialize LinUCB state:

$$V_1 = \lambda I_d, \qquad b_1 = 0, \qquad \widehat{\theta}_1 = 0$$

Set HANDOVER $\leftarrow$ False

for $t = 1, \ldots, T$ **do**

  Observe action set $\mathcal{A}_t$

  Compute LinUCB proposal

$$a_t^{(0)} \in \arg\max_{a \in \mathcal{A}_t} \left\{ \langle \widehat{\theta}_t, x_{t,a} \rangle + \alpha_t \|x_{t,a}\|_{V_t^{-1}} \right\}$$

  Set $u_t^{(0)} \leftarrow \|x_{t,a_t^{(0)}}\|_{V_t^{-1}}$

  for $i = 1, \ldots, m$ **do**

    Compute frozen proposal

$$a_t^{(i)} \in \arg\max_{a \in \mathcal{A}_t} \langle \widehat{\theta}_i, x_{t,a} \rangle$$

    Set $u_t^{(i)} \leftarrow \|x_{t,a_t^{(i)}}\|_{(V^{(i)})^{-1}}$

  end for

  if HANDOVER $=$ True **then**

    $A_t \leftarrow a_t^{(0)}$

  else

    Choose $i_t^\star \in \arg\min_{i \in [m]} u_t^{(i)}$

    if $u_t^{(i_t^\star)} < u_t^{(0)}$ **then**

      $A_t \leftarrow a_t^{(i_t^\star)}$

    else

      $A_t \leftarrow a_t^{(0)}$

    end if

  end if

  Observe reward r_t

  Update LinUCB:

$$V_{t+1} \leftarrow V_t + x_{t,A_t} x_{t,A_t}^\top, \qquad b_{t+1} \leftarrow b_t + r_t x_{t,A_t}, \qquad \widehat{\theta}_{t+1} \leftarrow V_{t+1}^{-1} b_{t+1}$$

  Compute $U_t^{(i)}$ for all $i \in [m]$ and $L_t^{(0)}$

  if $L_t^{(0)} \geq \max_{i \in [m]} U_t^{(i)}$ for K consecutive rounds **then**

    HANDOVER $\leftarrow$ True

  end if

end for

G Pre-Runtime Estimates: Definitions and Consistency

This appendix gives the formal definitions behind Section 3.8 and the elementary consistency statements that justify estimating them on a held-out sample. Throughout, $\mathcal{D} = \{(\mathcal{A}_s, \{x_{s,a}\}_{a \in \mathcal{A}_s}, \{r_{s,a}\}_{a \in \mathcal{A}_s})\}_{s=1}^n$ is a labeled held-out sample of n contexts drawn from the same distribution as the online stream, with a reward label for every candidate action (for binary Precision@1 this is the ground-truth label), and a_s^{E} is the expert's proposal on context s. Nothing in the online trajectory is used.

G.1 Definitions

Gap estimates. Let $\widehat{\theta}_{\mathrm{or}}$ be the ridge estimate fitted on all $(x_{s,a}, r_{s,a})$ pairs of $\mathcal{D}$ with regularization λ, and $\widehat{\Delta}_{s,a} := \max_{a'} \langle \widehat{\theta}_{\mathrm{or}}, x_{s,a'} \rangle - \langle \widehat{\theta}_{\mathrm{or}}, x_{s,a} \rangle$. For binary reward with a single correct action one may instead use the label directly, $\widehat{\Delta}_{s,a} = \mathbf{1}\{a \neq a_s^\star\}$.

Expert Gram matrix. $\widehat{M}_{\mathsf{E}} := \sum_{s=1}^{n} x_{s,a_s^{\mathsf{E}}} x_{s,a_s^{\mathsf{E}}}^{\top}$. For a target number τ of routed rounds, $(\tau/n)\widehat{M}_{\mathsf{E}}$ is an unbiased estimate of $\mathbb{E}[M_\tau]$ when the expert is followed for τ rounds on contexts from the same distribution.

Width-limited regret under a design. For a positive definite V and radius β,

$$\widehat{\Phi}(V) := \frac{1}{n} \sum_{s=1}^{n} \max_{a \in \mathcal{A}_s} \min\bigl\{\widehat{\Delta}_{s,a},\, 2\beta \|x_{s,a}\|_{V^{-1}}\bigr\}.$$

We evaluate it at $V = \lambda I$ (cold start) and at $V = \lambda I + (\tau/n)\widehat{M}_{\mathsf{E}}$ (after τ expert rounds). The difference $\widehat{\Phi}(\lambda I) - \widehat{\Phi}(\lambda I + (\tau/n)\widehat{M}_{\mathsf{E}})$ is the estimated reduction, at a single future round, in the regret LinUCB can still incur, attributable to the expert's data. It is zero whenever the expert's proposals lie in directions orthogonal to every competitor whose width exceeds its gap (Lemma 3), which is the failure mode the principal-angle proxy detects coarsely.

Acquisition regret. $\widehat{C}_{\mathsf{E}}(\tau) := \tau \cdot \frac{1}{n} \sum_s \widehat{\Delta}_{s,a_s^{\mathsf{E}}}$.

Tail check. $\widehat{\mathrm{acc}}_{\mathsf{E}} := \frac{1}{n} \sum_s \mathbf{1}\{a_s^{\mathsf{E}} = a_s^\star\}$ and $\widehat{\mathrm{acc}}_{\mathrm{or}} := \frac{1}{n} \sum_s \mathbf{1}\{\arg\max_a \langle \widehat{\theta}_{\mathrm{or}}, x_{s,a} \rangle = a_s^\star\}$. The learner's online accuracy is at most about $\widehat{\mathrm{acc}}_{\mathrm{or}}$ (up to the usual generalization gap), so $B_\tau > 0$ requires $\widehat{\mathrm{acc}}_{\mathsf{E}} < \widehat{\mathrm{acc}}_{\mathrm{or}}$ by a margin.

Net-value rule. With $n_{\mathrm{fut}} = T - \tau$ future rounds,

$$\widehat{N}(\tau) := n_{\mathrm{fut}} \bigl[\widehat{\Phi}(\lambda I) - \widehat{\Phi}(\lambda I + (\tau/n)\widehat{M}_{\mathsf{E}})\bigr] - \widehat{C}_{\mathsf{E}}(\tau).$$

A handoff is predicted to multiply when $\widehat{N}(\tau)$ is a large positive fraction of $n_{\mathrm{fut}}\widehat{\Phi}(\lambda I)$ for some moderate τ and the tail check passes. We use "large" to mean at least one third; this threshold is provisional. Two caveats bound what the rule can certify. First, $\widehat{\Phi}(\lambda I)$ is the regret cold LinUCB could incur at round one, not what it incurs on average over its own trajectory, so a positive $\widehat{N}$ is not sufficient. Second, $\widehat{N}$ counts only the saving on *future* rounds and therefore misses the concurrent channel of Proposition 1: on the instance of Theorem 2 the saving is carried by A_τ (cold pays for its exploration inside the routed window at the higher price), the warm widths on the bad arms stay near the gap, $\widehat{\Phi}$ barely moves, and $\widehat{N}(\tau_0) < 0$ even though the handoff provably multiplies. So a negative $\widehat{N}$ argues against a tail-channel (V_τ) multiplier but does not exclude a concurrent-channel (A_τ) one; on the instance of Proposition 9, where the same price ratio is collected as V_τ, $\widehat{\Phi}$ falls below the gap on every bad direction and $\widehat{N}(\tau_0)$ is positive once the tail is long enough. The estimate that covers both channels at once is a *dry run*: replay the held-out contexts (resampled to a horizon T') with rewards drawn from the labels or the ridge fit, simulate cold LinUCB and LinUCB warm-started from $\lambda I + (\tau/n)\widehat{M}_{\mathsf{E}}$, and compare $\widehat{R}_{T'}(\mathsf{C})$ against $\widehat{C}_{\mathsf{E}}(\tau) + \widehat{R}_{[\tau+1,T']}(\mathrm{warm})$. The dry run uses no LLM calls beyond the held-out proposals already collected and no online rounds; its cost is one bandit simulation per candidate τ.

What these estimators do on our data, and why. Neither predicts the size of the observed multiplier, and Section 3.4 says why. On all twelve configurations $\widehat{\Phi}(\lambda I)$ saturates at 1.0000: with 13,330 actions in 768 dimensions the width term $2\beta\|x\|_{(\lambda I)^{-1}}$ exceeds the unit gap for every held-out context, so the clipping in (6) is active everywhere, $\widehat{\Phi}$ cannot move, and $\widehat{N}(\tau) = -\widehat{C}_{\mathsf{E}}(\tau) < 0$ for every expert and every τ we tried ($\tau \in \{1000, 2000, 5000, 10000\}$). The dry run, run at the full horizon $T' = 50{,}000$ over ten seeds with all 13,330 actions scored, does reproduce the observed ordering at $\tau = 1600$ (small 1.021 ± 0.029, base 1.057 ± 0.087, large 1.084 ± 0.082; large over small is significant, paired $t = 2.91$) and its cold-start accuracy matches the online runs (8.46% against 8.39%), but it underestimates the magnitude. The dry run's multiplier is a ratio of regrets, so it is to be compared with the observed regret ratios: its excesses over one, 2.2%, 5.7% and 8.4%, stand against observed excesses of 7.0%, 18.7% and 26.6%, a shortfall by a factor of about 3.2 that is nearly the same for all three experts. It ranks the experts correctly and misses their magnitude by a common factor.

Both failures have the same cause. The estimators are built from $\widehat{M}_{\mathsf{E}}$, that is from coverage, and by Corollary 3 coverage cannot distinguish experts on this data—the price ratios, measured post hoc, are 1.73, 1.59 and 1.58, ordered opposite to the multipliers. What distinguishes them is the transfer, and by Proposition 6 the transfer is the ridge projection of the residuals *at the actions the expert chooses*, which is not a function of $\widehat{M}_{\mathsf{E}}$ and is not available before the expert is run. Predicting the multiplier of a given expert before running it therefore remains open, and Proposition 7 bounds what the second channel can deliver without identifying which expert will realize it.

G.2 Consistency

Lemma 14 (Concentration of the held-out quantities)**.** *Suppose contexts are i.i.d. and $\widehat{\Delta}_{s,a} = \Delta_{s,a}$ (labels are exact). Then for any fixed V, $\widehat{\Phi}(V)$ is an average of n i.i.d. variables in $[0, \Delta_{\max}]$ with mean $\mathbb{E}[\Phi_t(V)]$, so $|\widehat{\Phi}(V) - \mathbb{E}\Phi_t(V)| \le \Delta_{\max}\sqrt{\log(2/\delta)/(2n)}$ with probability $1-\delta$ (Hoeffding); the same holds for $\widehat{C}_{\mathsf{E}}(\tau)/\tau$ and for the two accuracies. For the warm design, $\|\frac{1}{n}\widehat{M}_{\mathsf{E}} - \mathbb{E}[xx^\top]\|_{\mathrm{op}} \le CL^2\sqrt{\log(2d/\delta)/n}$ with probability $1-\delta$ (matrix Bernstein), and since $V \mapsto \|x\|_{V^{-1}}$ is Lipschitz on $\{V \succeq \lambda I\}$ with constant $\|x\|_2/(2\lambda^{3/2})$ in operator norm, $\widehat{\Phi}(\lambda I + (\tau/n)\widehat{M}_{\mathsf{E}})$ is within $O\big(\beta L\tau\lambda^{-3/2}L^2\sqrt{\log(2d/\delta)/n}\big)$ of its population counterpart.*

Proof. The scalar statements are Hoeffding's inequality applied to bounded i.i.d. summands; $\Phi_t(V) \in [0, \Delta_{\max}]$ because it is a maximum of minima each bounded by a gap. The matrix statement is the matrix Bernstein inequality for i.i.d. PSD summands of operator norm at most L^2. For the Lipschitz claim, $\|x\|^2_{V^{-1}} - \|x\|^2_{V'^{-1}} = x^\top V^{-1}(V'-V)V'^{-1}x$, bounded by $\|x\|_2^2\|V'-V\|_{\mathrm{op}}/\lambda^2$, and $|\sqrt{u}-\sqrt{v}| \le |u-v|/(2\sqrt{\min\{u,v\}})$ with $\min\{u,v\} \ge \|x\|_2^2/(\lambda + L^2\tau)$; the stated constant follows after simplification. Plugging into the definition of $\widehat{\Phi}$ and using that $\min\{\cdot,\cdot\}$ and $\max$ are 1-Lipschitz completes the proof. □

When labels are not exact, $\widehat{\Delta}$ carries the error of the ridge fit; this affects the first argument of the minimum only and is the usual price of a gap-dependent quantity.

G.3 Relation to the Four Proxies

The four statistics of Table 2 are cheaper than $\widehat{\Phi}$ and each bounds one ingredient of it:

- *Effective rank* $\widehat{r} = \mathrm{tr}(\widehat{\Sigma})/\|\widehat{\Sigma}\|_{\mathrm{op}}$ of the context covariance. By the elliptical potential bound, cold LinUCB can have at most $O(\widehat{r}\log T)$ rounds at which some played width exceeds a fixed level, so a small $\widehat{r}$ means the cold-start widths, and with them $\Phi_t(V_t^{\mathsf{C}})$, fall quickly on their own.
- *Coherence* $\widehat{\alpha}$, the median over actions of the minimum pairwise cosine. By Lemma 4 the cosine is the transfer coefficient at a cold start up to the factor $\|x\|^2/(\lambda + \|x\|^2)$; positive coherence means cold observations shrink the widths of other arms from the first round.
- *Gap tail*, the fraction of held-out contexts whose estimated runner-up gap is below η. Small gaps cap the first argument of the minimum in Φ_t; a heavy tail means many rounds are cheap whatever the widths are, which by itself argues *against* a large multiplier.
- *Subspace mismatch*, the largest principal angle between the top-$\widehat{r}$ eigenspaces of $\widehat{M}_{\mathsf{E}}$ and of the context covariance. An angle near 90° means the expert's data covers directions that carry little of the online context mass, so by Lemma 3 it cannot reduce the widths that matter. The statistic is a coarse indicator that $\widehat{\Phi}$ supersedes; it is not a deployment rule.

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spaces, and the round-level update rule for the LinUCB sufficient statistics. The held-out sample size for diagnostic computation (5,000 contexts) and the Router's routing rule are also specified. Per-dataset configurations are detailed in Sections 4.2 and 4.3, including the WikiLinks c-sweep results (Table 4). The provisional threshold of the net-value rule is stated in Appendix G.

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Justification: A brief broader-impact paragraph appears at the end of Section 5. Positive impacts include reducing unnecessary LLM inference in sequential decision systems by identifying when cold-start LinUCB suffices, which lowers compute cost and energy use. Potential negative impacts include the risk that LLM-guided contextual-bandit systems may inherit biases from pretrained models or logged data if deployed in recommendation, ranking, or other consequential decision-making pipelines without auditing. The work is primarily theoretical and diagnostic, so concrete deployment harms depend on follow-on applications not studied here.

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